Class 6

Physics

(Solutions)

Table of Content

1.	Units and Measurements	1 – 19
2.	Kinematics	20 – 48
3.	Force and Laws of Motion	49 – 68
4.	Work, Energy, and Power	69 – 95
5.	Heat	96 – 109
6.	Electricity	110 – 117
7.	Magnetism	118 – 126

1.Units and Measurements

Solutions

LEVEL-I

1. Correct option: (A)

To know and understand a physical situation completely, one is likely to measure the quantities such as distance, speed, time, mass, acceleration, force etc. which are called the physical quantities. Physical quantity is that in terms of which laws of physics can be expressed and which can be measured directly or indirectly.

2. Correct options: (A), (B), (C)

Fundamental quantities are the quantities whose measurement is done with the independent units. The quantities in option: (D) Length, time and mass are measured by independent units such as kilogram, meter and second respectively. In all the other options velocity is there which is expressed in terms of derived units which is meters per second. So all the other sets cannot enter into the list of fundamental quantities in any system of units.

3. Correct option: (B)

Mass is a physical quantity as it is used to measure a physical situation. Rest all are the emotional situations

4. Correct option: (**A)**

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. For example, quantities Length, time and mass are measured independently and hence they are fundamental physical quantities.

5. Correct option: (B)

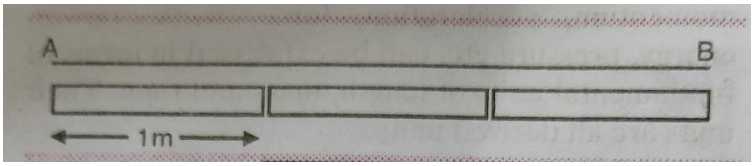
Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. For example, velocity is a derived physical quantity which depends on fundamental quantities distance(length) and time.

6. Correct option: (B)

Measurement is basically a comparison process. It involves the selection of a unit of measurement and comparing the quantity with standard unit. The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod we have to take length as the standard quantity

7. **Correct option: (A)**

In comparing the quantity with the standard unit, we have to find the number of times this unit is contained in the quantity. For example, suppose we have to measure the length of a wire AB and we select meter as the unit of measurement. We place the meter rod along the wire AB and find that it is contained 3 times



in AB. Thus 3 is the numerical value of the length AB, when m is the unit of measurement. We write AB= 3 meter

8. Correct option: (B)

The SI unit of temperature as per the International System of Units is Kelvin which is represented by the symbol K.

9. Correct option: (A)

The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod we have to take meter as the standard unit.

10. Correct option: (B)

In general magnitude of a physical quantity (P)= numerical value × size of it's unit.
For example. If length of a rod = 3 m
Then 3 = numerical value
1m = size of the unit

11. Correct option: (A)

In general magnitude of a physical quantity (P)= numerical value(N) × size of it's unit(U).
Or P=NU
For example. If length of a rod = 3 m
Then 3 = numerical value
1m = size of the unit

12. Correct option: (C)

As we know
physical quantity (P)= numerical value(N) × size of it's unit(U)
Further we know that magnitude of a quantity remains the same, whatever may be the units of it's measurement. Hence we may write

$$P = N_1U_1 = N_2U_2$$

Where
U_1, U_2 are the two unts of measurement of the same quantity and N_1 and N_2 are their respective numerical values.
If $U_1 > U_2$, then $N_1 < N_2$ which means bigger is the unit , smaller is the numerical value and vice – versa. For example,
Length of rod = 3 m = 300 cm
Here, unit(m) is bigger and numerical value 3 is small
Unit (cm) is small and numerical value 300 is bigger.
Hence numerical value and size of unit are inversely proportional to each other.

13. Correct option: (A)

$$\text{Physical quantity} = PR = QS$$

Where
R , S are the two units of measurement of the same quantity and P and Q are their respective numerical values.
If $R > S$, then $P < Q$ which means bigger is the unit , smaller is the numerical value and vice – versa. For example,
Length of rod = 3 m = 300 cm
Here, unit(m) is bigger and numerical value 3 is small
Unit (cm) is small and numerical value 300 is bigger.

14. Correct option: (B)

Energy is a derived physical quantity because it's measurement depends on other fundamental quantities like distance, time, mass etc

15. Correct option: (D)

As we know
physical quantity (P)= numerical value(N) × size of it's unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of it's measurement. Hence we may write

$$P = n_1U_1 = n_2U_2$$

Where

U_1, U_2 are the two unts of measurement of the same quantity and n_1 and n_2 are their respective numerical values.

16. Correct option: (A)

In general magnitude of a physical quantity (P)= numerical value(N) × size of it's unit(U)

Or P=NU

For example. If length of a rod = 3 m

Then 3 = numerical value

1m = size of the unit

17. Correct option(A)

The mole is a unit of amount of substance in the International System of Units

It has the symbol(mol).

Amount of substance (n) = $\frac{\text{Given weight (g)}}{\text{molecular weight (gmol}^{-1})}$

18. Correct option: (C)

Measurement is basically a comparison process. It involves the selection of a unit of measurement and comparing the quantity with standard unit. The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod we have to take length as the standard quantity.

19. Correct option: (B)

1 m = 100 cm

0.5 m = 0.5 × 100 cm

$= \frac{5}{10} \times 100$ cm

0.5 m = 50 cm

20. Correct option(B)

The kilogram is the base unit of mass in the International System of Units (SI). It's unit symbol is kg

Hence

Shortest form of kilogram is Kg.

LEVEL-II

1. Correct option: (B)

1 meter = 100 centimeters

1 kilometer = 100000 centimeters

1 mile = 160934 centimeters

Hence, for smaller length we use centimeter as it is the smallest unit among all the other three options.

2. Correct option: (C)

Strength of Electric current is measured in Amperes, Hence SI unit of strength of electric current is Ampere.

It's symbol is (A).

3. Correct option: (D)

The SI unit of luminous intensity is Candela. It's symbol is (cd).

4. Correct option: (D)

The mole is a unit of amount of substance in the International System of Units
It has the symbol(mol).

Amount of substance (n) = $\frac{\text{Given weight (g)}}{\text{molecular weight (gmol}^{-1})}$

5. Correct options: (B), (C)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. For example, quantities Length, time and mass are measured independently and hence they are fundamental physical quantities.

Volume and Force are derived quantities as their measurement depends on other fundamental quantities

6. Correct option: (A), (B), (C)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities :Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter. These quantities can be measured independently.

Length can't be a derived quantity in any system as it is a fundamental quantity.

7. Correct options: (A), (B), (D)

M.K.S system makes use of meter, kilogram and second as the fundamental units of mass, length and time respectively.

However, these three fundamental quantities were not sufficient to obtain the units of quantities in other branches of physics. So, a new system of units named SI system was Introduced in which there are seven

fundamental quantities :Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter and two Supplementary physical quantities.

8. Correct options: (A), (B)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter. These quantities can be measured independently.

Density and Area are derived quantities as their measurement depends on other physical quantities

9. Correct options: (C), (D)

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. For example, measurement of Acceleration and Density depends on other physical quantities.

10. Correct options: (A), (D)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter. Quantity of work done and amount of force acting are derived quantities as their measurement depends on other physical quantities.

11. Correct options: (A), (B), (C)

Fundamental quantities are the quantities whose measurement is done with the independent units. The quantities Length, time and mass are measured by independent/fundamental units such as kilogram, meter and second respectively. Newton is a derived unit for derived physical quantity (force).

12. Correct options: (A), (D)

Mass of a body is the quantity of matter it contains.

SI unit of mass is Kilogram.

1 kilogram = 1000000 milligrams

Hence, kilogram and milligram are units used to measure mass. Kilometer and millimeter are used to measure distance.

13. Correct option: (D)

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities.

Impulse is a derived quantity which is measured by the total change in linear momentum.

Force is a derived quantity which is measured by product of mass and acceleration.

14. Correct option: (C)

As it is clear from the given formula that speed depends on the distance /length covered by an object in a given time. Hence it depends on length and time.

15. Correct option: (D)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities.

Diameter is measured in terms of units of length i.e meters and we know that length is a fundamental quantity. Hence, diameter is a fundamental quantity.

16. Correct option: (A)

As we know

physical quantity (P)= numerical value(N) × size of it's unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of it's measurement. Hence we may write

$$P = N_1U_1 = N_2U_2$$

Where

U_1, U_2 are the two unts of measurement of the same quantity and N_1 and N_2 are their respective numerical values.

If $U_1 > U_2$, then $N_1 < N_2$ which means larger is the unit , lesser is the numerical value and vice – versa. For example,

Length of rod = 3 m = 300 cm

Here, unit(m) is larger and numerical value 3 is lesser

Unit (cm) is lesser and numerical value 300 is larger

17. Correct option: (A)

As we know

physical quantity (P)= numerical value(N) × size of it's unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of it's measurement. Hence we may write

$$P = N_1U_1 = N_2U_2$$

Where

U_1, U_2 are the two unts of measurement of the same quantity and N_1 and N_2 are their respective numerical values.

If $U_1 > U_2$, then $N_1 < N_2$ which means larger is the unit , lesser is the numerical value and vice – versa. For example,

Length of rod = 3 m = 300 cm

Here, unit(m) is larger and numerical value 3 is lesser

Unit (cm) is lesser and numerical value 300 is larger

Hence numerical value and size of unit are inversely proportional to each other.

18. Correct option: (B)

1 kilogram = 1000 grams or 10^3 grams

now , 1 gram = 1000 mg

so , 1 kilogram = 1000000 mg or 10^6mg

Hence 1kilogram can't be expressed as 10000 grams.

19. (a) – (Q),(T) (b) - (R) (c) - (S) (d) - (P)

(a) SI unit of Temperature is Kelvin or Celsius

(b) SI unit of amount of substance is Mole

(c) SI unit of luminous intensity is Candela

(d) SI unit of Electric charge is Coulomb

20. (a) – (Q), (S) (b) - (R) (c) - (Q), (S) (d) - (R)

(a) Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities.

General quantities are also known as fundamental quantities.

Thermodynamic temperature is a fundamental and general quantity.

(b) Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. For example, Density $=\frac{\text{mass}}{\text{volume}}$

Hence, density depends on mass and volume so it is a derived quantity.

(c) Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities.

General quantities are also known as fundamental quantities.

Mass is a fundamental and general quantity.

(d) Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. Volume depends on other physical quantities for it's measurement.

Supplementary physical quantities are two additional quantities in SI system. Two supplementary quantities are plane angle and solid angle.

Multiples and Sub-Multiples
LEVEL-I

1. Correct option: (A)

$1\text{cm} = \frac{1}{100}\text{m}$

$300\text{ cm} = \frac{300}{100}\text{m}$

300 cm = 3 m

2. Correct option: (A)

1 gram = 10^6 micrograms

Or

1 microgram = 10^{-6}grams

3. Correct option: (B)

(nm) is the symbol for nanometer

$1\text{ cm} = 10^7\text{nm}$

$3\text{cm} = 3 \times 10^7\text{ nm}$

$= 0.3 \times 10^8\text{ nm}$

4. Correct option: (A)

$1J = 10^7 ergs$

Joule is the SI(MKS) unit of energy and ergs is the cgs unit of energy.

5. Correct option: (B)

1 min = 60 s

40 min = 40 ×60 s

= 2400 s

6. Correct option: (A)

Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.

as velocity of light in vacuum $= 3 \times 10^8\ ms^{-1}$

$1year = 365 \times 24 \times 60 \times 60$ seconds

$so, 1\ light\ year = 3 \times 10^8 \times 365 \times 24 \times 60 \times 60$ meter

$1\ light\ year = 9.46 \times 10^{15} m$

7. Correct option: (D)

1 tone = 1000 kg

$= 10^3$ kg

8. Correct option: (C)

$1\ nm = 10^{-9} m$

$400\ nm = 400 \times 10^{-9} m$

9. Correct option: (A)

1 km = 1000 m

1h = 3600 s

$1kmph = \frac{1000}{3600}\ ms^{-1}$

$1kmph = \frac{5}{18}\ ms^{-1} - - - - - - - - (1)$

Given

$$x\ ms^{-1} = y\ kmph$$

$$\frac{x}{y} = \frac{kmph}{ms^{-1}}$$

From equation (1)

$$\frac{x}{y} = \frac{\frac{5}{18}\ ms^{-1}}{ms^{-1}}$$

$$\frac{x}{y} = \frac{5}{18}$$

10. Correct option: (D)

1 meter = 100 cm

$1\ cm = \frac{1}{100} m$

= 0.01 m

11. Correct option: (B)

$1\ cm = 10^7 nm$

$3cm = 3 \times 10^7\ nm$

$= 0.3 \times 10^8\ nm$

12. Correct option: (B)

1 quintal = 100 kg

Kg is a smaller unit of mass as compared to quintal.

13. Correct option: (D)
1 day = 24 hours
1 hour = 60 minutes
1minute = 60 seconds
Hence, 1 day = 24×60×60 seconds
= 86,400 seconds
14. Correct option: (D)
Both option(A) and (B) are the advantages of using scale. It can be easily taken from one place to another and It does not change during measurements.
15. Correct option: (A)
1 centimeter = 10 millimeter
Millimeter is a smaller unit of length as compared to centimeter
16. Correct option: (D)
1 centimeter = 10 millimeters
Or
1 millimeter = 1/10 centimeter
17. correct option: (A)
The diameter of the sun = 1.39×10^{9} m
18. Correct option: (B)
1 quintal = 100 kg
1quintal = 10^{2}kg
19. Correct option: (D)
Diameter of hydrogen atom is approximately 1.06×10^{-10} m. It is defined by the diameter of the spherical
electron cloud around the nucleus.
20. Correct option: (B)
1 km = 1000 m
Meter (m) is smaller unit of length as compared to km
21. Correct option: (A)
1 m = 10^{9} nm
Or
1 nm = 10^{-9}m
(nm) is the symbol for nanometer. It is a smaller unit than meter
22. Correct option: (D)
1nm=10^{-9}m
1m=10^{10}A°
So, 1nm=$10^{-9}\times10^{10}$A°
1 nm =10 A°
23. Correct option: (B)
1μm=10^{-6}m
1m=10^{10}A°
1μm=$10^{-6}\times10^{10}$A°
1μm=10^{4} A°
24. Correct option: (B)
A century is a period of 100 years
So, 1 century = 100 years
25. Correct option: (D)
1 day = 24 hours

1 hour = 60 minutes
1minute = 60 seconds
So, 1 day =24×60 minutes
=1440 min
Hence, 1 day = 24×60×60 seconds
= 86,400 seconds

26. Correct option: (A)

1μ g $=10^{-6}$ g
5μ g $=5\times10^{-6}$ g----(1)
$as, 1\text{g} = 10^{-3}\text{kg}$
So, from equation 1
5μ g $=5\times10^{-6}\times10^{-3}$kg
$=5\times10^{-9}$ kg

27. Correct option: (D)

1 picometer $= 10^{-12}$m
$1\text{m} = 10^{2}\text{cm}$
$so, 1 \text{ picometer} = 10^{-12}\times10^{2}$ cm
1 picometer $= 10^{-10}$ cm

28. Correct option: (A)

1 metric ton is also known as 1 tone
1 metric ton = 1000 kg

29. Correct option: (B)

1 hectometer (hm) = 100 m
Meter (m) is a smaller unit of length as compared to hectometer

30. Correct option: (C)

1 millisecond $= 10^{-3}$seconds
1microsecond $= 10^{-6}$seconds
1 nanosecond $= 10^{-9}$seconds
1shake $= 10^{-8}$seconds

LEVEL-II

1. Correct options(A), (B)

1 metric ton is also known as 1 tone
1 metric ton= 10 quintal
And we know
1 quintal = 100 kg
So,
1 metric ton =10×100 kg
= 1000 kg

2. Correct options: (A), (B), (C), (D)

1μ g $=10^{-6}$ g
5μ g $=5\times10^{-6}$ g----(1)
$as, 1\text{g} = 10^{-3}\text{kg}$
So, from equation 1
5μ g $=5\times10^{-6}\times10^{-3}$kg
$=5\times10^{-9}$ kg - - - - (2)
Also, $1\text{g} = 10^{3}\text{mg}$

So, from equation 1

$5\mu g = 5\times10^{-6}\times10^{3}mg$

$=5\times10^{-3}$ mg

Also, $1kg = 10^{-3}$tonne

So, from equation 2

$5\mu g = 5\times10^{-9}\times10^{-3}$tonne

$= 5\times10^{-12}$tonne

3. Correct option(A), (B)

$1g = 10^{3}mg$

$1\ mg = 10^{-3}\ g$

Also, $1g = 10^{6}\mu g$

So,

$1mg = 10^{-3}\times10^{6}\mu g$

$1mg = 10^{3}\ \mu g$

4. Correct option(A), (B), (C), (D)

1 day = 24 hours

1 hour = 60 minutes

1minute = 60 seconds

So,

1 day =24×60 minutes

=1440 min

Hence, 1 day = 24×60×60 seconds

= 86,400 seconds

1day = 1/365 of an year

5. Correct options(A), (D)

$1\ \mu s = 10^{-15}$ giga second

$1\ \mu s = 100$ shake

6. Correct options(B), (C), (D)

Multiples: For expressing larger measurements, we use Some prefixes with the units known as multiples. For example

$1\ km = 10^{3}m$

Submultiples : For expressing smaller measurements, we use Some prefixes with the units known as submultiples. For example

1 milli metre = 10^{-3}m

1 deca metre= 10 m

1 milli metre = 10^{-3}m

1 centimeter = 10^{-2} m

1 micrometer = 10^{-6}m

7. Correct option: (B)

Submultiples : For expressing smaller measurements, we use Some prefixes with the units known as submultiples. For example

1 milli metre = 10^{-3}m

Multiples: For expressing larger measurements, we use Some prefixes with the units known as multiples. for example.

1 decameter= 10 m

1 hectometer=100 m

1 kilometer= 1000 m

8. Correct option: (A), (B), (C)

1 decagram $= 10^{-2}$ kg

1 milligram $= 10^{-6}$ kilogram

1 hectogram $= 10^{-1}$ kilogram

9. Correct option: (D)

1 tera meter $=10^{12}$ m

Meter(m) is a smaller unit as compared to tera meter

10. Correct option: (A)

1 m $= 10^{9}$ nm

Or

1 nm $= 10^{-9}$m

(nm) is the symbol for nanometer. It is a smaller unit than meter

11. Correct option: (D)

1 Exa meter= 10^{18}m

Meter(m) is a smaller unit as compared to Exa meter

12. (a)—(Q),(S) (b)--- (T) (c) --- (P) (d)----(R)

(a)Shake is an informal metric unit of time equal to 10^{-8} seconds.

(b)Pound is a unit of measuring weight and is approximately equal to 0.453 kg.

(c)The micron is also known as micrometer and it is non SI unit of length equal to $10^{-6}m$.

(d) Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.

1 light year $= 9.46 \times 10^{15}$m

13. (a)---(T) (b)----(R) (c)----(Q) (d)-----(p)

Multiples: For expressing larger measurements, we use Some prefixes with the units known as multiples. For example

1 gigahertz(GHz) $= 10^{9}$ Hz

1 megahertz(MHz) $= 10^{9}$Hz

Submultiples: For expressing smaller measurements, we use Some prefixes with the units known as submultiples. For example

1 micro metre $= 10^{-6}$m

1 centi metre $= 10^{-2}$m

14. (a) -------(Q) (b)-------(P) (c)-------(S) (d)--------(R)

(a) Chandrasekhar limit is the maximum mass of a stable white dwarf star. Current value of the Chandrasekhar limit is about 2.765 x 10^{30} kg.

(b) Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.

1 light year $= 9.46 \times 10^{15}$m

(c) A lunar month is the duration between successive new moons, it has a mean period of 29.53059 days (29 days 12 hours and 44 minutes).

(d) The torr having symbol(Torr), is the unit of pressure. 1 torr = 133.32 pascal.

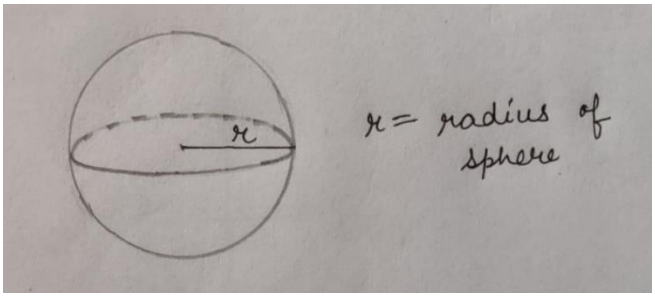
Measurements
LEVEL-I

1. Correct option: (B)

Volume of the sphere $=\frac{4\pi}{3}r^3$

Here r= radius of the sphere

So, it is clear from the formula that volume of sphere depends on radius of the sphere.



2. Correct option: (C)

We know ,

Volume of room =length x breadth x height

Density $=\frac{\text{mass}}{\text{volume}}$----------(1)

Given: density= 1.3 kg/ m^3

Dimensions of the room= 5m x 4m x 2.5m

Length = 5m

Breadth= 4m

Height= 2.5m

Hence,

Volume = 5 x 4 x 2.5

= 50 m^3

From equation 1

$1.3 = \frac{\text{mass}}{50}$

Or

Mass= 1.3 x 50

= 65 kg

3. Correct option: (B)

Given : length= 2.5m

Breadth= 1.5 m

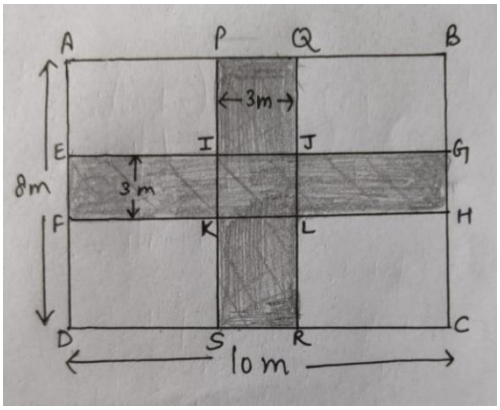
Height= 1.0 m

Volume of rectangular box = length x breadth x height

= 2.5 x 1.5 x 1.0

=3.75 m^3

4. Correct option: (D)



Given: length of rectangular field = 10m

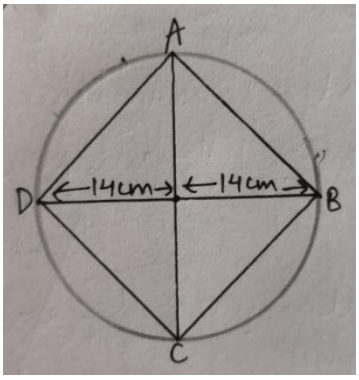
Breadth = 8 m

Width of the paths (PQRS and EGHF)= 3 m

Length of Path PQRS =8m
Breadth = 3m
Area of PQRS= length x breadth
= 8m x 3m
= 24 m^2
Length of Path EGHF =10m
Breadth = 3m
Area of EGFH= length x breadth
= 10m x 3m
= 30m^2
Side of square IJLK= 3m
Area of square IJLK= side x side
= 3m x 3m
= 9 m^2
Total area occupied by the paths = area of (rectangle PQRS + rectangle EGHF – square IJLK)
= (24+30-9) m^2
=45m^2

5. Correct option: (A)

To get maximum area of the square sheet , it's diagonal (DB) is equal to the diameter of the circle
Given: radius of the circle= 14cm
Diameter (DB) = 2r
= 2 x 14 cm
= 28 cm
Diagonal of square (DB)= diameter of the circle
√2 x side = 28 cm
Side = $\frac{28}{\sqrt{2}}$ cm
Area of the square ABCD = side x side
= $\frac{28}{\sqrt{2}}$ cm x $\frac{28}{\sqrt{2}}$ cm
= $\frac{28\times28}{2}$ cm^2
= 392 cm^2
Area of the circle = πr^2
=$\frac{22}{7}\times14\times14$
=616 cm^2
Area of remaining cardboard = Area of circle – area of square ABCD
= 616-392 cm^2
= 224 cm^2



6. Correct option: (B)

Given: radius = 2.1 m

And π= 22/7

Area of circle = πr^2

$=\frac{22}{7} \times 2.1 \times 2.1$

$=13.86\ m^2$

7. Correct option(B)

Given : radius = 4 cm

Height= 21mm

$=\frac{21}{10} cm = 2.1\ cm$

And $\pi = \frac{22}{7}$

Volume of cone $=\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 4 \times 4 \times 2.1$

$= 35.2\ cm^3$

8. Correct option: (B)

Given: radius of the cylinder = 5.5 cm

Height of the cylinder = 14 cm

Side of cube = $l\ cm$

Also π= 22/7

When cylinder is recast into cube it's volume remains same

Volume of cube = volume of cylinder

$\text{Side}^3 = \pi r^2 h$

$l^3 = \frac{22}{7} \times 5.5^2 \times 14$

$l^3 = \frac{22}{7} \times 5.5 \times 5.5 \times 14$

$l^3 = 1331\ cm^3$

$l = 11$cm

9. Correct option: (A)

1hectare = 10000 m^2

50hectare = 50×10000 m^2

$= 5 \times 10^5\ m^2$

10. Correct option: (A)

Area of Triangle = $\frac{1}{2}$ base × height

$= \frac{1}{2} \times 5 \times 8$

$= 20\ m^2$

11. Correct option: (A)

Given: radius of circle (r) = 200m

Area of circle = $\pi\ r^2$

$= \pi \times 200^2$

$= \pi \times 200 \times 200$

$= 40000\ \pi$

12. Correct option: (C)

Given: side of square = 10 cm

Perimeter of square = 4×side

= 4×10 cm

= 40 cm

13. Correct option: (A)

Given: radius of circle (r) =7 cm

Area of circle = πr^2

$= \pi \times 7^2$

$= \pi \times 7 \times 7$

$= 49\ \pi cm^2$

14. Correct option: (A)

Given : area = $120 cm^2$

breadth (b) = 6 cm

To find: length (L)

Area of rectangle =length ×breadth

120 = L×6

120/6 = L

20 = L

Or length = 20 cm

15. Correct option: (B)

Given : length (L) = 9 cm

breadth (b) = 6cm

Perimeter = 2(length + breadth)

= 2(9+6)

=2(15)

= 30 cm

16. Correct option: (C)

(A) A meter scale is a one meter long scale used to measure length.

(B) Graph paper or coordinate paper is a writing paper printed with fine lines making up a regular grid. These lines help in plotting graphs.

(C) A measuring jar is used primarily to measure the volume of liquids. A measuring jar has scale markings at varying heights used to measure fixed volume of liquids.

(D) A physical balance is used to measure mass of a body.

17. Correct option: (B)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence, C.G.S unit of volume is cm^3

18. Correct option: (B)

1 ml = 1 cm^3

The reason for this is, volume of liquid that is present in 1 ml of a liquid is equal to the volume of water that fits in a unit cube of dimensions (1 cm length x 1 cm breadth x 1 cm height)

19. Correct option: (B)

Volume is the amount of space occupied by any liquid.

20. Correct option: (B)

Length (l)=10m

Breadth(b)=7m

Height(h)=5m

Total surface area of cuboid = 2(lb+bh+hl)

=2(10x7+7x5+5x10)

=2(70+35+50)

=2(155)

= 310 m^2

21. Correct option: (D)

(A) A measuring jar is used primarily to measure the volume of liquids. A measuring jar has scale markings at varying heights used to measure fixed volume of liquids.

(B) A pipette is a laboratory instrument. It is used to measure out or transfer small quantities of liquid in volumes of milliliters or microliters.

(C) A burette is a graduated glass tube with tap at one end. It is used to deliver known volumes of a liquid. It is usually used in titrations.

LEVEL-II

1. Correct option(B)

Given : length = 4 m

Breadth = 3m

We know

Area of rectangle = length x breadth

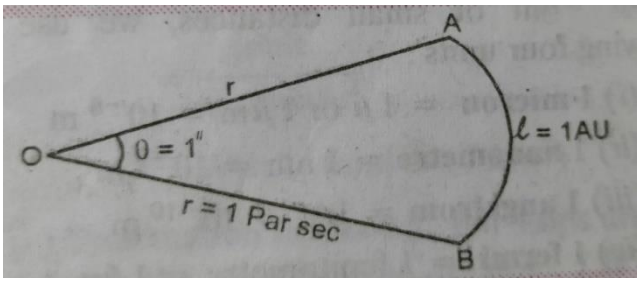
= 4m x 3m

= 12 m^2

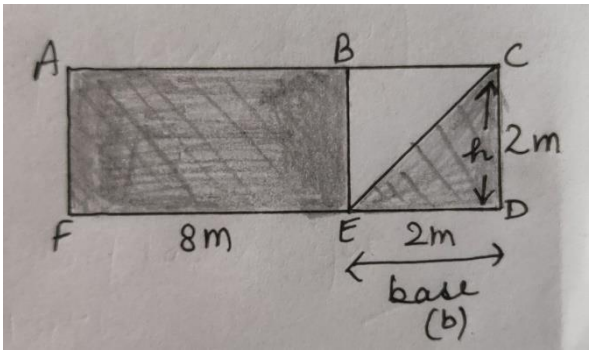
2. Correct option(A)

Parallactic second (parsec) is a unit of long distances. One parsec is the radius of a circle at the center of which an arc of the circle,1 astronomical unit (1 AU= $1.496 \times 10^{11} m$) long subtends an angle of 1 second (1")

1 parsec = $3.1 \times 10^{16} m$



3. Correct option(B)



Area of rectangle ABEF = Length × breadth

= 8m × 2m

= 16m^2

Area of triangle CDE = $\frac{1}{2}$ base × height

= $\frac{1}{2} \times 2 \times 2$

= 2 m^2

Area of shaded region = area of (rectangle ABEF + Triangle CDE)

= (16 + 2) m^2

= 18 m^2

4. Correct option(B)

Volume of cuboid = length x breadth x height

5. Correct option(C)

1 decagram = 10 g

6. Correct options: (A), (B), (D)

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 day = 24 hours

1 hour = 60 minutes

1minute = 60 seconds

So, 1 day =24×60 minutes

=1440 min

Hence, 1 day = 24×60×60 seconds

= 86,400 seconds

7. Correct option(A)

$1\text{cm} = 10^{-2}m$

$1\text{cm} \times 1\text{ cm} = 10^{-2} \times 10^{-2}m$

$1\text{cm}^2 = 10^{-4}m^2$

8. Correct option: (B)

Area of a place is equal to the number of units square contained in it.

9. Correct options(A),(B),(D)

Volume is the amount of space occupied by any substance.

M.K.S is the system of units based on measuring lengths in meters ,mass in kilograms, time in seconds.

The M.K.S unit of volume is m^3

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence, The C.G.S unit of area is cm^2

10. Correct options: (A), (B), (C)

(A) Area of rectangle =length ×breadth

(B) Area of square = (length^2)

(C) A scale is divided into 100 equal parts and each part is equal to 1 cm.

Further 1 cm is divided into 10 equal parts with each part corresponding to 1mm.

11. Correct options: (A),(B),(C)

$1\ cm = 10^{-2}m$

$1\ cm^3 = 10^{-6}\ m^3$

1 liter= 1000 ml

We know 1ml=1 cm^3

1liter = 1000 cm^3

1liter= 1000 x $10^{-6}\ m^3$

1 liter =$10^{-3}\ m^3$

0r 1 liter = $\frac{1}{1000}\ m^3$

1000 liter = 1 m^3

12. Correct options: (A),(B),(C),(D)

Volume is the amount of space occupied by any liquid.

SI unit of volume is m^3

$1m^3 = 1000000cm^3$

1cc is also known as 1 cm^3

And 1 cm^3 = 1 ml

13. Correct option(C)

Volume of cuboid = length x breadth x height

=10 cm x 8 cm x 6 cm

= 480 cm^3

14. Correct option(B)

Given: side of the cube = 10 cm

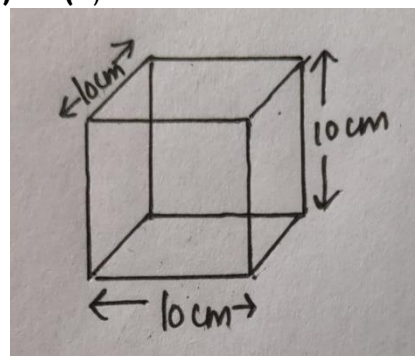
We know Volume of cube = side^3

$= 10^3$

= 10 x 10 x 10 cm^3

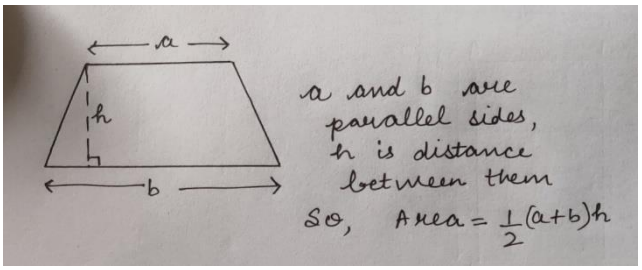
=1000 cm^3

15. (a)----(Q) (b)----(S) (c)----(R) (d)-----(P)



(a) Area of trapezium $=\frac{1}{2}\times$ (sum of parallel sides) $\times$ (distance between them)

(b) Given: diagonal of square = $\sqrt{8}m$



= $2\sqrt{2}m$

= $\sqrt{2}\times 2$ m-------(1)

Also, diagonal of square = $\sqrt{2}\times side$ ----------(2)

comparing equation 1 and 2

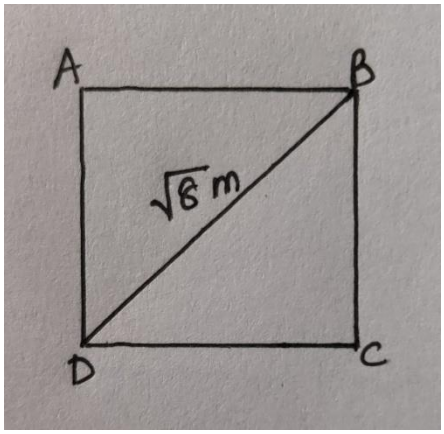
we get $\sqrt{2}\times 2 = \sqrt{2}\times side$

Or Side =2 m

Area of square = side x side

= 2 m x 2 m

= 4 m^2



(c) Total surface area of cube = 6 x $side^2$

=6 x 10^2

= 6 x 10cm x 10cm

= 600 cm^2

(d) we know

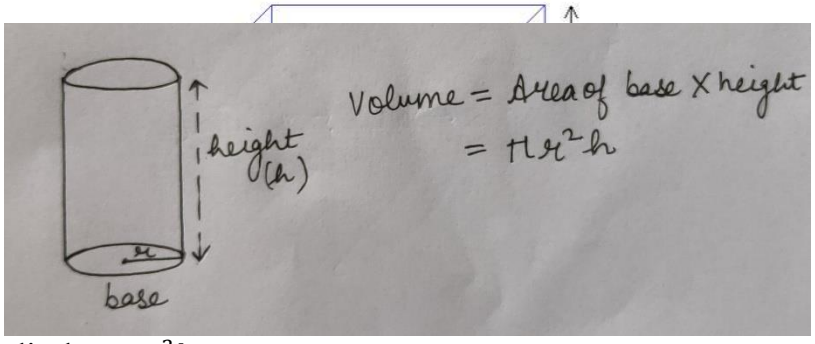
diagonal of square = $\sqrt{2} \times side$

= $\sqrt{2} \times 10\ cm$

=$10\sqrt{2}\ cm$

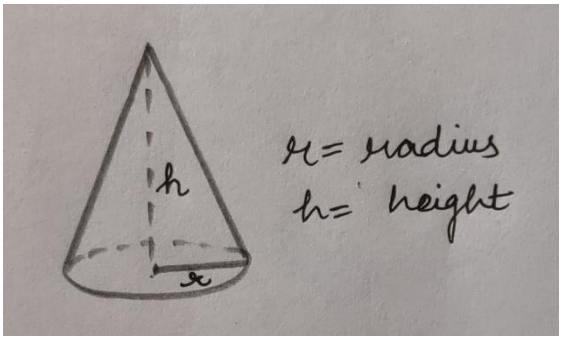
16. (a)----(S) (b)----(R) (c)----(Q) (d)-----(P)

(a) Volume of cuboid = length x breadth x height

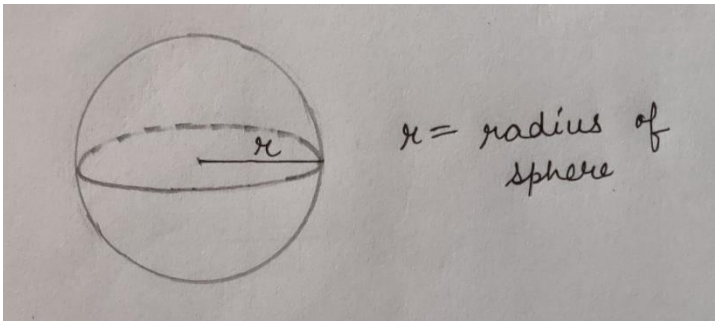


(b) Volume of cylinder =$\pi r^2 h$

(c) Volume of cone = $\frac{1}{3}\pi r^2 h$



(d) Volume of sphere = $\frac{4}{3}\pi r^3$



2. Kinematics
Solutions

LEVEL-I

1. Correct option(B)

Rest and motion are relative: It means an object in one situation can be at rest but in another situation the same object can be in motion. For illustration, a person sitting in a moving train is at rest with respect to its fellow passengers but is in motion with respect to the objects outside the train.

2. Correct option(A)

Periodic motion: A motion that repeats itself after equal intervals of time is known as periodic motion. The movement of the moon around the earth is repeated after every 30 days and hence 30 days is the time interval of the periodic motion of the moon around the earth.

3. Correct option(B)

Rectilinear motion: It is also known as linear motion. It is that motion in which a particle or an object /body moves along a straight line or well-defined straight path. For example, the motion of train along a straight railway track, an object dropped from a height above the ground etc.

4. Correct option(A)

Periodic motion: A motion that repeats itself after equal intervals of time is known as periodic motion. The oscillations (To and from motion) of a simple pendulum of a wall- clock repeats itself after a definite interval of time. That time interval is different for different simple pendulums.

5. Correct option(A)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time. Hence motion of a potter's wheel is an example of rotatory motion.

6. Correct option(B)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

Hence javelin thrown by an athlete is an example of curvilinear motion as it travels along a curved path.

7. Correct option(C)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

A windmill converts wind power into rotational energy using its blades. Hence, windmill is an example of rotatory motion.

8. Correct option(A)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well defined

limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Hence the needle of the sewing machine is in oscillatory motion.

9. Correct option(B)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

10. Correct option(A)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

Hence, pedals of bicycle possess rotatory motion.

11. Correct option(D)

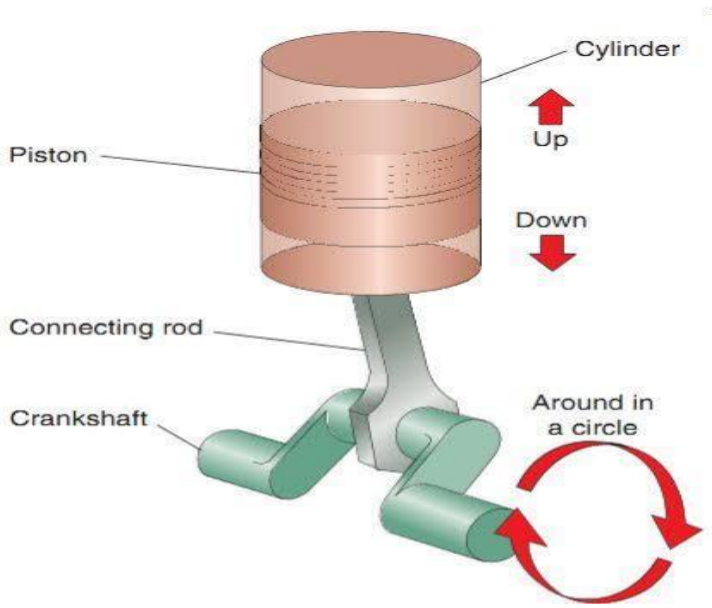
Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Hence, pendulum of a wall-clock is an example of oscillatory motion.

12. Correct option(A)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Hence, motion of piston of an engine is an example of oscillatory motion as it moves up and down between two fixed points about a mean position.



13. Correct option(B)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

If in the oscillatory motion, the amplitude is very small, the motion of body is said to be vibratory motion.

Hence, the motion of strings of a violin is an example of vibratory/ oscillatory motion.

14. Correct option(A)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

When a car takes a turn, it follows a curved path. Hence, a car taking a turn executes curvilinear motion.

15. Correct option(A)

Scalar quantities: These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

16. Correct option(B)

SI system is an internationally accepted system of units for measurements.

According to this system unit of length is meter.

As distance is measured in metres.so, SI unit of distance is meter.

17. Correct option(A)

Displacement: It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

Here, initial position is A and final position is C. So, displacement is the line joining Initial and final position of the body.

18. Correct option(C)

Displacement: It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the given figure, the person is starting from P and returning to P.

As displacement is the difference between the final and initial positions of the object in a given time.

Here, initial position is P and final position is also P.

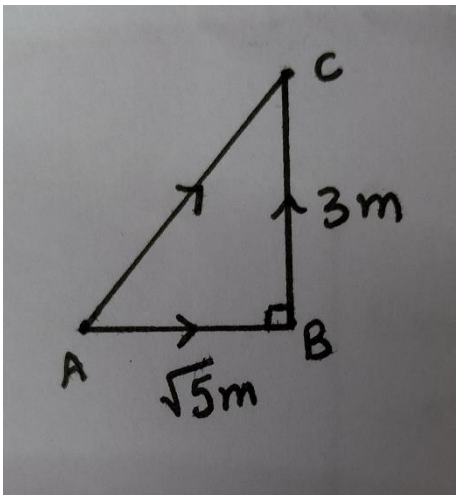
So, displacement is zero.

19. Correct option(A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the given figure, the object is moving from point A to B to C.

Here, initial position is A and final position is C.



When we join A to C, we get the shortest distance AC= displacement
Triangle ABC is a right-angled triangle
AB = base = $\sqrt{5}m$
BC=perpendicular=3m
AC= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = \left(\sqrt{5}\right)^2 + (3)^2$
$(AC)^2 = 5 + 9$
$(AC)^2 = 14$
AC = $\sqrt{14}\,m$
Hence, displacement (AC)= $\sqrt{14}\,m$

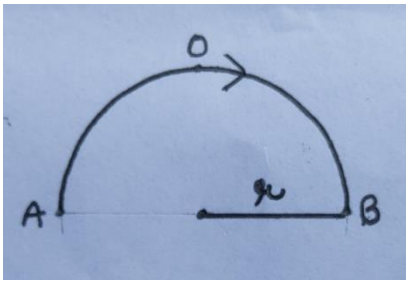
20. Correct option(B)

An odometer is an instrument used for measuring the distance travelled by a vehicle such as car.

21. Correct option(B)

Distance: It is the actual path traversed by the object in the given time.
Distance travelled along a circular path is calculated with the help of circumference of the circle.
Distance travelled by the particle when it completes full circle = circumference of the circle
Distance travelled by the particle when it completes half circle (AOB)

$$= \frac{(circumference\ of\ the\ circle)}{2}$$
$$= \frac{2\pi r}{2}$$
$$= \pi r$$



22. Correct option(D)

Distance is the actual path traversed by the object in the given time. Distance travelled by an object can never be negative.

Whereas displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

The displacement of an object in a time can be equal to or less than distance travelled but never greater than the distance travelled.

Displacement ≤ distance

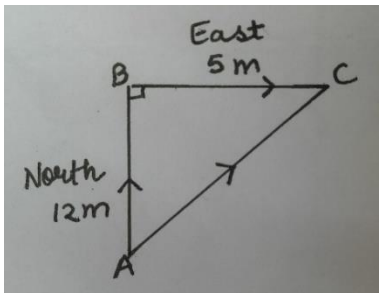
Numerical ratio($\frac{\text{displacement}}{\text{distance}}$)

Can be equal to 1 or less than 1. Since, the denominator(distance) is either greater than or equal to the numerator(displacement).

23. Correct option(B)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the question, let the horse is running from point A to B to C.



Here, initial position is A and final position is C.

When we join A to C, we get the shortest distance AC= displacement

Triangle ABC is a right-angled triangle

AB = base = $12m$

BC=perpendicular=5m

AC= hypotenuse

Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(AC)^2 = (AB)^2 + (BC)^2$

$(AC)^2 = (12)^2 + (5)^2$

$(AC)^2 = 144 + 25$

$(AC)^2 = 169$

$AC = \sqrt{169}m$

$AC = 13m$

Hence, displacement (AC)= $13m$

24. Correct option(B)

Displacement: It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.

In the given question, we are starting from home and returning to home.

As displacement is the difference between the final and initial positions of the object in a given time.

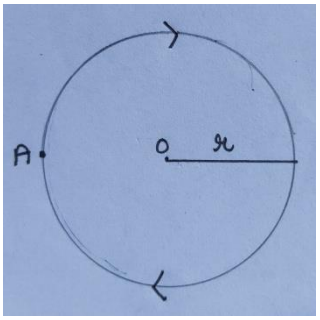
Here, initial position is home and final position is also home.

So, displacement is 0.

25. Correct option(B)

• Distance: It is the actual path traversed by the object in the given time.
Distance travelled by the particle when it completes 1 revolution (A to A)
= circumference of the circle
= $2\pi r$ (r= radius of the circle)

• Displacement is the difference between the final and initial positions of the object in a given time.
In a circular motion when the particle completes 1 revolution it returns to its initial position(A). So, initial and final position are same.
Hence, displacement of the particle is 0.

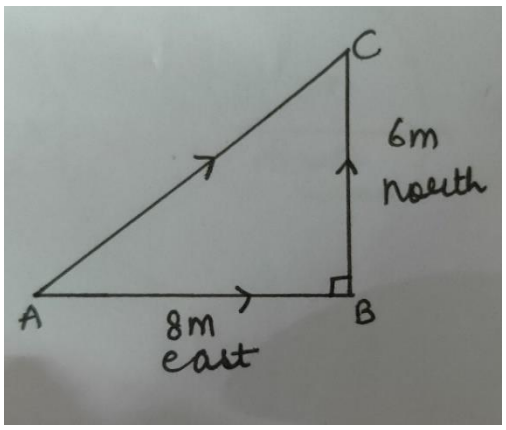


26. Correct option(B)

Distance: It is the actual path traversed by the object in the given time.
Distance travelled along a circular path is calculated with the help of circumference of the circle.
Distance travelled by the particle when it completes 1 revolution
= circumference of the circle
= $2\pi R$ (R= radius of the circle)
Distance travelled by the particle when it completes 5 revolutions = $5\times 2\pi R$
= $10\pi R$

27. Correct option(D)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.
In the given question, let the particle is moving from point A to B to C.



Here, initial position is A and final position is C.
When we join A to C, we get the shortest distance AC= displacement
Triangle ABC is a right-angled triangle
AB = base = 8m
BC=perpendicular=6m

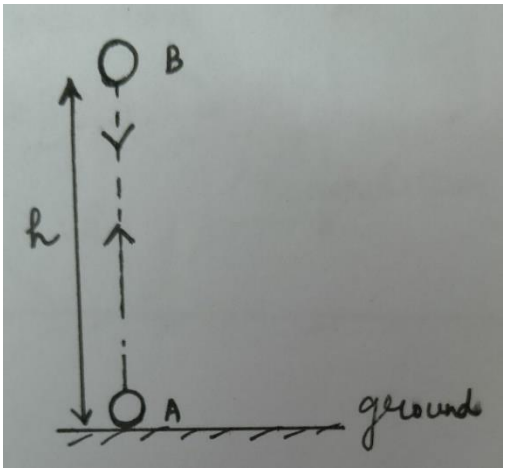
AC= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = (8)^2 + (6)^2$
$(AC)^2 = 64 + 36$
$(AC)^2 = 100$
AC = $\sqrt{100}\, m$
AC= 10m
Hence, displacement (AC)= 10 m

28. Correct option(D)

According to the question distance covered by the particle is zero, which means it does not change its position in a given time or it is at rest.
We know that
Displacement is the difference between the final and initial positions of the object in a
given time. Here, object is at rest so it's initial and final positions are same. Hence, displacement must be zero.

29. Correct option(B)

• Distance: It is the actual path traversed by the object in the given time.
Distance travelled by the body upwards (AB) = h
Distance travelled by the body while returning to ground (BA) = h
Total distance travelled during journey= 2h
• Displacement: It is the difference between the final and initial positions of the object in a
given time.
In the given question, body is thrown from the ground and returns back to the ground (A to A)
As displacement is the difference between the final and initial positions of the object in a given time.
Here, initial position is ground and final position is also ground.
So, displacement is 0.



30. Correct option(C)

• Distance: It is the actual path traversed by the object in the given time.
Distance travelled by the particle when it completes 1 revolution
= circumference of the circle
= 2 πR (R= radius of the circle)

$= \pi d$ (diameter(d)= 2R)

$= 5\pi\, m$ (given d= 5m)

Distance travelled by the particle when it completes 3 revolutions= $3 \times 5\pi\, m$

$= 15\pi\, m$

- Displacement is the difference between the final and initial positions of the object in a given time.

In a circular motion when the particle completes 3 revolution it returns to its initial position. So, initial and final position are same.

Hence, displacement of the particle is 0.

31. Correct option(D)

Galileo experimented with various pendulums and he found that a pendulum of a given length takes always the same time to complete one oscillation. Simple Pendulum executes oscillatory motion which is periodic means it repeats itself after fixed intervals of time. Hence, simple pendulum can be used to measure time.

32. Correct option(B)

1 minute = 60 seconds

40 minutes = 40 ×60 seconds

= 2400 seconds

33. Correct option(B)

A leap year is a calendar year that contains an additional day to keep the calendar synchronized with the astronomical year or seasonal year. this extra day is added to the end of shortest month, February. So, February has 29 days in leap year.

34. Correct option(C)

1 second = $\frac{1}{60}$ minutes

90 seconds = $\frac{90}{60}$ minutes

= 1.5 minutes

35. Correct option(B)

1 second = 1000mlliseconds

3 seconds = 3000ms

LEVEL-II

1. Correct option(B)

Circular motion is that motion in which a particle or a point mass is moving in a circle.

Circular motion is special case of rotatory motion as in this motion the distance from the axis of rotation is always fixed.

A spinning top around its own axis executes circular motion.

1. Correct option(A)

An object is said be at rest if it does not change its position with time, with respect to its surroundings. For example, a book lying on a table, a person sitting on a chair are at rest.

2. Correct option(C)

An object is said to be in motion if it changes its position with time, with respect to its surroundings. For example, a bird flying in air, a man walking on the road are in motion.

4. Correct option(A)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

A football kicked from the ground into the air travels a curved path and hence it is an example of curvilinear motion.

5. Correct option(B)

Circular motion is that motion in which a particle or a point mass is moving in a circle.

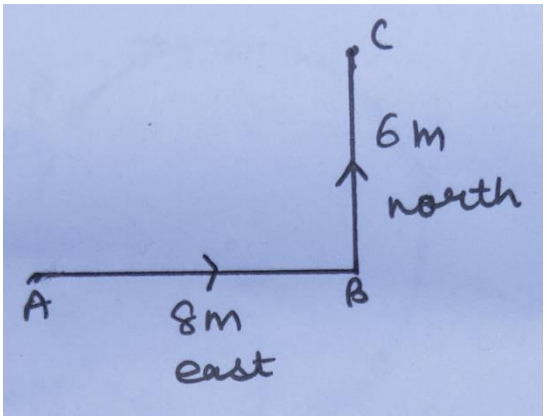
Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

Circular motion is special case of rotatory motion as in this motion the distance from the axis of rotation is always fixed. In rotatory motion a body is involved so, there are a number of particles which may have different radius from the axis of rotation.

Hence, option (B) is correct.

6. Correct option(B)

Distance is the actual path traversed by the object in the given time.



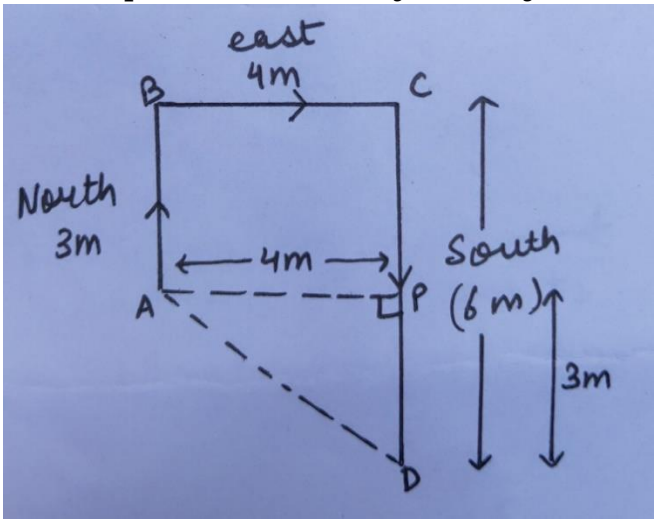
Distance travelled towards east (AB) =8m

Distance travelled towards north (BC)= 6m

Total distance travelled = 8m +6m = 14m

7. Correct option(A)

- Distance is the actual path traversed by the object in the given time.

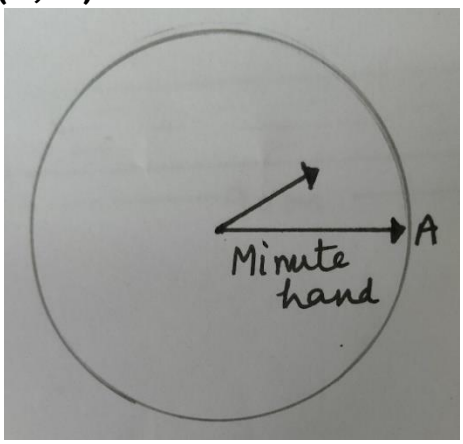


Distance travelled towards north (AB) = 3m
Distance travelled towards east (BC) = 4m
Distance travelled towards south (CD) = 6m
Total distance travelled = 3m +4m+6m
= 13m

- Displacement is the difference between the final and initial positions of the object in a given time.

It is the shortest distance between two positions of the object.
As shown in the figure
Shortest distance /displacement= AD
Drop a perpendicular AP on CD
In right angled triangle APD
PD = base = 3m
AP=perpendicular=4m
AD= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(\mathrm{AD})^2 = (\mathrm{PD})^2 + (\mathrm{AP})^2$
$(\mathrm{AD})^2 = (3)^2 + (4)^2$
$(\mathrm{AD})^2 = 9 + 16$
$(\mathrm{AD})^2 = 25$
$\mathrm{AD} = 5\ m$
Hence, displacement (AD)= $5\ m$

8. Correct option (A, D)



In one hour, minute's hand of the clock moves in a circle and completes one revolution.

- Displacement is the difference between the final and initial positions of the object in a given time.

In a circular motion, when the minute's hand completes 1 revolution it returns to its initial position(A). So, initial and final position are same.
Hence, displacement of the minute's hand is 0.

- Distance is the actual path traversed by the object in the given time.

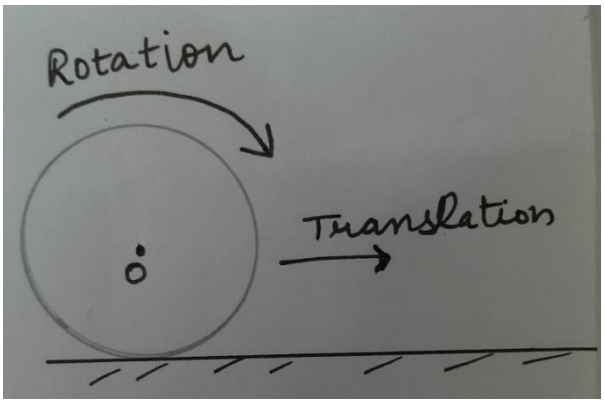
Distance travelled along a circular path is calculated with the help of circumference of the circle.
Distance travelled by the minute's hand after one hour when it completes 1 revolution (A to A)
= circumference of the circle
= $2\ \pi R$ (R= radius of the circle)

9. Correct option (A, B)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

Translatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

Hence, a rolling ball executes both rotatory and translatory motion.



10. Correct option (A, B, C)

Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

Displacement is positive when the final position is at a greater distance than the initial position.

Displacement is negative when the final position is at a smaller distance than the initial position.

Displacement is zero when the final position is same as the initial position.

11. Correct option (A, B, C, D)

Periodic motion: A motion that repeats itself after equal intervals of time is known as periodic motion.

For example:

(A) Hands of a wrist watch come to their starting position after a fixed interval of time and then repeat the same motion with same interval of time. Hence, it is a periodic motion.

(B) The rotation of earth around its own axis repeats itself after 24 hours and hence it is a periodic motion.

(C) The strings of a musical instrument vibrate when plucked. vibration repeats itself after certain period of time. so, it represents periodic motion.

(D) Rotation of earth around the sun repeats itself after 365 days and 6 hours. hence, it is a periodic motion.

12. Correct option (A, B, C, D)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

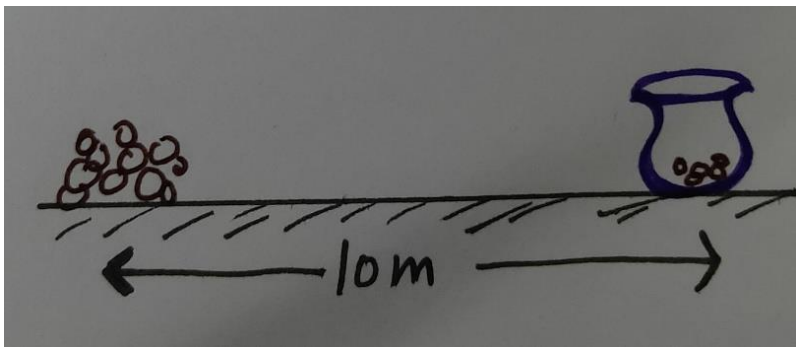
For example: spinning top, wind mill, fan, wheels of a car all execute rotatory motion

13. Correct option(B)

Translatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

Hence, a car moving on a straight level road possess translatory motion.

14. Correct option(A)



Distance travelled by the crow in one trip= distance from pebbles to pot +distance from pot to pebbles = 10m+10m

=20m

Distance travelled by the crow in 5 trips = 5 x Distance travelled by the crow in one trip

= 5 x 20m

= 100m

15. Correct option(B)

Distance is the actual path traversed by the object in the given time.

Distance covered in 1 round (A to A) = 200m

Distance covered in 4 rounds A to A) = 4 x 200m

= 800m

16. Correct option(D)

Displacement is the difference between the final and initial positions of the object in a given time.

In the given question after taking 4 rounds of the track the athlete returns back to its initial position that is starting point(A). So, initial and final position are same.

Hence, displacement of the athlete is 0.

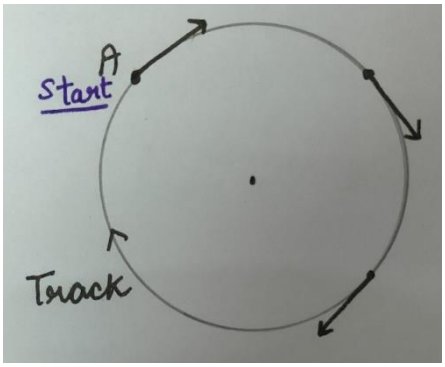
17. Correct options (A, B)

When an athlete is moving along a circular track. He executes uniform circular motion.

Uniform circular motion is that motion in which an object has a constant speed but its direction is changing along the tangent to the circle. Hence, velocity keeps on changing.

An athlete also moves with a constant speed along a circular track but changes his direction continuously. So, an athlete executes both uniform and circular motion.

18. Correct option(B)



Distance is the actual path traversed by the object in the given time.
Distance covered in 1 round = 200m
Distance covered in 4 rounds = 4 x 200m
= 800m
Displacement is the difference between the final and initial positions of the object in a given time.
In the given question after taking 4 rounds of the track the athlete returns back to its initial position that is starting point. So, initial and final position are same.
Hence, displacement of the athlete is 0.
Hence, distance and displacement are not equal.

19. a- R; b- Q; c- S; d- P

(a) Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Swinging cradle is an example of oscillatory motion because it moves between two fixed

point about a mean position.

(b) Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

Spinning top around its own axis is an example of rotatory motion.

(c) Motion of fishes in water is a random motion.

(d) Rectilinear motion: It is also known as linear motion. It is that motion in which a

particle or an object /body moves along a straight line or well defined straight path. So, a car moving in a straight line is an example of rectilinear motion.

20. a-Q; b-T; c-S; d-P

(a) Distance is the actual path traversed by the object in the given time.
(b) Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object.
(c) Distance travelled by the particle when it completes 1 rotation of circular motion

= circumference of the circle
= 2 πr (r= radius of the circle)

(d) Displacement is the difference between the final and initial positions of the object in a given time.
In a circular motion when the particle completes 1 rotation it returns to its initial position. So, initial and final position are same.
Hence, displacement of the particle is 0.

21. a- S; b- P; c- Q; d- R

1 hour = 60 minutes
1 minute = 60 seconds
So, 1 hour = 60 x 60 seconds
1 hour = 3600 seconds

(a) 1 day = 24 hours
(c) 1 decade = 10 years
(d) 1 century = 100 years

Speed, Velocity and Acceleration
LEVEL-I

1. Correct option (C)

Given: speed = 72km/h= 20m/s
Time = 2s

1km/h = $\frac{5}{18}$ m/s

72km/h= 72 x $\frac{5}{18}$ m/s

72km/h = 20m/s

As Speed = $\frac{\text{distance travelled}}{\text{time taken}}$ = 20m/s

20m/s= $\frac{\text{distance travelled}}{2}$

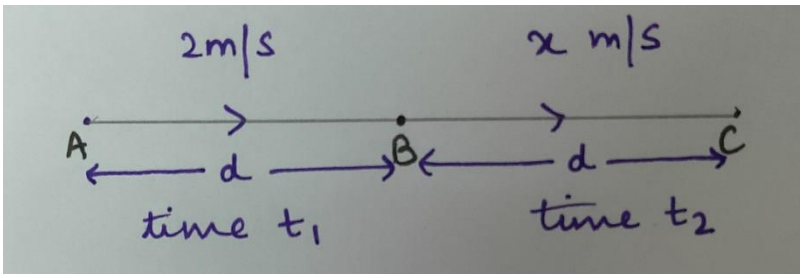
Distance travelled = 2s x 20m/s
= 40m

2. Correct option(A)

Speed of an object in motion is defined as the ratio of the total path length/distance covered and the corresponding time taken by the object,

Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

3. Correct option(B)



Let the total distance of the journey = 2d
(a)For the first half of the journey (AB):
Given: speed = 2 ms^{-1}
Let distance = d
Time = t_1

Speed $= \frac{\text{distance travelled}}{\text{time taken}}$

Time $= \frac{\text{distance travelled}}{\text{speed}}$

$t_1 = \frac{d}{2} \quad ----(1)$

(b) For the remaining half of the journey (BC):

Let speed = x ms^{-1}

Distance = d

Time = t_2

Time $= \frac{\text{distance travelled}}{\text{speed}}$

$t_2 = \frac{d}{x} \quad ----(2)$

(b) Average speed $= \frac{\text{total distance}}{\text{total time}}$

Given: average speed = 3 ms^{-1}

Average speed $= \frac{d+d}{t_1+t_2}$

From equation 1 and 2

$3\ ms^{-1} = \frac{2d}{\frac{d}{2}+\frac{d}{x}}$

$3\ ms^{-1} = \frac{2d}{d\left(\frac{1}{2}+\frac{1}{x}\right)}$

$3\ ms^{-1} = \frac{2}{\left(\frac{1}{2}+\frac{1}{x}\right)}$

$3\ ms^{-1} = \frac{2}{\frac{(x+2)}{2x}}$

$3\ ms^{-1} = \frac{2x\ 2x}{x+2}$

$3(x + 2) = 4x$

$3x + 6 = 4x$

$6 = 4x - 3x$

$6 = x$

Or $x = 6\ ms^{-1}$

Hence, speed with which the object travels remaining distance is 6 ms^{-1}.

4. Correct option(C)

Displacement = 200m –west

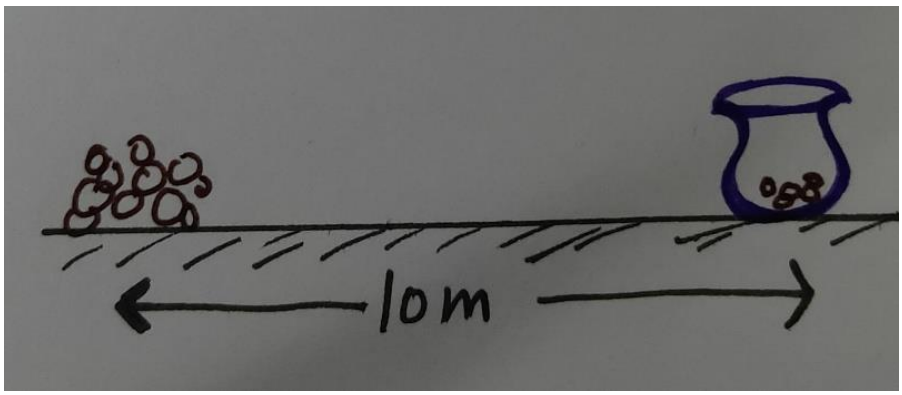
Time= 2 s

Velocity $= \frac{\text{displacement}}{\text{time interval}}$

Velocity $= \frac{200\text{m}}{2\text{ s}}$

Velocity= 100 m/s

5. Correct option(A)



Distance travelled by the crow in one trip= distance from pebbles to +distance from pot to pebbles

= 10m+10m

=20m

Distance travelled by the crow in 5 trips = 5 x Distance travelled by the crow in one trip

= 5 x 20m

= 100m

Total distance = 100m

Total time= 80s

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Average speed = $\frac{100}{80}$

= 1.25 m/s

6. Correct option(B)

An odometer is an instrument used for measuring the distance travelled by a vehicle such as car.

7. Correct option(C)

Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

= $\frac{10\text{m}}{2s}$

= $5ms^{-1}$

8. Correct option(B)

Given: distance between the stations =20 km

Speed of the train= $60kmh^{-1}$

Average speed = $\frac{\text{total distance}}{\text{total time}}$

$60kmh^{-1}$ = $\frac{20\text{km}}{\text{time}}$

time = $\frac{20\text{km}}{60\text{kmh}^{-1}}$

time= $\frac{1}{3}\ h$

1 h = 60 minutes

So, time= $\frac{1}{3} \times 60\ minutes$

= 20 minutes

So, time taken by the train to reach the next station is 20 minutes.

9. Correct option(A)

When an object is moving with a variable speed, then the average speed of the object is that constant speed with which the object covers the same distance in a given time as it does while moving with variable speed during the given time.

Average speed = $\frac{\text{total distance}}{\text{total time}}$

10. Correct option(A)

An object is said to be moving with a uniform speed, if it covers equal distances in equal intervals of time. For example, if a car is moving on a straight road from place A to B, covers a distance of 80 km in 8 hours, and covers 10 km in every one hour, then the car is moving with a uniform speed =10km/h.

Uniform Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Given: uniform speed =10km/h
time = 2 h
Uniform Speed x time taken =distance travelled
10km/h x 2 h = distance
Distance = 20 km ------(1)
When an object is moving with a variable speed, then we have to calculate average speed. For example, if a car is moving on a straight road from place A to B and covers successive 20 km distances in timings 1 h, 1.5h, 2h and 0.75 h, the speed of the car at different locations is different, so we calculate the average speed.

Average speed = $\frac{\text{total distance}}{\text{total time}}$

given: time = 2h
from equation 1
total distance = 20km

Average speed = $\frac{20\text{ km}}{2\text{h}}$

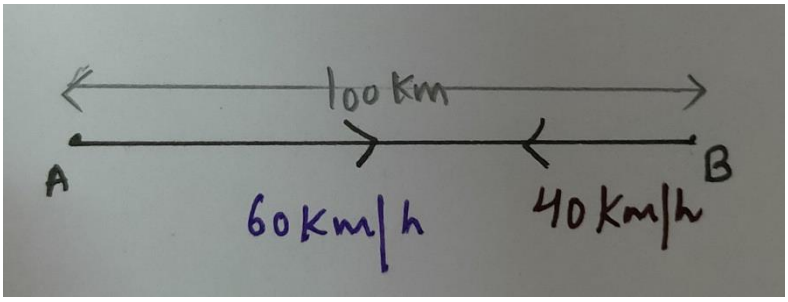
= 10km/h

Which is same as the uniform speed, when the speed is uniform, average speed is equal to the uniform speed.

11. Correct option(B)

Vector quantities are those quantities which have both magnitude(size) and direction.
Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.
For example, if velocity = 20 km/h towards north
Then 20 km/h is the magnitude(speed) and north is the direction.

12. Correct option (48km/h**)**



Given: speed during going (AB) = 60km/h
Distance travelled = 100km

Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Time taken while going = $\frac{\text{distance}}{\text{speed}}$

= $\frac{100\text{ km}}{60\text{km/h}}$

= $\frac{5}{3}$ h

Speed while returning (BA) = 40km/h
distance travelled = 100km

Time taken to return = $\frac{\text{distance}}{\text{speed}}$

= $\frac{100\text{ km}}{40\text{km/h}}$

= $\frac{5}{2}$ h

Total distance = 100km+100km

= 200km

total time taken=$\frac{5}{3}$h + $\frac{5}{2}$h

= $\frac{25}{6}$h (L.C.M of 3 and 2 = 6)

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Average speed = $\frac{200}{\frac{25}{6}}$

= $\frac{200\times6}{25}$

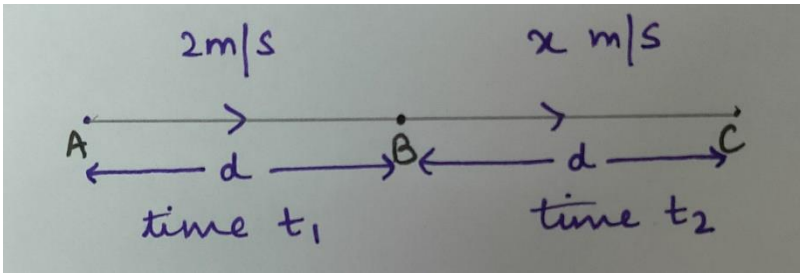
Average speed for the whole journey = 48km/h

13. Correct option(A)

Instantaneous speed: when the speed of an object is variable/changing, then the object possesses different speeds at different instants. The speed of an object at a given instant of time is called its instantaneous speed.

Instantaneous speed of a vehicle is measured by speedometer.

14. Correct option(B)



Let the total distance of the journey = 2d

(a) For the first half of the journey:

Given: speed = 2 ms^{-1}

Let distance = d

Time = t_1

Speed =$\frac{\text{distance travelled}}{\text{time taken}}$

Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_1 = \frac{d}{2}$ ----(1)

(b) For the remaining half of the journey:

Let speed = x ms^{-1}

Distance = d

Time = t_2

Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_2 = \frac{d}{x}$ ----(2)

(c) Average speed = $\frac{\text{total distance}}{\text{total time}}$

Given: average speed = 3 ms^{-1}

Average speed = $\frac{d+d}{t_1+t_2}$

From equation 1 and 2

3 ms^{-1} = $\frac{2d}{\frac{d}{2}+\frac{d}{x}}$

3 ms^{-1} = $\frac{2d}{d\left(\frac{1}{2}+\frac{1}{x}\right)}$

$$3\ ms^{-1} = \frac{2}{\left(\frac{1}{2}+\frac{1}{x}\right)}$$

$$3\ ms^{-1} = \frac{2}{\frac{(x+2)}{2x}}$$

$$3\ ms^{-1} = \frac{2x\ 2x}{x+2}$$

$$3(x + 2) = 4x$$

$$3x + 6 = 4x$$

$$6 = 4x - 3x$$

$$6 = x$$

Or

$$x = 6\ ms^{-1}$$

Hence, speed with which the object travels remaining distance is 6 ms^{-1}.

15. Correct option(A)

Given: initial velocity(u) of car =10m/s

Final velocity(v) of car= 0m/s (because, car comes to rest)

Time taken = 5 s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$$a = \frac{v-u}{t}$$

here, a= acceleration

u= initial velocity

v = final velocity

$$a = \frac{0-10}{5}\ m/s^2$$

$$a = -2\ m/s^2$$

acceleration =-2m/s^2

16. Correct option(B)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

The C.G.S unit of velocity = cm/s

The C.G.S unit of time = s

The C.G.S unit of acceleration = $\frac{cm}{s \times s}$

=cm/s^2

17. Correct option(A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

As it is given in the question that velocity of particle is uniform or constant means, velocity is not changing. so, change in velocity is zero.

Hence, acceleration of the particle is also zero.

18. Correct option(A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

If the velocity is increasing, then change in velocity will be positive and acceleration will be positive.

If the velocity is decreasing, then change in velocity will be negative and acceleration will be negative.

Negative acceleration is also called retardation.

Hence, decrease in velocity per unit time is called **retardation**.

19. Correct option(C)

Given: initial velocity(u) of scooter =0m/s (because, scooter is starting from rest)

Final velocity(v) of scooter= 36km/h =10m/s

1km/h = $\frac{5}{18}$ m/s

36km/h = 36 x $\frac{5}{18}$ m/s

= 10m/s

Time taken = 10s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{10-0}{10}$ m/s^2

a= 1 m/s^2

acceleration =1 m/s^2

20. Correct option(D)

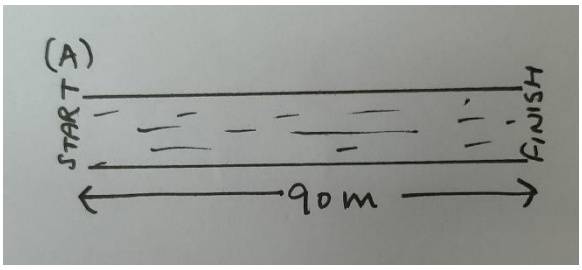
Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

velocity of a body changes when either speed(20km/h) changes or the direction(north) changes or both changes.

21. Correct option(B)



- calculation of average speed:

Length of the pool = 90 m

Given:

Distance travelled in going from one end to other of pool and returning back (A to A) =180m

Time taken in going from one end to other of pool and returning back=1min =60s

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Average speed = $\frac{180\text{ m}}{60\text{s}}$

= 3ms^{-1}

- calculation of average velocity:

Average Velocity =$\frac{\text{total displacement}}{\text{total time}}$

Displacement is the difference between the final and initial positions of the object in a given time.

As, Usha is returning to the same point (A), her initial and final position are same.

Hence, displacement is 0.

So, Average velocity is also 0.

22. Correct option(C)

An object is said to be in motion, if it changes its position with respect to surroundings with the passage of time and the rate of change of motion of an object is also called speed.

If this rate of change of motion is in specified direction, it is called velocity.

23. Correct option(A)

An object is said to be moving with a uniform acceleration if its velocity changes by equal amounts in equal intervals of time.

24. Correct option(B)

Given: acceleration of car =3 m/s^2

Initial velocity(u) of car =0m/s (the car is starting from rest)

Final velocity(v) of car= 54m/s

To find: time taken to acquire velocity of 54m/s

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a = \frac{v-u}{t} \qquad \textit{here,}\ a = \text{acceleration}$$

u= initial velocity

v = final velocity

$$3 = \frac{54-0}{\text{time taken}}$$

$$\text{Time taken} = \frac{54}{3}\text{s}$$

Time taken = 18 s

25. Correct option(C)

Speed = 0.06 m/s

$1\text{m/s} = \frac{18}{5}$ km/h

$0.06\text{m/s} = 0.06 \times \frac{18}{5}$ km/h

$= \frac{6\times18}{100\times5}$ km/h

= 0.216 km/h

26. Correct option(B)

Acceleration of an object in motion is defined as the ratio of change in velocity and the corresponding time taken by the object.

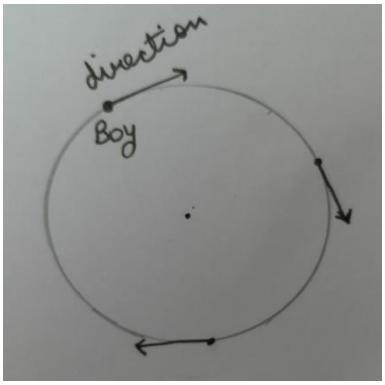
$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

If the velocity is increasing, then change in velocity will be positive and acceleration will be positive.

27. Correct option(C)

As the boy is moving with a constant speed on merry-go-round, he is moving in a circular motion.

Although his speed is constant but direction is changing. So, his velocity is



changing because velocity along a circular path is directed along the tangent to the circle. Thus, change in his velocity leads to acceleration. Hence, boy is in accelerated motion.

28. Correct option(A)

Given:

Initial velocity (u) of aero plane = 270km/h = 75m/s

$1\text{km/h} = \frac{5}{18}\text{ m/s}$

$270\text{km/h} = 270 \text{ x } \frac{5}{18}\text{ m/s}$

$= 75\text{m/s}$

Final velocity (v) of aero plane = 0 km/h (aero plane is coming to rest)

$= 0\text{m/s}$

Displacement (s)= 1000m

Using the third equation of motion

$v^2 - u^2 = 2as$ Here, a= acceleration

u= initial velocity

v = final velocity

s= displacement

$0^2 - 75^2 = 2 \times a \times 1000$

$-5625 = 2000a$

$-\frac{5625}{2000} = a$

$a = -2.8125\ \text{m}/s^2$

as negative acceleration is also called retardation

so, retardation of the aero plane $=2.8125\ \text{m}/s^2$

29. Correct option(C)

$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

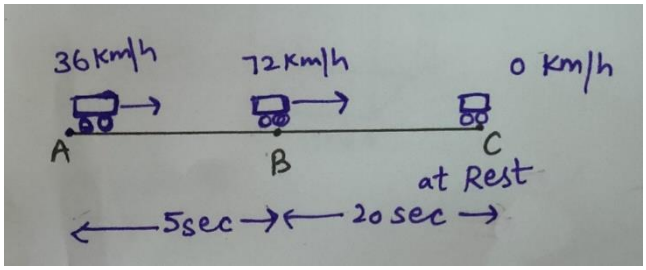
v = final velocity

If the velocity is increasing(v>u) then change in velocity will be positive and acceleration will be positive.

If the velocity is decreasing(v<u) then change in velocity will be negative and acceleration will be negative.

Negative acceleration is also called **retardation**.

30. Correct option(B)



(a) Given: initial velocity(u) of car =36km/h = 10m/s

Final velocity(v) of car= 72km/h= 20m/s

$1\text{km/h} = \frac{5}{18}\text{ m/s}$

$36\text{km/h} = 36 \text{ x } \frac{5}{18}\text{ m/s}$

= 10m/s

72km/h = 72 x $\frac{5}{18}$ m/s

= 20m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

a = $\frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

a = $\frac{20-10}{5}$ ms^{-2}

a= 2 ms^{-2}

acceleration =2 ms^{-2}

(b) when the car stops in 20 seconds:

initial velocity(u) of car =72km/h = 20m/s

Final velocity(v) of car= 0km/h=0m/s (car stops in 20 seconds)

1km/h = $\frac{5}{18}$ m/s

72km/h = 72 x $\frac{5}{18}$ m/s

= 20m/s

Time taken = 20s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

a = $\frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

a = $\frac{0-20}{20}$ m/s^2

a= -1 ms^{-2}

acceleration = - 1 ms^{-2}

Negative acceleration is also called **retardation**.

So, retardation= 1 ms^{-2}

LEVEL-II

1. Correct option(A)

speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Distance travelled by the particle when it goes from point A to O to B on a semicircle

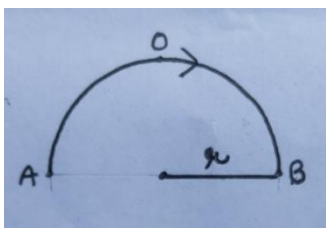
= circumference of the semicircle

= πr (r= radius of the circle)

= 2π m (r = 2m)

time taken= 2 s

speed = $\frac{2\pi}{2}$ m/s



= π m/s

2. Correct option(D)

speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Distance travelled by the particle when it completes one rotation along a circular path

= circumference of the circle

= $2\pi r$ m (r= radius of the circle)

time taken= t s

speed = $\frac{2\pi r}{t}$ m/s

3. Correct option(A)

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Distance travelled by the particle when it goes from point A to B on a semicircle

= circumference of the semicircle

= πr (r= radius of the circle)

= π m (r = 1m)

time taken= 1 s

Average speed = $\frac{\pi}{1}$ m/s

= π m/s

4. Correct option(C)

Given: initial velocity(u) of car= 108km/h =30m/s

Final velocity(v) of car= 72km/h =20m/s

1km/h = $\frac{5}{18}$ m/s

108km/h = 108 x $\frac{5}{18}$ m/s

= 30m/s

72km/h= 72 x $\frac{5}{18}$ m/s

=20m/s

Time taken = 4 s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

a = $\frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

a = $\frac{20-30}{4}$ m/s^2

a= -2.5 m/s^2

acceleration =-2.5 m/s^2

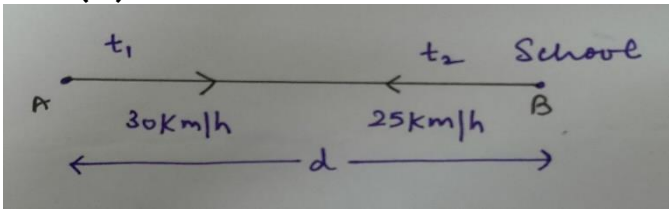
deceleration = 2.5 m/s^2

Negative acceleration is also called **retardation/deceleration**.

5. Correct option(A)

An object is said to be moving with a uniform acceleration if its velocity changes by equal amounts in equal intervals of time.

6. **Correct option(B)**



(a) While driving to school (AB):
Given: average speed = 30km/h
Distance= d
Let time taken = t_1
Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_1 = \frac{d}{30}$ - - - - - - **(1)**

(b) While returning from school (BA):
Given: average speed = 25km/h
Distance= d
Let time taken = t_2
Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_2 = \frac{d}{25}$ - - - - - - **(1)**

(c) Average speed for the round trip (ABA)

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Total distance = d + d
=2d

Total time = $t_1 + t_2$

Average speed = $\frac{2d}{t_1+t_2}$

Average speed = $\frac{2d}{\frac{d}{30}+\frac{d}{25}}$

Average speed = $\frac{2d}{d\left(\frac{1}{30}+\frac{1}{25}\right)}$

$= \frac{2}{\frac{1}{30}+\frac{1}{25}}$

$= \frac{2}{\frac{5+6}{150}}$ (Lcm of 30 and 25=150)

$= \frac{2x\ 150}{11}$

Average speed = 27.27 km/h
Hence, average speed for the round trip is 27.27 km/h

7. Correct option(A)

Given: initial velocity(u) =0m/s (because, body is starting from rest)
Final velocity(v)= 54km/h =15m/s

1km/h = $\frac{5}{18}$ m/s

54km/h = 54 x $\frac{5}{18}$ m/s
= 15m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration
u= initial velocity
v = final velocity

$a = \frac{15-0}{5}\ m/s^2$

$a = 3\ m/s^2$

acceleration =3 m/s^2

8. Correct option(C)

Given: initial velocity(u) =5m/s

Final velocity(v) =35m/s

Time taken = 5 s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{35-5}{5}\ m/s^2$

$a = 6\ m/s^2$

acceleration =6 m/s^2

9. Correct option (C, D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

If the velocity is increasing (v>u) then change in velocity will be positive and acceleration will be positive.

If the velocity is decreasing(v<u) then change in velocity will be negative and acceleration will be negative.

Negative acceleration is also called **retardation**.

10. Correct option (A, B)

(A) Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

(B) Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Hence, rate of change of displacement is called velocity.

(C) Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Hence, if displacement is zero then velocity is also zero.

(D) m/s is the unit of velocity.

11. Correct option (A, B, C)

(A) Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

(B) Velocity =$\frac{\text{displacement}}{\text{time interval}}$

SI unit of displacement is m

SI unit of time is s

So, SI unit of velocity is m/s.

(C) Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Hence, if displacement is 0 then velocity is also 0.

(D) velocity is a vector quantity.

12. Correct option (B)

Distance is the actual path traversed by the object in the given time. Distance travelled by an object can never be negative.

As Average speed = $\frac{\text{total distance}}{\text{total time}}$

So, average speed of an object can never be negative.

Whereas displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

The displacement of an object in a time can be equal to or less than distance travelled but never greater than the distance travelled.

As, Average velocity =$\frac{\text{total displacement}}{\text{time interval}}$

So, average velocity of an object in a time can be equal to or less than average speed but never greater than the average speed.

Displacement ≤ distance

Average Velocity ≤ average speed

Numerical ratio($\frac{\text{average velocity}}{\text{average speed}}$)

Can be equal to 1 or less than 1. Since, the denominator(speed) is either greater than or equal to the numerator(velocity).

13. Correct option (A, B, C, D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Velocity is a vector quantity as it has both magnitude(speed) and direction.

So, Acceleration is a vector quantity as it depends on the change in velocity.

Velocity can change in following ways:

- Its magnitude (speed) changes.
- Its direction changes.
- Its magnitude(speed) change but direction does not change.
- Its direction changes but magnitude(speed) does not change.

So, a body can do acceleration due to any of the above reasons.

14. Correct option (A, B, C)

Vector quantities are those quantities which have both magnitude(size) and direction.

(a) Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. Hence it is distance measured in a particular direction.so it is a vector quantity.

(b) Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Velocity is also a vector quantity as it depends on displacement. Velocity is the speed of an

object in a particular direction.

(c) Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Acceleration is a vector quantity as it depends on the change in velocity.

(d) Distance is the actual path traversed by the object in the given time. It has just the magnitude and no direction so it is a scalar quantity.

15. Correct option(B)

Given: velocity of the car= 45 km/h = 750m/min

As, 1 hour = 60 minutes

1 km = 1000 m

45 km/h = $\frac{45x1000}{60}$ m/min

= 750m/min

Time = 1 minute

Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Displacement= velocity x time

= 750 m/min x 1 min

= 750 m

Or distance covered in 1 minute in a particular direction = 750m

16. Correct option(A)

Given: velocity of the car= 45 km/h = 12.5m/s

As, 1 hour = 3600 seconds

1 km = 1000 m

1km/h = $\frac{5}{18}$ m/s

45 km/h = $\frac{5}{18}$ x 45 m/s

= 12.5 m/s

Time = 1 second

Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Displacement= velocity x time

= 12.5 m/s x 1 second

= 12.5m

Or distance covered in 1 second in a particular direction = 12.5m

17. Correct option(C)

Displacement is the shortest distance between the initial and final positions of the object in a given time. It is represented by a vector drawn from initial to final position of the object, during the given interval of time. It is a vector quantity as it possesses both magnitude and direction.

As the aero plane flies a distance of 324km towards south

So, displacement = 324km-south

18. Correct option(A)

Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Given: displacement = 324 km-south

Time interval = 20 minutes

1 minute = $\frac{1}{60}$hour

20minutes=$\frac{20}{60}$ hour

= $\frac{1}{3}$ hour

Velocity=$\frac{324}{\frac{1}{3}}$ km/h

= 324x 3 km/h

=972 km/h

19. Correct option(C)

Given: displacement = 90 km-east

Time = $1\frac{1}{2}$hr = $\frac{3}{2}$hr

Velocity = $\frac{\text{displacement}}{\text{time interval}}$

Velocity = $\frac{90\times2}{3}$ km/hr

$=\frac{90}{\frac{3}{2}}$km/hr

$=\frac{90\times2}{3}$

=60 km/hr

20. Correct Matching: a-Q; b-S; c-R; d-T

(a) 1km= 1000m

1hr= 3600s

1km/hr. = $\frac{1000}{3600}$ m/s

= $\frac{5}{18}$ m/s

(b) Instantaneous speed: when the speed of an object is variable/changing, then the object possesses different speeds at different instants. The speed of an object at a given instant of time is called its instantaneous speed.

Instantaneous speed of a vehicle is measured by speedometer.

(c) Average velocity: For a given motion of an object, the average velocity is that uniform velocity with which the object will cover the same displacement in a given interval of time as it does with actual variable velocity during the given interval of time

Average velocity = $\frac{\text{Total displacement}}{\text{Total time taken}}$

SI unit of Average velocity is m/s.

(d)Uniform speed: An object is said to be moving with a uniform speed, if it covers equal distances in equal intervals of time. For example, if a car is moving on a straight road from place A to B, distance of 80 km in 4 hours, and covers 20 km in every one hour, then the car is moving with a uniform speed =20km/h.

SI unit of speed is m/s.

3. Force and Laws of Motion

Solutions

LEVEL - I

1. Correct option: (B)

Solution: Inertia of a body is a measure of its resistance to any changes in its state of motion. Higher the mass of the object, greater the resistance to the change in state of motion of the object. Therefore, inertia has direct dependence on mass.

2. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net force acting on a body of mass m is:

$F_{net} = ma$

where F_{net} is the total external force on the body, and a is its acceleration.

Now, if the body is stationary, its acceleration is zero. Therefore, the total force acting on it should be zero as well.

Therefore, if a body is stationary, then the combination of forces acting on it balances each other.

3. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Given, an object is accelerating.

Now, if the resultant force on it begins to decrease, its resulting acceleration will also begin to decrease. The object will continue accelerating.

If its velocity changes direction, then also the object will continue accelerating as change in velocity leads to change in acceleration.

If the resultant force on the object is zero, then the acceleration of the object is also zero. The object will stop accelerating.

If the resultant force is at right angles to its direction of motion, then the speed of the object in the direction of motion cannot change. But the direction of velocity can change, and the object will still be accelerating.

Therefore, an object will continue accelerating until resultant force on it is zero.

4. Correct option: (D)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Given, a body is acted upon by a constant force.

Then, from $\vec{F}_{net} = m\vec{a}$, the acceleration of the body will also be constant.

Therefore, if a body is acted upon by a constant force, then it will have a uniform acceleration.

5. Correct option: (B)
Solution: We know, inertia is a measure of the ability of an object to resist changes in the state of motion of the object.
Inertia of rest is the ability of an object to resist changes in its state of rest.
An athlete runs some distance before taking a long jump. This is because he overcomes inertia of rest. If the athlete just stands in one spot and jump, his body will resist the change in its state of rest. But if he runs some distance before jumping, then he will be already moving before jumping, and thus can jump to greater length than if he were just standing.
Therefore, an athlete runs some distance before taking a long jump to over comes inertia of rest.

6. Correct option: (B)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Given, passengers standing in a bus are thrown outwards when the bus takes a sudden turn.
This is because, when the bus turns, there is an external force on the bus (from the road), which can change the state of motion of the bus. The feet of the person also come to rest. But the upper part of the person's body continues to move in forward direction due to inertia of direction. They then therefore try to maintain their original direction of motion owing to inertia of direction. As a result, they are thrown outwards.
Therefore, passengers standing in a bus are thrown outwards when the bus takes a sudden turn this happens because of inertia of direction.

7. Correct option: (C)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion.
Hence, Newton's first law of motion describes inertia.

8. Correct option: (C)
Solution: We know, inertia of a body is a measure of its resistance to any changes in its state of motion.
Therefore, the tendency of the body to continue in its state of rest or uniform motion in a straight line, even when some external force acts on it, is called inertia.

9. Correct option: (C)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Given, a knife is sharped by placing it on rotating iron disk, the sparks move tangentially.
Now, when the knife is put in contact with the rotating iron disk, sparks get generated due to the friction between the rotating knife and the disk. Theses sparks move tangentially to the rotating disk. This is because, the disk was originally rotating, so the sparks generated resist any changes in their direction of motion owing to the inertia of direction. They therefore move in their original direction even when they have been separated from the disk.
Consequently, a knife is sharped by placing it on rotating iron disk, the sparks move tangentially to the disc due to inertia of direction.

10. Correct option: (C)
Solution: Inertia of a body is a measure of its resistance to any changes in its state of motion. Higer the mass of the object, greater the resistance to the change in state of motion of the object. That is, inertia is directly proportional to the mass.
Given, there are two spheres of equal mass, made of iron and wood.
Since both the spheres have equal mass, they have the same inertia.

11. Correct option: (B)
Solution: The weight of an object is defined as the gravitational force of attraction of earth acting on the object.
Therefore, gravitational force of attraction of earth acting on a body is weight.

12. Correct option: (A)
Solution: We know, the weight of an object W is defined as the gravitational force of attraction of the earth acting on the object, and is given by:
$W = mg$
where m is the mass of the object and g is the acceleration due to gravity.

13. Correct option: (A)
Solution: We know, the weight of an object W is defined as the gravitational force of attraction of the earth acting on the object.
Its direction is always toward the centre of the earth, that is, vertically downward on the surface of the earth.

14. Correct option: (C)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion.
Hence, Newton's first law of motion describes inertia.

15. Correct option: (B)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of motion is the ability of the body to resist changes in its state of motion.
Given, when a fan switched off, it continuous to move. This is because of inertia of motion, owing which the fan tries to maintain its original state of motion.

16. Correct option: (B)
Solution: Inertia of a body is a measure of its resistance to any changes in its state of motion. Higher the mass of the object, greater the resistance to the change in state of motion of the object. Therefore, inertia has direct dependence on mass.

17. Correct option: (B)
Solution: Inertia of a body is a measure of its resistance to any changes in its state of motion. Higer the mass of the object, greater the resistance to the change in state of motion of the object. That is, inertia is directly proportional to the mass.
Given, two bodies having masses 1 kg and 4 kg. Since inertia is directly proportional to the mass, 4kg body would have more inertia.

18. Correct option: (C)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
This means, if body is at rest, it will continue to remain in rest unless some external force act on it.
Also, if the body is in a state of motion in a straight line, it will continue to move in a straight line until some external force act on it.

19. Correct option: (B)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Now, when a running car stops suddenly, the passengers are jerked in forward direction.
This is because, when the car is stopped, there is an external force on the car from the road, which brings it to rest. Inside the car, the lower part of the body in contact with seat experience a force and comes to rest but the upper part of the body continues to be in state of motion. As a result, they are jerked in forward direction.
Therefore, when a running car stops suddenly, the passengers are jerked move forward.
20. Correct option: (B)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of this resistance to any changes in its state of motion. It is directly dependent on the mass of an object. Mass is the physical cause that object has inertia to motion.
Therefore, the physical cause which changes or tends to change the state of rest (or) the state of motion of a body is mass.
21. Correct option: (A)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of this resistance to any changes in its state of motion. Hence Newton's first law of motion is also called as Law of inertia.
22. Correct option: (C)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of this resistance to any changes in its state of motion.
Now, a rider on horse falls when horse starts running all of a sudden. This is because Inertia of rest keeps the upper part of body at rest whereas lower part moves forward with the horse.
23. Correct option: (C)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Now, when a train stops suddenly passengers in the running train feel an instant jerk in the forward direction.
This is because, when the train is stopped, there is an external force on the train from the rail, which brings it to rest. Inside the train, the lower part of the body in contact with seat experience a force and comes to rest but the upper part of the body continues to be in state of motion. As a result, they are jerked in forward direction.
Therefore, when a train stops suddenly passengers in the running train feel an instant jerk in the forward direction because upper part of the body continues to be in state of motion whereas lower part of the body in contact with seat remain at rest.
24. Correct option: (D)
Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.

Inertia of a body is a measure of this resistance to any changes in its state of motion. Inertia is the property by virtue of which the body is unable to change by itself the state of rest and uniform linear motion.

25. Correct option: (B)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.

Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.

Now, a man getting down a running bus fall forward.

This is because, when the person gets down from a running bus, his feet come to rest due to a force of friction between the surface of the road and his feet. But due to inertia of motion, the upper part of his body continues to be in motion in the forward direction. As a result, the man falls in forward direction.

LEVEL-II

1. Correct option: (D)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Given, a person sitting in an open car moving with constant velocity throws a ball vertically up into air.

Now, once the ball is released upwards, a downward gravitational force start acting on it. But there is no horizontal force acting on the body. Hence the horizontal velocity of the ball cannot change. Therefore, both the boy and the ball have same horizontal velocity. The ball at the end of the journey falls exactly in the hand of the person who threw the ball.

Therefore, if a person sitting in an open car moving with constant velocity throws a ball vertically up into air the ball falls exactly in the hand which threw it up.

2. Correct option: (A)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.

Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.

Given, passengers standing in a bus are thrown outwards when the bus takes a sudden turn.

This is because, when the bus turns, there is an external force on the bus (from the road), which can change the state of motion of the bus. The feet of the person also come to rest. But the upper part of the person's body continues to move in forward direction due to inertia of direction. They then therefore try to maintain their original direction of motion owing to inertia of direction. As a result, they are thrown outwards.

Therefore, passengers standing in a bus are thrown outwards when the bus takes a sudden turn this happens because of inertia of direction.

3. Correct option: (D)

Solution: We know, inertia is a measure of the ability of an object to resist changes in the state of motion of the object.

Inertia of rest is the ability of an object to resist changes in its state of rest.

An athlete runs some distance before taking a long jump. This is because by running he gives himself large inertia of motion. If the athlete just stands in one spot and jump, his

body will resist the change in its state of rest. But if he runs some distance before jumping, then he will be already moving before jumping, and thus can jump to greater length than if he were just standing.
Therefore, an athlete runs some distance before taking a long jump because by running he gives himself large inertia of motion.

4. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.
Now, if the forces are unbalanced, we have a non-zero external force acting on the body. These forces then, in accordance to Newton's law of motion can accelerate the object.
Therefore, unbalanced forces can move a stationary body or they can stop a moving body.

5. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net force acting on a body of mass m is:

$F_{net} = ma$

where F_{net} is the total external force on the body, and a is its acceleration.

Now, if the body is stationary, its acceleration is zero. Therefore, the total force acting on it should be zero as well.
Therefore, if a body is stationary, then the combination of forces acting on it balances each other.

6. Correct option: (B)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of this resistance to any changes in its state of motion.
Now, a rider on horse falls backwards when horse starts running all of a sudden. This is because Inertia of rest keeps the upper part of body at rest whereas lower part moves forward with the horse.
Therefore, a rider on a horse fall backwards when the horse suddenly starts, this due to inertia of rider.

7. Correct option: (C)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Now, when a car (or) bus starts suddenly, the passengers fall backward.
This is because, when the car or bus starts suddenly, there is an external force on the car from the road, which brings it to the motion. Inside the car, the lower part of the body of the person in contact with seat experience a force and comes to motion but the upper part of the body continues to be in state of rest. As a result, they fall in backward direction.
Therefore, when a car (or) bus starts suddenly, the passengers fall backward. This is due to the fact that because of inertia of rest.

8. Correct option: (A)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.

Now, when a running car stops suddenly, the passengers are jerked in forward direction.
This is because, when the car is stopped, there is an external force on the car from the road, which brings it to rest. Inside the car, the lower part of the body in contact with seat experience a force and comes to rest but the upper part of the body continues to be in state of motion. As a result, they are jerked in forward direction.
Therefore, when a running car stops suddenly, the passengers are jerked move forward due to inertial of motion.

9. Correct options: (A, B)

Solution:
We know, a force is a kind of push or pull interaction between two objects.
Given, water in a bucket from the well is being pulled out. We have to list all the forces acting on the bucket.
Now, the earth pulls the bucket vertically downward. This force is weight of the bucket.
There is also tension force acting on the bucket via the rope attached to it, which is used to pull the bucket upward.
Therefore, while pulling the water in a bucket from well, the forces acting on the bucket are weight and tension.

10. Correct options: (A, B, C, D)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.
A force can brings about acceleration in the object. Hence, a force can produce stationary object into motion and also moving object to rest. Further, a force can also change direction of moving body.
Also, force can change dimensions of an object. For example, applying a force to the dough, changes its shape.

11. Correct options: (A, B, C, D)

Solution: We know, a force is a kind of push or pull interaction between two objects.
Different kinds of forces are muscular force, mechanical force, electrostatic force and magnetic force.

12. Correct options: (C)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia is the measure of tendency of the body to resist changes in its state of motion. Inertia is directly related to the mass of the body, Higher the mass, greater the inertia, and hence lighter body can change its state more easily than heavier body.

13. Correct option: (A, B, C)

Solution: We know, Newton's first law of motion is also called Galileo's law of inertia.
Inertia of body may be defined as the tendency of a body to opposite any change in state of rest (or) uniform motion.
The quantity of motion of the moving the body depends on its mass and velocity. It is directly proportional to mass of the body.
The tendency of a body to oppose any change in its direction of motion by itself is known as inertia of direction.

14. Correct option: (A)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.

Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Now, person in a car tends to fall back when it is suddenly starts.
This is because, when the car starts suddenly, there is an external force on the car from the road, which brings it to the motion. Inside the car, the lower part of the body of the person in contact with seat experience a force and comes to motion but the upper part of the body continues to be in state of rest. As a result, the person falls in backward direction.
Therefore, when a car starts suddenly, the passengers fall backward. This is due to the fact that because of inertia of rest.

15. Correct option: (C)

Solution: We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force.
Inertia of a body is a measure of its resistance to any changes in its state of motion. Inertia of direction is the ability of the body to resist changes in its direction of motion.
Given, a stone tied to a string is whirled around against in a circle. When the string is cut, the stone flies off tangentially.
This is because, the stone was originally rotating, so it will resist any changes in its direction of motion owing to the inertia of direction. The stone therefore move in its original direction even when the string is cut, which is the tangential direction to the circle.
Therefore, if a stone tied to a string is whirled around against in a circle and he string is cut, then the stone flies off tangentially due to inertia of direction.

16. Correct match: a) -> T), b) -> P), c) -> R), d) -> Q)

Solution: We know, muscular force is used when we push a lawn roller.
Also, mechanical force used to run a motor car engine.
A stone falls downwards due to the gravitational force acting on it.
Charged comb attracts small pieces of paper due to the electrostatic force between the charged comb and small pieces of paper.
Therefore, the correct match is a) -> T), b) -> P), c) -> R), d) -> Q)

Momentum

LEVEL - I

1. Correct option: (A)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.
Therefore, Force= mass $\times$ acceleration

2. Correct option: (A)

Solution: We know, the weight of an object W is defined as the gravitational force of attraction of the earth acting on the object, and is given by:

$W = mg$

where m is the mass of the object and g is the acceleration due to gravity.

We also know, the Newton's second law of motion states that the net force acting on a body of mass m is:

$F_{net} = ma$

where F_{net} is the total external force on the body, and a is its acceleration.

When the mass of the object is 1 kg, then:

$ma = mg$

$a = g$

$a = 9.8$

Hence, 9.8 N = 1 kgf

3. Correct option: (B)

Solution: We know, the momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Therefore, the product of mass and velocity of a body is called momentum.

4. Correct option: (C)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Linear momentum is a measure of total quantity of motion contained in a body.

Therefore, the total quantity of motion contained in a body is called linear momentum.

5. Correct option: (B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Therefore, the linear momentum of a body is denoted by the letter $\vec{p}$.

6. Correct option: (B)

Solution: We know, Newton's second law of motion states that the external force acting on an object is equal to the rate of change of the momentum of the object:

$\vec{F}_{ext} = \frac{\delta \vec{p}}{\delta t}$

where $\delta \vec{p}$ is the change in the momentum of the object in the small-time interval δt .

Since unit of force is Newton N, and unit of times is second s, we see the unit of momentum is Ns.

Therefore, Ns is the unit of momentum.

7. Correct option: (A)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, a toy car of mass 200 g moving with a speed of 5 m/s.

Therefore, the magnitude of momentum of the toy car is:

$p = mv$

$p = (0.2)(5)$

$p = 1\text{kgm/s}$

Therefore, the momentum of a toy car of mass 200 g moving with a speed of 5 m/s is 1 kg m/s.

8. Correct option: (A, B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Now, if the mass is reduced to half, the magnitude of the new momentum is:

$p' = (m/2)v$

$p' = p/2$

If velocity is reduced to half, the magnitude of the new momentum is

$p' = mv/2$

$p' = p/2$

Further, if the mass is reduced to half and velocity quadrupled, then the magnitude of new momentum is:

$p' = (m/2)(4v)$

$p' = 2p$

Therefore, to reduce the momentum of a given body to half of its original value, the mass must be reduced to half or velocity must be reduced to half.

9. Correct option: (D)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, two bodies A and B of same mass are moving with velocities v and $3v$ respectively.

Hence, the ratio of their momentum is:

$$\frac{p_1}{p_2} = \frac{m_1 v_1}{m_2 v_2}$$

$$\frac{p_1}{p_2} = \frac{mv}{m3v}$$

$$\frac{p_1}{p_2} = \frac{1}{3}$$

Therefore, if two bodies A and B of same mass are moving with velocities v and $3v$ respectively, then the ratio of their momentum is 1:3.

10. Correct option: (B)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Also, for constant acceleration a the distance covered s is related to the initial velocity u through:

$v^2 = u^2 + 2as$

Given, a car and lorry are moving with same momentum, and the same brake force is applied.

Now, let the mass of the car is m_c and the mass of the lorry be m_l. Then the acceleration of the car is:

$$a_c = \frac{F}{m_c}$$

Similarly, the acceleration of the lorry is:

$$a_l = \frac{F}{m_l}$$

Since when the car comes to rest, its final velocity is zero, therefore:

$$v^2 = u^2 + 2as$$

$$0 = u_c^2 - 2a_c s_c$$

$$s_c = \frac{u_c^2}{2\frac{F}{m_c}}$$

$$s_c = \frac{m_c u_c^2}{2F}$$

Here the negative acceleration signifies the braking force.
Similarly, the distance covered by the lorry before coming to rest is:

$$v^2 = u^2 + 2as$$

$$0 = u_l^2 - 2a_l s_l$$

$$s_l = \frac{u_l^2}{2\frac{F}{m_l}}$$

$$s_l = \frac{m_l u_l^2}{2F}$$

Now, we are also given car and lorry are moving with same momentum, hence:

$$m_l u_l = m_c u_c$$

Now, the ratio of distance covered by the car to the distance covered by the lorry is:

$$\frac{s_c}{s_l} = \frac{\frac{m_c u_c^2}{2F}}{\frac{m_l u_l^2}{2F}}$$

$$\frac{s_c}{s_l} = \frac{u_c}{u_l}$$

Since the mass of car is less than the mass of the lorry, therefore, its initial velocity is greater. Therefore, lorry comes to rest in shorter distance.

11. Correct option: (B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$$\vec{p} = m\vec{v}$$

where $\vec{v}$ is the velocity of the body.
Therefore, when the speed of a moving object is doubled, its momentum is doubled.

12. Correct option: (B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$$\vec{p} = m\vec{v}$$

where $\vec{v}$ is the velocity of the body.
We know, Newton's second law of motion states that the external force acting on an object is equal to the rate of change of the momentum of the object:

$$\vec{F}_{ext} = \frac{\delta \vec{p}}{\delta t}$$

where $\delta \vec{p}$ is the change in the momentum of the object in the small-time interval δt.
Given, the force of 15 N acts upon a mass of 15 kg.
Hence, the velocity it generates in 8 seconds is:

$$F_{ext} = \frac{\delta p}{\delta t}$$

$$F_{ext}\delta t = \delta(mv)$$

$$\delta v = \frac{F_{ext}\delta t}{m}$$

$$\delta v = \frac{15(8)}{15}$$

$$\delta v = 8\,m/s$$

Therefore, if the force of 15 N acts upon a mass of 15 kg, then the velocity it generates in 8 seconds is 8 m/s.

13. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$$\vec{F}_{net} = m\vec{a}$$

where $\vec{a}$ is the acceleration of the body.

Therefore, Newton second law measures magnitude of force.

14. Correct option: (B)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$$\vec{F}_{net} = m\vec{a}$$

where $\vec{a}$ is the acceleration of the body.

When m=1kg and a=1 m/s^2, then F=1N.

Therefore, 1 N is the force which produces an acceleration of 1 m/s^2 on a body of mass 1 kg.

15. Correct option: (B)

Solution: We know, relation between Newton and Dyne is:

$$1\text{N} = 10^5 \text{ dyne}$$

16. Correct option: (A)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$$\vec{p} = m\vec{v}$$

where $\vec{v}$ is the velocity of the body.

Therefore, momentum is directly proportional to mass.

17. Correct option: (B)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction.

This is a statement about the Law of conservation of momentum, which states that the for an isolated system, the total linear momentum is conserved.

18. Correct option: (B)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction.

When a horse pulls a cart, he exerts a force on the cart and the ground. As a result, the cart and the ground also push on the horse. The force exerted by the ground on the horse is what helps the horse to move forward.

Therefore, when a horse pulls a cart the force that helps the horse to move forward is the force exerted by the ground on the horse.

19. Correct option: (D)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction.

When we jump out of a boat standing in water, we pull the boat opposite to the direction of jump. Therefore, when we jump out of a boat standing in water, it moves opposite to the direction of jump.

20. Correct option: (C)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction.

Therefore, according to Newton's third law to every action there is equal and opposite reaction.

21. Correct option: (A)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

Hence, the force exerted by first body on the second body is called action.

22. Correct option: (B)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

The action and reaction forces act on two different bodies. Hence, they never cancel each other.

23. Correct option: (A)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

Given, A body exerts a force of 10 N on the ground.

Then from the Newton's third law of motion, the ground also exerts the same force on the body in opposite direction.

Therefore, a body exerts a force of 10 N on the ground when the reaction force of the ground on the body is 10 N.

24. Correct option: (C)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

When a gun is fire, it exerts a force on the bullet. The bullet, in accordance to Newton's third law of motion also exerts equal and opposite force on the gun. As a result, the gun recoils.

Therefore, gun recoil can be explained by Newton's third law.

25. Correct option: (B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$$\vec{p} = m\vec{v}$$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 5 kg is moving with a velocity of 5 m/s.

Therefore, the momentum of the body will be:

$$p = mv$$

$$p = (5)(5)$$

$$p = 25\,kgm/s$$

26. Correct option: (A)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is: $\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Given, in a body, 10 m/s^2 acceleration is produced when a 20 N force is acted upon it.

Therefore, the mass of the body is:

$$F_{net} = ma$$

$$m = \frac{F_{net}}{a}$$

$$m = \frac{20}{10}$$

$$m = 2\,kg$$

Therefore, if in a body, 10 m/s^2 acceleration is produced when a 20 N force is acted upon it, then its mass is 2kg.

27. Correct option: (D)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$$\vec{F}_{net} = m\vec{a}$$

where $\vec{a}$ is the acceleration of the body.

Given, in a body, 2 m/s^2 acceleration is produced when a 24 N force is acted upon it.

Therefore, the mass of the body is:

$$F_{net} = ma$$

$$m = \frac{F_{net}}{a}$$

$$m = \frac{24}{2}$$

$$m = 12\,kg$$

Therefore, if in a body, 2 m/s^2 acceleration is produced when a 24 N force is acted upon it, then its mass is 12kg.

28. Correct option: (B)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is: $\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Given, a force of 2 kgf acts on body of mass 20 kg.

Now, $1kgf = 10\,N$.

Therefore, the acceleration of body is:

$$F_{net} = ma$$

$$a = \frac{F_{net}}{m}$$

$$a = \frac{2(10)}{20}$$

$$a = 1m/s^2$$

Therefore, if a force of 2 kgf acts on body of mass 20 kg, then its acceleration is $1m/s^2$.

29. Correct option: (B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 15 kg is moving with a velocity of 5 m/s.

Therefore, the momentum of the body will be:

$p = mv$

$p = (15)(5)$

$p = 75\,kgm/s$

Therefore, the momentum of the body of mass 15 kg moving with a velocity 5 m/s is 75 kgm/s.

30. Correct option: (A)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 10 kg is moving with a velocity of 10 m/s.

Therefore, the momentum of the body will be:

$p = mv$

$p = (10)(10)$

$p = 100\,kgm/s$

Therefore, the momentum of the body of mass 10 kg moving with a velocity 10 m/s is 100 kg m/s.

LEVEL - II

1. Correct option: (D)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Now, a force or a set of forces can accelerate the object. That is, the forces may change the speed of a body, and may change the direction of motion of a body.

A force can also change the shape of a body.

2. Correct option: (A)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Therefore, Newton's second law explains us definition of force.

3. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Given, an object is accelerating.

Now, if the resultant force on it begins to decrease, its resulting acceleration will also begin to decrease. The object will continue accelerating.
If its velocity changes direction, then also the object will continue accelerating as change in velocity leads to change in acceleration.
If the resultant force on the object is zero, then the acceleration of the object is also zero. The object will stop accelerating.
If the resultant force is at right angles to its direction of motion, then the speed of the object in the direction of motion cannot change. But the direction of velocity can change, and the object will still be accelerating.
Therefore, an object will continue accelerating until resultant force on it is zero.

4. Correct option: (B)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.
Therefore, According to Newton's third law, Action = -reaction

5. Correct option: (A)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$$\vec{p} = m\vec{v}$$

where $\vec{v}$ is the velocity of the body.
We know, Newton's second law of motion states that the external force acting on an object is equal to the rate of change of the momentum of the object:

$$\vec{F}_{ext} = \frac{\delta \vec{p}}{\delta t}$$

where $\delta \vec{p}$ is the change in the momentum of the object in the small-time interval δt.
Given, a constant force 12 N acts on a body for 4s.
Hence, the change in the linear momentum of the body is:

$$F_{ext} = \frac{\delta p}{\delta t}$$
$$\delta p = F_{ext}\delta t$$
$$\delta p = 12(4)$$
$$\delta p = 48 kgm/s$$

Therefore, a constant force 12 N acts on a body for 4s, then the change in the linear momentum of the body is 48 kg m/s.

6. Correct option: (C)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$$\vec{F}_{net} = m\vec{a}$$

where $\vec{a}$ is the acceleration of the body.
Given, a force produces an acceleration of 5 cms^{-2} when it acts on a body of mass 20g.
Therefore, the force is:

$$F_{net} = ma$$
$$F_{net} = 0.02(5\times10^{-2})$$
$$F_{net} = 10^{-3}\ N$$

7. Correct option: (B)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

The action and reaction forces act on two different bodies.

8. Correct option: (C)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

When a cannon fires, it exerts a force on the cannonball. The cannonball, in accordance to Newton's third law of motion also exerts equal and opposite force on the gun. As a result, the cannon recoils.

Therefore, a cannon after firing recoils due to Newton's third law of motion.

9. Correct options: (A, B, C, D)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

As we row a boat, we exert a force on the river water backwards. Consequently, the river water exerts a force in the forward direction, moving the boat.

As a man walks on the ground, he applies a force on the surface of the road backwards. The road then exerts and equal and opposite force on the person, making him move forward.

As a rocket burns fuel, the exhaust gases are thrown outward. The exhaust gases then exert an equal and opposite force on the rocket, propelling it upwards.

As a gun is fired, it exerts a force on the bullet, which also exert an equal and opposite force on the gun, which leads to the recoil of the gun.

Therefore, rowing a boat, a man walking on the road, rocket and gun all works on the principle of Newton's 3rd law.

10. Correct option: (A)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$$\vec{p} = m\vec{v}$$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 2 kg moving with a velocity 4m/s, after some time velocity become 8m/s.

Therefore, the change in momentum is given by:

$$\delta p = m\delta v$$

$$\delta p = 2(8-4)$$

$$\delta p = 8 kgm/s$$

Therefore, if a body of mass 2 kg moving with a velocity 4m/s, after some time velocity become 8m/s, then the change in momentum is 8 kg m/s.

11. Correct options: (B)

Solution: We know, momentum is a measure of quantity of motion contained by the body.

Also, change in momentum is the measure of impulse.

Impulse and acceleration act in the same direction to the change in momentum.

In the case of uniform circular motion, the speed of the body is constant but its velocity continuously changes direction. The linear momentum is not conserved in uniform circular motion.

12. Correct options: (C)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

We know, Newton's second law of motion states that the external force acting on an object is equal to the rate of change of the momentum of the object:

$$\vec{F}_{ext} = \frac{\delta \vec{p}}{\delta t}$$

where $\delta \vec{p}$ is the change in the momentum of the object in the small-time interval δt.

Given, a force of 0.6N acting on a particle increases its velocity from 5 m/s to 6 m/s in 2s.

The mass of the particle is therefore:

$$F_{ext} = \frac{\delta(mv)}{\delta t}$$

$$F_{ext} = \frac{m\delta(v)}{\delta t}$$

$$m = \frac{F_{ext}\delta t}{\delta(v)}$$

$$m = \frac{0.6(2)}{1}$$

$$m = 1.2\,kg$$

Therefore, if a force of 0.6N acting on a particle increases its velocity from 5 m/s to 6 m/s in 2s, then its mass is 1.2 kg.

13. Correct option: (B, C)

Solution: We know, Newton's second law of motion states that the external force acting on an object is equal to the rate of change of the momentum of the object:

$$\vec{F}_{ext} = \frac{\delta \vec{p}}{\delta t}$$

where $\delta \vec{p}$ is the change in the momentum of the object in the small-time interval δt.

Hence the rate of change in momentum of a body is proportional to the net force applied on it and in the direction of applied force.

14. Correct option: (A)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

Given, a man applies a force of 5N to push a car.

The force applied by the man on the car is therefore action.

15. Correct option: (A)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

Given, a man applies a force of 5N to push a car.

Therefore, the car also pushes the man by an equal and opposite force which is the reaction.

The value of reaction is therefore 5 N.

16. Correct option: (A)

Solution: We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.

Given, a man applies a force of 5N to push a car.

The value of action is therefore 5 N.

17. Correct option: (C)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 3 kg moving a velocity 10 m/s after sometime its velocity becomes 20 m/s.

Therefore, Initial momentum of the body is:

$p = mv$

$p = (3)(10)$

$p = 30\,kg\,m/s$

Therefore, if a body of mass 3 kg moving a velocity 10 m/s after sometime its velocity becomes 20 m/s, then its initial momentum is 30 kg m/s.

18. Correct option: (D)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 3 kg moving a velocity 10 m/s after sometime its velocity becomes 20 m/s.

Therefore, final momentum of the body is:

$p = mv$

$p = (3)(20)$

$p = 60\ kg\,m/s$

Therefore, if a body of mass 3 kg moving a velocity 10 m/s after sometime its velocity becomes 20 m/s, then its initial momentum is 60 kg m/s.

19. Correct option: (B)

Solution: We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.

Given, a body of mass 3 kg moving a velocity 10 m/s after sometime its velocity becomes 20 m/s.

Therefore, change in momentum of the body is:

$\delta p = \delta mv$

$p = (3)(20-10)$

$p = 30\ kg\,m/s$

Therefore, if a body of mass 3 kg moving a velocity 10 m/s after sometime its velocity becomes 20 m/s, then change in its momentum is 30 kg m/s.

20. Correct match: a) -> Q), b) -> S), c) -> P), d) -> R)

Solution: We know, the Newton's second law of motion states that the net external force $\vec{F}_{ext}$ acting on a body of mass m is:

$\vec{F}_{net} = m\vec{a}$

where $\vec{a}$ is the acceleration of the body.

Therefore, the unit of force is kg m/s^2

This second law also explain force quantitatively.

We know, Newton's first law of motion states that the object tends to maintain their state of motion unless acted upon by an external force. This measure of resistance to change in state of motion is called as inertia.
We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.
Any object in motion has momentum.
Therefore, the correct match is: a) -> Q), b) -> S), c) -> P), d) -> R)

21. Correct match: a) -> S), b) -> Q), c) -> R), d) -> T)

Solution: We know, the weight of an object W is defined as the gravitational force of attraction of the earth acting on the object, and is given by:

$W = mg$

where m is the mass of the object and g is the acceleration due to gravity.

We know, Newton's third law states that for every action by a body, there is always equal and opposite reaction where action is the force exerted by the first body on the second body.
We know, the linear momentum $\vec{p}$ of a body of mass m is defined as:

$\vec{p} = m\vec{v}$

where $\vec{v}$ is the velocity of the body.
Therefore, change in momentum is $mv - mu$ where u is the initial velocity and v is the final velocity.
As we row a boat, we exert a force on the river water backwards. Consequently, the river water exerts a force in the forward direction, moving the boat. Rowing of boat thus works on the principle of Newton's third law.
Therefore, the correct match is: a) -> S), b) -> Q), c) -> R), d) -> T)

4.Work Energy and Power
Solutions

LEVEL-I

1. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved in the direction of applied force.

Therefore, mathematical Expression for work is

Force $\times$ Distance moved in the direction of applied force

2. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved in the direction of applied force.

Hence, work is said to be done when a force produces motion such that we have non zero value in the above equation for work.

3. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Given, an agent is moving positively charged body towards another fixed positive charge.

Since the body is moving in the direction of the force applied by the agent, the work done by the agent is positive.

4. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Therefore, work done when the body is displaced in the opposite direction of applied force is negative.

5. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a body moves along a circular path.

Now, for a body moving in uniform circular motion, the force is always acting toward the centre and the displacement is always in the tangential direction to the circle. That is, the angle between the force and the displacement is 90°.

Therefore, the work done is:

$W = Fd\cos 90^\circ$

$W = 0$

Therefore, a body when it moves along a circular path, the work done on the body is zero.

6. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Therefore, work done when the body is displaced in the direction of applied force is positive.

7. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, the force fails to displace the body. Therefore, the displacement of the body is zero. Consequently, the work is also zero.

Therefore, the work is done by the force if it fails to displace the body is zero.

8. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Given, brakes are applied on a moving vehicle.

Since the braking force acts in the opposite direction of motion, work done by the braking force is negative.

9. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Given, a bucket full of water is drawn up by a person.

As the bucket is drawn upwards, the displacement of the bucket is in upward direction. The gravitational force acting on the bucket is vertically downwards. Therefore, the work done by the gravitational force is negative.
Hence, a bucket full of water is drawn up by a person. In this case the work done by the gravitational force is negative because the force and displacement are in opposite direction.

10. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved in the direction of applied force.

Now, $W = 1$ when $F = 1$ and $d = 1$.

Therefore, 1 joule is the work done when a force of one Newton moves an object through a distance of one metre in the direction of force.

11. Correct option: (B)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a suitcase of mass 10 kg is at a height of 2 m from the ground.

In order to move the suitcase to this height, work need to be done against the gravitational force. Now, the work done by the gravitational force as the suitcase is moved to a height of 2 m from the ground is:

$W = mgh\cos 180^\circ$

$W = (10)(9.8)(2)(-1)$

$W = -196\,J$

where negative sign shows gravitational force act in the opposite direction of the displacement.

Therefore, the work done by external agent to move the suitcase to 2m, will be opposite of the work done by the gravity.

The work done in holding a suitcase of mass 10 kg at a height of 2 m from the ground is 196 J.

12. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved in the direction of applied force.

Therefore, work is measured as the product of force and displacement.

13. Correct option: (D)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a load of 100 kg is pulled up by 5m.

Now, the work done by the gravitational force as load is moved to a height of 5 m from the ground is:

$W = mgh\cos 180^\circ$

$W = (100)(9.8)(5)(-1)$

$W = -4900\,J$

where negative sign shows that the displacement and gravitational force are in opposite direction.
Therefore, the work done by an external agent to pull the load by 5m will be opposite of the work done by the gravity.
Therefore, if a load of 100 kg is pulled up by 5m, the work done is 4900 J.

14. Correct option: (A)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.
Given, a body is moved horizontally along a frictionless surface.
In this case the direction between the displacement and the gravitational force is perpendicular to each other. Therefore, the work done by the gravity is:

$W = mgh\cos 90^\circ$

$W = 0$

Therefore, the work done against gravity when a body is moved horizontally along a frictionless surface is zero.

15. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved in the direction of applied force.
Work is a scalar quantity. It has no sense of direction attached to it.

16. Correct option: (B)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.
To raise a body, we need to do work against the force of gravity.
Now, work done by gravity when a body of mass m is raised to a height h above the ground is:

$W = mgh\cos 180^\circ$

$W = -mgh$

Hence work needed to lift a body of mass m to a height h above the ground is mgh.

17. Correct option: (D)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.
Given, a body of mass 100 kg is lifted up by 10 m.
Now work done by the gravity is:

$W = mgh\cos 180^\circ$

$W = -(100)(10)(10)$

$W = -10000\,J$

Therefore, if a body of mass 100 kg is lifted up by 10 m, then work done by the gravity is $-10000\,J$.

18. Correct option: (B)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a boy weighing 50 kg climbs up vertical height of 100 m.

Now work done by the gravity is:

$W = mgh\cos 180^\circ$

$W = -(50)(10)(10)$

$W = -5000\ J$

Now, the as the boy climbs the hill, he does work against the gravitational force. Therefore, the work done by him is equal and opposite to that of the work done by the gravity.

Therefore, if a boy weighing 50 kg climbs up vertical height of 100 m, then work done by him in Joules is 5000.

19. Correct option: (A)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a 5 kg mass falls through 400 cm.

The work done on it by the earth's gravitational force is therefore:

$W = mgh\cos 0^\circ$

$W = 5(9.8)(4)$

$W = 196\ J$

Therefore, if a 5 kg mass falls through 400 cm, then the work done on it by the earth's gravitational force is 196 J.

20. Correct option: (D)

Solution: We know, work W done by a constant force F is given by: $W = Fd$

where d is the distance moved in the direction of applied force.

Given, a force of 1 N moves a body through a distance 4 m in its own direction.

Therefore, the work done by the force is:

$W = (1)(4)$

$W = 4\ J$

Therefore, the work done when force of 1 N moves a body through a distance 4 m in its own direction is 4 J.

21. Correct option: (A)

Solution: We know, work W done by a constant force F is given by: $W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, an engine pulls a train 1 km over a level track. The frictional resistance is $5\times10^5\ N$ .The work done by the train is:

$W = Fd\cos 180^\circ$

$W = (5\times10^5)(1000)(-1)$

$W = -5\times10^8\ J$

Therefore, if an engine pulls a train 1 km over a level track and the frictional resistance is $5\times10^5\ N$, then work done by the train is $-5\times10^8\ J$.

22. Correct option: (C)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, man does 2500 J of work in climbing a tree 5 m tall.

Now work done by the gravity as the man climbs the tree is:

$W = mgh\cos180^\circ$

$W = -(m)(10)(5)$

$W = -50m\ J$

Now, the as the boy climbs the tree, he does work against the gravitational force. Therefore, the work done by him is equal and opposite to that of the work done by the gravity. Therefore, his mass is:

$2500 = 50m$

$m = 50\,kg$

Therefore, if a man does 2500 J of work in climbing a tree 5 m tall then is mass is 50kg.

23. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, work done by a force in moving an object through a distance 20 cm is 24.2 J.

The magnitude of force is:

$W = Fd$

$24.2 = (F)(0.2)$

$F = 121N$

Therefore, work done by a force in moving an object through a distance 20 cm is 24.2 J, then magnitude of force is 121 N.

24. Correct option: (C)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a person of mass 50 kg climbs a tower of height 72 m.

Now work done by the gravity as the person climbs the tower is:

$W = mgh\cos180^\circ$

$W = -(50)(9.8)(72)$

$W = -35280\,J$

Now, the as the boy climbs the tree, he does work against the gravitational force. Therefore, the work done by him is equal and opposite to that of the work done by the gravity: $35280\,J$

Therefore, a person of mass 50 kg climbs a tower of height 72 m, then the work done is 35280 J.

25. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved in the direction of applied force.

Given, a force of 2 N moves a body through a distance 10 cm in the direction of force.

Therefore, the work done by the force is:

$W = (2)(0.1)$

$W = 0.2J$

Therefore, the work done by the force of 2 N moving a body through a distance 10 cm in the direction of force is 0.2 J.

26. Correct option: (A)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a weight lifter is lifting weights of mass 200 kg up to a height 2 m.

Now work done by the gravity as the person lifts the weight is:

$W = mgh\cos 180^\circ$

$W = -(200)(9.8)(2)$

$W = -3920\,J$

Now, the as the boy climbs the tree, he does work against the gravitational force. Therefore, the work done by him is equal and opposite to that of the work done by the gravity: $3920\,J$

Therefore, if a weight lifter is lifting weights of mass 200 kg up to a height 2 m, then the work done is 3920 J.

27. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a coolie carries a load of 50 kg on his head and walks on a level road up to 100 m.

Now, the force exerted by the coolie on the load is the normal force, acting perpendicular upwards. The displacement of the collie is in the horizontal direction. The angle between the force and the displacement is therefore in perpendicular direction.

Therefore, the work done is:

$W = Fd\cos 90^\circ$

$W = 0$

Therefore, if a coolie carries a load of 50 kg on his head and walks on a level road up to 100 m, then the work done by him is 0.

28. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Therefore, the work done by a force on a body depend upon the displacement of the body, the angle between the force vector and the displacement vector, and since the force acting on a body depends on its mass, the work done also depends on the mass of the body.

Therefore, the work done by a force on a body does not depend upon the initial velocity of the body.

29. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a body is moved through a distance of 3 m

Now, if a body is pushed over an inclined plane, then work done is:

$W = Fd\cos\theta$

$W = 3F\cos\theta$

where θ is the angle of inclination.

And, if a body is lifted vertically upward, then work done is:

$W = Fd$

$W = 3F$

When the body is pushed over smooth rollers, then work done is:

$W = 3F_1$

When pushed on a plane, horizontal surface, then work done is:

$W = 3F_2$

Now, when the body is pushed over rollers, we don't have to do work against the friction force, so less work need to be done.

And when the body is pushed in a plane horizontal surface, work is needed to be done against friction force.

Therefore, the maximum work is done when a body is lifted vertically upward.

30. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Now, if a body is pushed over an inclined plane, then work done is:

$W = Fd\cos\theta$

$W = 3F\cos\theta$

where θ is the angle of inclination.

And, if a body is lifted vertically upward, then work done is:

$W = Fd$

$W = 3F$

When the body is pushed over smooth rollers, then work done is:

$W = 3F_1$

When pushed on a plane, horizontal surface, then work done is:

$W = 3F_2$

Now, when the body is pushed over rollers, we don't have to do work against the friction force, so less work need to be done.

And when the body is pushed in a plane horizontal surface, work is needed to be done against friction force.

Therefore, the minimum work is done when pushed over smooth rollers.

LEVEL-II

1. Correct option: (A)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a 5 kg mass falls through 400 cm.

Now work done by the gravity is:

$W = mgh\cos 0^\circ$

$W = (5)(9.8)(4)$

$W = 196J$

Therefore, if a 5 kg mass falls through 400 cm, then work done by the gravity is 196 J.

2. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a girl is carrying a school bag of 3 kg mass on her shoulder and moves 200 m on a level road.

Now, the force exerted by the girl on the bag is the normal force, acting perpendicular upwards. The displacement of the girl is in the horizontal direction. The angle between the force and the displacement is therefore in perpendicular direction.

Therefore, the work done is:

$W = Fd\cos 90^\circ$

$W = 0$

Therefore, a girl is carrying a school bag of 3 kg mass on her shoulder and moves 200 m on a level road, then the work done by her is 0.

3. Correct option: (B)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a girl weighing 400 N climbs a vertical ladder. She climbed up to a height 2m.

Now work done by the gravity is:

$W = mgh\cos 180^\circ$

$W = -(400)(2)$

$W = -800\,J$

Now, the as the girl climbs the ladder, she does work against the gravitational force. Therefore, the work done by her is equal and opposite to that of the work done by the gravity: $800\,J$

Therefore, if a girl weighing 400 N climbs a vertical ladder up to a height 2m, then the work done is 800 J.

4. Correct option: (C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a man moves on straight horizontal road with a block of mass 2 kg in his hand. He covers a distance of 40 m with an acceleration of $0.5 m/s^2$.

We know, the force F on an object of mass m is:

$F = ma$

where a is the acceleration of the object.

Hence work done by the man on the block during the motion is:

$W = Fd\cos 0^\circ$

$W = mad$

$W = 2(0.5)(40)$

$W = 40J$

Therefore, if a man moves on straight horizontal road with a block of mass 2 kg in his hand and he covers a distance of 40 m with an acceleration of $0.5 m/s^2$, then work done by the man on the block during the motion is 40 J.

5. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a box is pushed through 4.0 m across a floor offering 100 N resistance.

The work done by the resisting force is

$W = Fd\cos 180^\circ$

$W = (100)(4)(-1)$

$W = -400\,J$

Therefore, a box is pushed through 4.0 m across a floor offering 100 N resistance, then work done by the resisting force in magnitude is 400J.

6. Correct option: (C)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, A body of mass 10 kg is moving at a constant speed of 20m s over a frictionless surface.

In this case the direction between the displacement and the weight is perpendicular to each other. Therefore, the work done by the weight is:

$W = mgh\cos 90^\circ$

$W = 0$

Therefore, if a body of mass 10 kg is moving at a constant speed of 20m s over a frictionless surface, then the work done by the weight is 0.

7. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a force of 5 N is applied on a 20 kg mass at rest.
We know, the force F on an object of mass m is:

$$F = ma$$

where a is the acceleration of the object.
Therefore, the acceleration of the object is:

$$a = \frac{F}{m}$$

$$a = \frac{5}{20}$$

$$a = \frac{1}{4} m/s^2$$

Therefore, the displacement of the mass at the end of third second is:

$$s = ut + \frac{1}{2}at^2$$

$$s = 0 + \frac{1}{2}(\frac{1}{4})(3)^2$$

$$s = \frac{9}{8}m$$

Now, the displacement of the mass at the end of fourth second is:

$$s = ut + \frac{1}{2}at^2$$

$$s = 0 + \frac{1}{2}(\frac{1}{4})(4)^2$$

$$s = \frac{16}{8}m$$

Therefore, the displacement of the body in third second is:

$$d = \frac{16}{8} - \frac{9}{8}m$$

$$d = \frac{5}{8}m$$

Hence the work done in the third second is:

$$W = Fd\cos 0^\circ$$

$$W = 5\left(\frac{5}{8}\right)$$

$$W = \frac{25}{8}J$$

8. Correct option: (A)
Solution: We know, work is a scalar quantity, it can be added by ordinary rules of arithmetic.
Given, the work done in carrying a body from A to B is 20 J and from B to C is 15 J.
Now, the work done in moving A to C is just the sum of the work done in carrying a body from A to B and then from B to C: $20 + 15 = 35J$.
Therefore, if the work done in carrying a body from A to B is 20 J and from B to C is 15 J, then the work done in moving A to C is 35J.

9. Correct options: (A, C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Therefore, the work done by a force on a body depends on magnitude of the force and distance through which the body moves.

10. Correct options: (A, C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Also, since the friction force always act in the opposite direction to the displacement, the work done by friction force is always negative.

11. Correct options: (A, B, D)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Now, when a stone tied to string is whirled in a circle, the force is always acting toward the centre and the displacement is always in the tangential direction to the circle. That is, the angle between the force and the displacement is 90°.

Therefore, the work done is:

$W = Fd\cos 90^\circ$

$W = 0$

Further, when the object moves in the direction in which force is applied, the work done by the force is positive.

And when the object moves in opposite direction in which the force is applied, the work done by the force is negative.

Hence, when brakes are applied on a moving vehicle, work done by the braking force is negative.

And if a body falling freely under the action of gravity has positive work done by the gravitational force.

Similarly, work done by the flying aeroplane is positive as the force of engine and the displacement are in the same direction.

12. Correct options: (A, C)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Therefore, no work is done by a force on an object if the force is always perpendicular to its velocity and the object is stationary but the point of application of force acts on the object.

13. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a force of 10 N moves an object through a distance of 1 m in the direction of the force.

The work done is

$W = Fd\cos 0^\circ$

$W = (10)(1)$

$W = 10\,J$

Therefore, if a force of 10 N moves an object through a distance of 1 m in the direction of the force, then work done by the force is 10J.

14. Correct option: (C)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, 200 kg of water through a vertical height of 6 metres is meant to be lifted.

Now work done by the gravity on the water is:

$W = mgh\cos 180^\circ$

$W = -(200)(10)(6)$

$W = -12000\,J$

Now, the as the agency lifts the water, work is done against the gravitational force. Therefore, the work done by the agency is equal and opposite to that of the work done by the gravity: $12000\,J$

Therefore, if 200 kg of water through a vertical height of 6 metres is meant to be lifted, then the work done is 12000 J.

15. Correct option: (D)

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a porter lifts a luggage of 15kg from the ground and puts it on his head 1.5m above the ground.

Now work done by the gravity on the luggage:

$W = mgh\cos 180^\circ$

$W = -(15)(10)(1.5)$

$W = -22.5\,J$

Now, the as the potter lifts the luggage, work is done against the gravitational force. Therefore, the work done by the potter is equal and opposite to that of the work done by the gravity: $22.5\,J$

Therefore, if a porter lifts a luggage of 15kg from the ground and puts it on his head 1.5m above the ground, then the work done is 22.5 J.

16. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a man weighing 50 kg supports a man of 20 kg on his hand and he is stationary.

Since there is no displacement of the man, the work done by him is zero.

17. Correct option: (B)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a man weighing 50 kg supports a man of 20 kg on his hand and moves a distance of 30 m on the smooth horizontal road.

Now, the force exerted by the man on another man is the normal force, acting perpendicular upwards. The displacement of both the men is in the horizontal direction. The angle between the force and the displacement is therefore in perpendicular direction.

Therefore, the work done is:

$W = Fd\cos 90^\circ$

$W = 0$

Therefore, a man weighing 50 kg supports a man of 20 kg on his hand and moves a distance of 30 m on the smooth horizontal road, then the work done by him is 0.

18. Correct option: 600 J.

Solution: We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.

Given, a man weighing 50 kg supports a man of 20 kg on his hand and moves a distance of 30 m vertically upwards.

Now work done by the gravity on the man:

$W = mgh\cos 180^\circ$

$W = -(20)(10)(30)$

$W = -600J$

Now, the as the potter lifts the luggage, work is done against the gravitational force. Therefore, the work done by the potter is equal and opposite to that of the work done by the gravity: $600\,J$

Therefore, if a man weighing 50 kg supports a man of 20 kg on his hand and moves a distance of 30 m vertically upwards, then the work done is 600 J.

19. Correct match: a) ->S), b) -> R), c) -> Q), d) ->P), e) -> P)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

When the object moves in the direction in which force is applied, the work done by the force is positive.

When the object moves in opposite direction in which the force is applied, the work done by the force is negative.
Therefore, Work done by a force when it displaces the block through a distance s in its own direction is:

$W = Fs\cos 0^\circ$

$W = Fs$

Now, work done by gravity on an object thrown upwards is negative. This is because the displacement of the object and the gravitational force are in opposite direction.
Work done by the frictional force when a body is displaced through a distance s

$W = Fs\cos 180^\circ$

$W = -Fs$

Work done by a force when it displaces a body through a distance in a direction perpendicular to its own direction is:

$W = Fd\cos 90^\circ$

$W = 0$

Also, work done in pushing a rock and not being able to move it is zero.
Therefore, the correct match is a) ->S), b) -> R), c) -> Q), d) ->P), e) -> P).

20. Correct match: a) ->P), b) -> T), c) -> S), d) ->R)

Solution: We know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.
We know, 1 Joule is related to the erg by:

$1\text{J} = 10^7\,\text{erg}$

Further, 1erg is the amount of work done when a force of one dyne acting on an object of mass 1 g displaces it by 1 cm, thus:

$1\text{erg} = 1\text{gcm}^2\text{s}^{-2}$

We know, the work W done by gravitational force on mass of an object m is:

$W = mgh\cos\theta$

where h is the distance moved by the object and θ is the angle between the gravitational force and the displacement.
Now, when the body is lifted, work is done against the gravitational force. Therefore, the work done in lifting is equal and opposite to that of the work done by the gravity

$W = -mgh\cos 180^\circ$

$W = mgh$

$W = \text{Weight of the body} \times \text{Vertical distance}$

If a body falls freely under the action of gravity, then the work done is positive as both the force and the displacement are in the same direction.
Therefore, the correct match is a) ->P), b) -> T), c) -> S), d) ->R)

Energy
LEVEL-I

1. Correct options: (C)

Solution: We know work is energy transferred to or from an object via a force acting on the object. The ability to do work is energy.

2. Correct option: (A)

Solution: We know, work W done by a constant force F is given by:

$W = Fd$

where d is the distance moved by the object in the direction of force.

If $F = 1N, d = 1m$, then $W = 1J$. Therefore, one joule is the product of force of 1 N and displacement of 1m.

3. Correct option: (B)

Solution: Energy is a scalar quantity. It can be added and subtracted using ordinary rules of arithmetic. It doesn't have a sense of direction.

4. Correct option: (B)

Solution: The SI unit of energy is Joule. One joule is the product of force of 1 N and displacement of 1m.

Different Forms of Energy

5. Correct option: (D)

Solution: We know, potential energy is energy that is associated with the configuration of a system in which a conservative force act. A force is conservative if the network it does on a particle moving between two points does not depend on the path taken by the particle.

Now, a compressed spring and a stretched elastic band, both have elastic potential energy, which is related to the state of compression or extension of an elastic object.

6. Correct option: (D)

Solution: Fuels possess chemical energy. When fuels burn, chemical energy is changed to heat energy.

7. Correct option: (D)

Solution: We know the potential energy associated with a system consisting of Earth and a nearby particle is gravitational potential energy.

Therefore, potential energy is stored in the water behind a dam on a hill – top.

8. Correct option: (A)

Solution: We know the potential energy associated with a system consisting of Earth and a nearby particle is gravitational potential energy. If the particle moves from height y_i to y_f height the change in the gravitational potential energy of the particle–Earth system is:

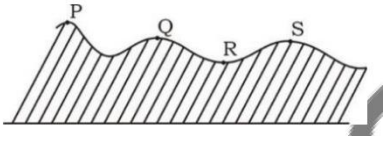
$\Delta U = mg(y_f - y_i)$

If we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$

Hence higher the height of the object, greater the potential energy of the object.

Given, a ball was rolled down a slope as shown in the diagram below. Sine the point P is at the greatest height from the ground, the ball has greatest potential energy at point P.



9. Correct option: The potential energy from the stretched elastic band was changed into kinetic energy and transferred from the marble to the ball

Solution: We know work is energy transferred to or from an object via a force acting on the object.

We know, potential energy is energy that is associated with the configuration of a system in which a conservative force act. A force is conservative if the network it does on a particle moving between two points does not depend on the path taken by the particle.

Further, kinetic energy is the energy associated with a moving object.

Given, Raj stretched the elastic band on his catapult and aimed a marble at his target which was a rubber ball. When he released the marble, it hit the ball and ball fell off the tree stump.

Now, elastic potential energy is related to the state of compression or extension of an elastic object. Therefore, the potential energy from the stretched elastic band was changed into kinetic energy and transferred from the marble to the ball.

10. Correct option: (B)

Solution: We know, kinetic energy is the energy associated with a moving object.

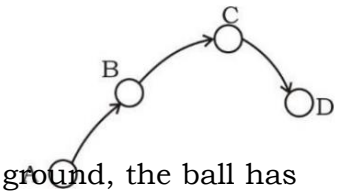
Now, when a substance is heated, the molecules inside the substance started moving more rapidly. Therefore, kinetic energy of the molecules increases.

11. Correct option: (A)

12. Correct option: (A)

Solution: If we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$



Hence higher the height of the object, greater the potential energy of the object. Given, a soccer ball was thrown in the air as shown in the diagram on right. Sine the point A is at the least height from the ground, the ball has least potential energy at point A.

13. Correct option: (B)

Solution: If we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$



We know, kinetic energy is the energy associated with a moving object.

Given, a boy drops a ball from a height as shown in the figure on the right.

Since point 3 is at the ground level, the ball doesn't possess any potential energy at this point.

Further, since the speed of the object is zero at the topmost point, it doesn't possess any kinetic energy at point 1.

The point at which the body possess both potential and kinetic energy is point 2.

14. Correct option: (D)

Solution: Given, a boy rolls a marble on level cement ground. After a while, it stops.

This is because the kinetic energy of the marble possessed was change into heat energy and sound energy. Further, there was friction between the moving marble and cement floor which transfers the kinetic energy of the marble into thermal and sound energy.

15. Correct option: (B)

Solution: We know, during photo synthesis, the light energy changes into chemical energy of the food.

Power

16. Correct option: (D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is: $P_{avg} = \frac{W}{\Delta t}$

Given, an iron of 2000 W uses energy 240000 J.

The time taken by the iron is:

$$\Delta t = \frac{W}{P_{avg}}$$

$$\Delta t = \frac{240000}{2000}$$

$$\Delta t = 120\,s$$

$$\Delta t = 2\,\text{min}$$

Therefore, if an iron of 2000 W uses energy 240000 J, then time taken by the iron is 2 minutes.

17. Correct option: (D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is: $P_{avg} = \frac{W}{\Delta t}$

Given, work done is 30 J in 5 sec by a machine.

Therefore, power of machine is:

$$P_{avg} = \frac{30}{5}$$

$$P_{avg} = 6W$$

Therefore, if a machine does work of 30 J in 5 sec, then its power is 6W.

18. Correct option: (C)

Solution: We know energy is the ability of doing work.

And work can be expressed as force$\times$ displacement

Unit of power is Watt.

And power is the amount of work done per unit time

The incorrect statement is thus unit of power is joule.

19. Correct option: (B)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is: $P_{avg} = \frac{W}{\Delta t}$

Given, the filament bulb of 100 w consumed energy of 1000 J

The time taken by the bulb is:

$$\Delta t = \frac{W}{P_{avg}}$$

$$\Delta t = \frac{1000}{100}$$

$$\Delta t = 10\,s$$

Therefore, if the filament bulb of 100 w consumed energy of 1000 J, then time taken by the bulb is 10 s.

20. Correct option: (C)

Solution: The S.I unit of power is Watt.

21. Correct option: 25W

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

We know, work W done by a constant force F is given by:

$$W = Fd\cos\theta$$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Given, a boy walks up the staircase of 20 m with a force of 600 N, the time taken by the boy is 8 min.

The power consumed by the body is:

$$P_{avg} = \frac{W}{\Delta t}$$

$$P_{avg} = \frac{Fd\cos 0}{\Delta t}$$

$$P_{avg} = \frac{(600)(20)}{(8)(60)}$$

$$P_{avg} = 25W$$

22. Correct option: (C)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Therefore, the work in relevance to time is related to power.

23. Correct option: (B)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Given, a fan keeps working for 30 min and uses energy of 90000 J

Therefore, power of fan is:

$$P_{avg} = \frac{90000}{30(60)}$$

$$P_{avg} = 50\ W$$

Therefore, if a fan keeps working for 30 min and uses energy of 90000 J, then its power is 50 W.

24. Correct option: (A)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

When $W = 1J$, $\Delta t = 1s$, then $P_{avg} = 1Watt$

Therefore, in symbols, one watt is equal to $1J/s$.

25. Correct option: (D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

KWH is the unit of energy consumed by an object of power 1 kilowatts running for an hour.

26. Correct option: (C)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Also, the work done to lift an object to a height h against gravity is:

$W = mgh$

Given, a man of mass 60 kg, climbs upstairs up to a height 10 m in 30 sec.

Therefore, power of man is:

$$P_{avg} = \frac{W}{\Delta t}$$

$$P_{avg} = \frac{mgh}{\Delta t}$$

$$P_{avg} = \frac{60(10)(10)}{30}$$

$$P_{avg} = 200W$$

Therefore, if a machine does work of 30 J in 5 sec, then its power is 6W.

27. Correct option: (C)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Also, the work done to lift an object to a height h against gravity is:

$W = mgh$

Given, a boy of mass 50 kg runs up a stair case of 45 steps in 9 seconds where the height of each step is 15 cm

Therefore, power of boy is:

$$P_{avg} = \frac{W}{\Delta t}$$

$$P_{avg} = \frac{mgh}{\Delta t}$$

$$P_{avg} = \frac{50(10)(45)(0.15)}{9}$$

$$P_{avg} = 375W$$

Therefore, if boy of mass 50 kg runs up a stair case of 45 steps in 9 seconds if the height of each step is 15 cm, then the power is 375 W.

28. Correct option: (B)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Also, the kinetic energy K of a moving object of mass m with speed v is:

$$K = \frac{1}{2}mv^2$$

Therefore, if a body of accelerates uniformly from rest to a speed v in t sec, the average power delivered is

$$P_{avg} = \frac{\frac{1}{2}mv^2}{t}$$

$$P_{avg} = \frac{1}{2t}mv^2$$

29. Correct option: (A)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is: $P_{avg} = \frac{W}{\Delta t}$

Also, the work done to lift an object to a height h against gravity is: $W = mgh$

Given, the power of motor pump is 0.39 kW.

Therefore, the amount of water pumped by it to a height of 10 m in 60 s is

$$P_{avg} = \frac{W}{\Delta t}$$

$$P_{avg} = \frac{mgh}{\Delta t}$$

$$m = \frac{P_{avg}\Delta t}{gh}$$

$$m = \frac{(390)(60)}{10(10)}$$

$$P_{avg} = 234\,kg$$

Therefore, if the power of motor pump is 0.39 kW, then the amount of water pumped by it to a height of 10 m in 60 s is 234 kg.

30. Correct option: (D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Also, the work done to lift an object to a height h against gravity is:

$$W = mgh$$

Given, a pump is lifting 2 kg water from an average depth of 10 m per second

Therefore, Power delivered by the pump is

$$P_{avg} = \frac{W}{\Delta t}$$

$$P_{avg} = \frac{mgh}{\Delta t}$$

$$m = \frac{(2)(10)(10)}{1}$$

$$P_{avg} = 200\ W$$

Therefore, a pump is lifting 2 kg water from an average depth of 10 m per second, then the power delivered by the pump is 200 W.

LEVEL-II

1. Correct option: (B)

Solution: We know, any moving object possess kinetic energy.

Given, a car stopped to avoid the crash with a van.

Now, as the car is stopped, it screeches and came to a stop. Therefore, its kinetic energy is converted into sound and thermal energy.

2. Correct option: (A)

Solution: We know, from conservation of energy, energy cannot be created or destroyed. It can only be converted or transferred from one form to another form.

3. Correct option: (C)

Solution: We know the potential energy associated with a system consisting of Earth and a nearby particle is gravitational potential energy. If the particle moves from height y_i to y_f height the change in the gravitational potential energy of the particle–Earth system is:

$$\Delta U = mg(y_f - y_i)$$

If we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$$\Delta U = mgh$$

Therefore, when an object is raised to a certain height above the ground it possesses gravitational potential energy.

4. Correct option: (B)

Solution: We know, the kinetic energy K of a moving object of mass m with speed v is:

$$K = \frac{1}{2}mv^2$$

Given, a shopping cart moves at a speed of 3 m/sec with a mass of 2 kg.

Therefore, the kinetic energy of the cart is:

$$K = \frac{1}{2}2(3^2)$$

$$K = 9\,J$$

Therefore, if a shopping cart moves at a speed of 3 m/sec with a mass of 2 kg, then its kinetic energy is 9 J.

5. Correct option: (B)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Since both work done and time are scalar quantities, hence power too is a scalar quantity.

6. Correct option: (B)

Solution: We know, one horsepower equal 746 Watts.

Therefore, 1 HP=746 W

7. Correct option: (B)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

8. Correct option: (B)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Given, 15 joules of work are done in a time of 15 sec.

Therefore, power is:

$$P_{avg} = \frac{15}{15}$$

$$P_{avg} = 1W$$

9. Correct option: (B, D)

Solution: We know, for a particle, a change in the kinetic energy equals the net work done on the particle:

$$K_f - K_i = W$$

Given, the kinetic energy of the particle continuously increases with time.

Therefore, the net work done on the particle must be positive.

Now, we know, work W done by a constant force F is given by: $W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Therefore, for the work to be positive, the resultant force on the particle must be at an angle less than 90° since cosine is positive in this interval.

Since kinetic energy K is related to the momentum p by: $K = \frac{p^2}{2m}$

a continuous increase in KE means the magnitude of its linear momentum is increasing continuously also.

We know the potential energy associated with a system consisting of Earth and a nearby particle is gravitational potential energy. If the particle moves from height y_i to y_f height the change in the gravitational potential energy of the particle–Earth system is:

$\Delta U = mg(y_f - y_i)$

If we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$

Now, for the network to be positive, its height above the ground level must continuously increase if gravity is the only force doing work.

10. Correct options: (B, C, D)

Solution: We know, for a particle, a change in the kinetic energy equals the net work done on the particle:

$K_f - K_i = W$

Given, the kinetic energy of the particle continuously increases with time.

Therefore, the net work done on the particle must be positive.

Now, we know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Therefore, when a force acts on a body, its kinetic energy increase if work is positive, its kinetic energy decreases if work is negative, its kinetic energy may become zero if work transfers all its kinetic energy.

11. Correct options: (A, B, C, D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$P_{avg} = \frac{W}{\Delta t}$

Hence, work done in unit time is called power and power is inversely proportional to time taken by the body.

Now, dimensional formula for power is:

$\frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$

Now, we know, work W done by a constant force F is given by:

$W = Fd\cos\theta$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Hence, power can be rewritten as:

$P = \frac{Fd\cos\theta}{\Delta t}$

$P = Fv\cos\theta$

$P = \vec{F}.\vec{v}$

12. Correct options: (A, B, D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

Now, we know, work W done by a constant force F is given by:

$$W = Fd\cos\theta$$

where d is the distance moved by the object and θ is the angle between the force and the displacement.

Hence, power can be rewritten as:

$$P = \frac{Fd\cos\theta}{\Delta t}$$

$$P = Fv\cos\theta$$

$$P = \vec{F}.\vec{v}$$

We know, one horsepower equal 746 Watts.

13. Correct options: (A)

Solution: We know, if we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$$\Delta U = mgh$$

Given, a box having a mass of 5.8 kg is lifted from the garage floor and placed on shelf. The box gains 145 J of P.E.

Therefore, the height of the shelf is:

$$\Delta U = mgh$$

$$h = \frac{\Delta U}{mg}$$

$$h = \frac{145}{5.8(9.8)}$$

$$h = 2.55\,m$$

Therefore, if a box having a mass of 5.8 kg when lifted from the garage floor and placed on shelf 145 J of P.E, then the height of the shelf is 2.55m.

14. Correct option: (B)

Solution: We know, if we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$$\Delta U = mgh$$

Given, a 30 kg child climbs 15 m up a tree.

Therefore, the potential energy of the child is:

$$\Delta U = mgh$$

$$\Delta U = 30(10)(15)$$

$$\Delta U = 4500\,J$$

Therefore, if a 30 kg child climbs 15 m up a tree, then the potential energy of the child is 4500J.

15. Correct option: (B)

Solution: We know, if we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$

Given, a concrete pole of mass 50 kg was lifted up at a height of 12 m

Therefore, the potential energy of the pole is:

$\Delta U = mgh$

$\Delta U = 50(10)(12)$

$\Delta U = 6000\,J$

Therefore, a concrete pole of mass 50 kg was lifted up at a height of 12 m, then the potential energy of the pole is 6000J.

16. Correct option: 600W

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

We know, if we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$

Given, a man M_1 of mass 90 kg runs up a stair case of length 10 m in 15 s.

Therefore, power of man M_1 is:

$$P_{avg} = \frac{mgh}{\Delta t}$$

$$P_{avg} = \frac{90(10)(10)}{15}$$

$$P_{avg} = 600W$$

17. Correct option: (D)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg} = \frac{W}{\Delta t}$$

We know, if we set the ground as the reference of zero potential energy, then gravitational potential energy of an object at a height h from the ground is:

$\Delta U = mgh$

Given, man M_2 of mass 90 kg runs up the same length of staircase in 20 s.

Therefore, power of man M_2 is:

$$P_{avg} = \frac{mgh}{\Delta t}$$

$$P_{avg} = \frac{90(10)(10)}{20}$$

$$P_{avg} = 450W$$

18. Correct option: $\frac{4}{3}$

Solution: We have above calculated the powers of man M_1 and $M_{2.}$

Hence, the ratio of the powers of man M_1 and $M_{2.}$ is:

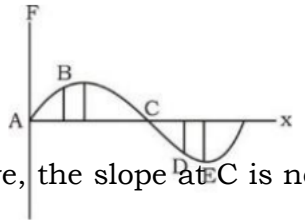
$$\frac{P_1}{P_2}=\frac{600}{450}$$

$$\frac{P_1}{P_2}=\frac{4}{3}$$

19. Correct match: a) ->P, b) ->P), c) ->Q), d) ->R)

Solution: We know, the force F is related to the negative slope of potential energy U as:

$$F=-\frac{dU}{dx}$$



Given, potential energy curve shown in fig on right.

We see, the slope at A is positive, the slope at B is also positive, the slope at C is negative and the slope at E is zero.

Therefore, the force at A is positive, the force at B is positive, the force at C is negative, and the force at E is zero.

Therefore, the correct match is: a) ->P, b) ->P), c) ->Q), d) ->R)

20. Correct match: a) ->S), b) ->R), c) ->Q), d) ->P)

Solution: We know, the power due to a force is the rate at which that force does work on an object. If the force does work W during a time interval Δt, the average power due to the force over that time interval is:

$$P_{avg}=\frac{W}{\Delta t}$$

Since both work done and time are scalar quantities, hence power too is a scalar quantity.

Now, dimensional formula for power is:

$$\frac{[ML^2T^{-2}]}{[T]}=[ML^2T^{-3}]$$

We know, one horsepower equal 746 Watts.

Therefore, 1 HP=746 W

Further, the cgs unit of power is erg/sec.

Therefore, the correct match is: a) ->S), b) ->R), c) ->Q), d) ->P)

5. Heat

Solutions

LEVEL – I

1. Correct option: (D) Energy.

According to the First Law of Thermodynamics, heat is a type of energy, and so thermodynamic processes are subject to the principle of energy conservation. This means that neither heat energy nor cold energy can be created or destroyed.

2. Correct option: (D) Cold, hot.

Heat is an energy form. It always flows from hotter to colder bodies. If there is a temperature difference between two systems that are in constant touch with one other, heat will always move from one system to the other until both systems are in equilibrium. Heat cannot flow on itself from a lower temperature body to a higher temperature body, according to the second law of thermodynamics.

3. Correct option: (C) 4.2 J.

We must determine how many joules one calorie equals.

A calorie is the amount of energy contained in the ingredients that comprise food. Because calories are a type of energy, they can be converted to joules using a heat measurement.

$^{o}F=\frac{5}{9}\times C^{o}+32^{o}$

$^{\circ}F=\frac{5}{9}\times -8^{o}+32^{o}$ 1 calorie = 4.2 joules

$^{o}F=17.6^{\circ}C$

4. Correct option: (C) Decreases.

The body's temperature rises only through the absorption of heat energy. The energy is returned to the environment in the case of reflection and transmission. Radiation causes a body to lose energy, causing its temperature to fall.

5. Correct option: (A) Temperature.

Temperature is measured with a thermometer.

Thermometer (Heat) (measuring gadget) It measures temperature in degrees Celsius or Fahrenheit.

6. Correct option: (A) Kelvin.

According to the International System of Units, the SI unit of temperature is Kelvin, which is denoted by the sign K. In the fields of science and engineering, the Kelvin scale is generally acknowledged or employed.

7. Correct option: (B) Scalar.

A scalar quantity is one that has simply magnitude, whereas a vector quantity has both magnitude and direction. Temperature is a scalar quantity since it is directionless at a location.

8. Correct option: (C) Temperature.

The degree of hotness or coolness of a body is measured by temperature.

Temperature rises as the kinetic energy of all molecules present in a thing rises.

A thermometer is an instrument used to measure the temperature of a body.

9. Correct option: (B) Temperature.

The direction of heat flow is determined by the temperature difference between the bodies. Heat is transferred from a hotter body to a colder one.

10. Correct option: (D) Both A and C.

When a substance is heated, its volume grows while its mass remains constant; thus, the density of the substance, defined as the ratio of mass to volume, decreases as temperature rises.

When heated, substance expands or contracts, causing their volume (either length or area) to alter and their density to vary, but their mass remains constant.

11. Correct option: (D) Between 30°C and 50°C.

The temperature will be just below. Because water expands when heated, one litre of water at $50^{\circ}\mathrm{C}$ has less mass than one litre of water at $30^{\circ}\mathrm{C}$. As a result, the $30^{\circ}\mathrm{C}$ litre contains somewhat more than the $50^{\circ}\mathrm{C}$ litre, and the average temperature will be slightly less than $40^{\circ}\mathrm{C}$.

12. Correct option: (C) Mercury.

Clinical thermometers contain mercury. Alcohol is utilised in laboratory thermometers.

13. Correct option: (A) Laboratory thermometer.

Because clinical thermometers cannot be used to monitor temperature outside of the human body, we require a different sort of thermometer for other applications. A laboratory thermometer, often known as a lab thermometer, is used to measure temperatures other than the temperature of the human body.

14. Correct option: (A) Thermometric substance.

A thermometric substance is a material or substance whose properties change with temperature and is utilised in the fabrication of thermometers. Mercury, alcohol, and other gases are common examples. Changes in characteristics, such as expansion or contraction, allow us to measure the temperature of a substance indirectly.

15. Correct option: (A) Substances expand on heating.

A thermometer works on the idea that a liquid expands when heated and shrinks when cooled. These liquid thermometers work on the thermal expansion principle.

16. Correct option: (C) Kelvin.

Each degree on the Fahrenheit temperature scale is smaller than on the Celsius, Kelvin, and Reaumur scales. As a result, the value of 1 degree Fahrenheit is the greatest of the supplied possibilities. Because the Fahrenheit scale has a smaller degree interval between each degree than the other scales, it has the biggest numerical value for a single degree.

17. Correct option: (C) 212.

We know that water's boiling point is $100^{\circ}\mathrm{C}$.

The conversion formula from Celsius to Fahrenheit is,

$$F = \frac{9}{5} \times C + 32$$

$$F = (1.8 \times 100) + 32 = 212\mathrm{F}$$

So, the boiling point of water on the Fahrenheit scale is 212F.

18. Correct option: (A) 273 K.

The freezing point or crystallisation point is the temperature at which a liquid change to a solid. The melting and freezing points of most substances are the same.

On the Kelvin scale, the melting point of ice is 273K.

19. Correct option: (D) -273°C.

In the Kelvin scale, the absolute zero temperature is $0\mathrm{K}$.

Let K be the kelvin temperature and C be the degree Celsius temperature.

We know that,

$K=273+C$

$0=273+C$

$C=-273^{\circ}C$

As a result, on the Celsius scale, absolute zero temperature is a $-273^{\circ}C$.

20. Correct option: (B) 180.

The lower fixed point on the Fahrenheit scale is 32 degrees, while the highest fixed point is 212 degrees. Because the difference between the two is 180, the number of divisions is also 180.

21. Correct option: (C) 318 K.

The conversion formula from Celsius to kelvin is,

$T=273+C$

$T=273+45$

$T=318K$

So, the $45^{\circ}C$ equal to 318K.

22. Correct option: (A) $37.77^{o}C$.

Given: $F=100^{\circ}$

The conversion formula from Fahrenheit to centigrade is,

$$^{\circ}C=\left[F-32\right]\times\frac{5}{9}$$

$$^{\circ}C=\left[100-32\right]\times\frac{5}{9}$$

$$^{\circ}C=68\times\frac{5}{9}$$

$$^{\circ}C=37.77^{\circ}$$

So, the $100^{\circ}F$ into centigrade is $37.77^{o}C$.

23. Correct option: (B) $12.5^{\circ}C$.

The conversion formula from Fahrenheit to Celsius is,

$^{\circ}C=1.25\times{}^{\circ}R$

$^{\circ}C=1.25\times{}^{\circ}10$

$^{\circ}C=12.5^{\circ}C$

So, $10^{\circ}R$ into Celsius is $12.5^{\circ}C$.

24. Correct option: (C) 546 K.

The conversion formula from Celsius to kelvin is,

$T=273+C$

$T=273+273$

$T=546K$

So, the $273^{\circ}C$ equal to 546K.

25. Correct option: (C) $-15^{o}C$.

The conversion formula from Fahrenheit to centigrade is,

$$^{\circ}C=\left[F-32\right]\times\frac{5}{9}$$

$$^{\circ}C=\left[5-32\right]\times\frac{5}{9}$$

$$^{\circ}C=15^{\circ}$$

So, the $5^{\circ}F$ into centigrade is $-15^{o}C$.

26. Correct option: (D)

27. Correct option: (A) $71.88^{o}C$.

Lower fixed point = -4

Upper fixed point = 92

Temperature on the thermometer = 65

Utilise the formula below to translate the temperature from the specified thermometer scale to Celsius (°C).

$$C=\frac{\text{Reading -Lower fixed Point}}{\text{Upper fixed point-Lower fixed point}}$$

$$C=\frac{65-(-4)}{92-(-4)}\times 100$$

$$C=71.88^{o}$$

So, the temperature is $71.88^{o}C$.

28. Correct option: (B) 3K.

The conversion formula from Celsius to kelvin is,

$$T=273+C$$

$$T=273-270$$

$$T=3K$$

So, the $270^{\circ}C$ equal to 3K.

29. Correct option: (C) $183.2^{\circ}F$.

The conversion formula from Celsius to Fahrenheit is,

$$^{o}F=\frac{5}{9}\times C^{o}+32^{o}$$

$$^{\circ}F=\frac{5}{9}\times 84^{o}+32^{o}$$

$$^{o}F=183.2^{\circ}C$$

So, the $84^{\circ}C$ into Fahrenheit scale is $183.2^{\circ}F$.

30. Correct option: (D) $43.33^{o}C$.

The conversion formula from Fahrenheit to centigrade is,

$$^{\circ}C=\left[F-32\right]\times\frac{5}{9}$$

$$^{\circ}C=\left[110-32\right]\times\frac{5}{9}$$

$$^{\circ}C=43.33^{\circ}C$$

So, the $110^{\circ}F$ into centigrade is $43.33^{o}C$.

31. Correct option: (C) $17.6^{\circ}F$

The conversion formula from Celsius to Fahrenheit is,

$$^{o}F=\frac{9}{5}\times C^{o}+32^{o}$$

$$^{\circ}F=\frac{9}{5}\times -8^{o}+32^{o}$$

$$^{o}F=17.6^{\circ}C$$

So, the $-8^{\circ}C$ into Fahrenheit scale is $17.6^{\circ}F$.

32. Correct option: (C) 312.4K.

The conversion formula from Fahrenheit to kelvin is,

$$K=(F+459.67)\times\frac{5}{9}$$

$$K=(103+459.67)\times\frac{5}{9}$$

$$K=312.4K$$

So, the $103^{\circ}F$ in kelvin is 312.4K.

33. Correct option: (C) 131°F.

The conversion formula from Celsius to Fahrenheit is,

$$^{o}F=\frac{5}{9}\times C^{o}+32^{o}$$

$$^{\circ}F=\frac{5}{9}\times 55^{o}+32^{o}$$

$$^{o}F=131^{\circ}F$$

So, the $55^{\circ}C$ into Fahrenheit scale is $131^{\circ}F$.

34. Correct option: (C) Radiation.

Radiation is the process of transferring heat without the use of any material medium.

As an example, radiation is a heat transmission mechanism in which heat is transferred or carried from one location to another in the form of rays or waves. It does not necessitate the use of any tangible medium. It is caused by electromagnetic radiation.

35. Correct option: (C) Thermal radiation.

Heat transmission in radiation happens via electromagnetic waves rather than particles. Radiation is a heat transfer process that does not require a medium. It refers to the transport of heat in waves, as it does not require the passage of molecules.

36. Correct option: (A) Solids.

Convection is not feasible in solids because for convection to occur, a substance's molecules must be free to move, as in liquids and gases. A solid's molecules are closely packed together, making it harder for molecules to move about and allow convection to occur.

37. Correct option: (B) Convection.

Convection is the primary means through which heat energy is transmitted in a liquid. Convection is the movement of heated particles within a fluid (liquid or gas), which results in heat transfer. This movement is aided in liquids by temperature differences, which cause warmer, less dense areas of the liquid to rise and cooler, denser regions to sink. This procedure generates a circulation pattern that distributes heat throughout the liquid. Convection is the predominant route of heat transport in liquids, while conduction and radiation also play a role.

38. Correct option: (A) Solids.

Heat is mostly transmitted through solids. Conduction is the process of direct heat transfer between particles within a substance, such as molecules vibrating against each other in a solid. Heat transmission in gases is mostly accomplished through convection, in which heated gas particles rise and colder particles sink, resulting in fluid motion that transmits heat. Another mechanism of heat transmission that can occur in both solids and gases is radiation, which involves the emission and absorption of electromagnetic waves. Conduction, on the other hand, is most prevalent in solids because to the near closeness of particles.

39. Correct option: (C) Good conductors of heat.

Substances that allow heat to easily move through them are referred to as excellent conductors, whereas those that do not are referred to as bad conductors. Aluminium, copper, and gold are a few examples of good heat conductors.

40. Correct option: (C) Radiation.

Radiation is the quickest technique of transferring heat from a hot body to a cold body.

We know that radiation travels at the speed of light and does not require any medium. It occurs in the form of electromagnetic waves.

Electromagnetic radiation, such as radio waves, microwaves, infrared, visible light, ultraviolet, x-rays, and gamma radiation, is included in this radiation. Particle radiation, such as alpha, beta, proton, and neutron radiation; and acoustic radiation, such as ultrasound, sound, and seismic waves.

We know that convection is the process through which heat is transported by the development of a warmed liquid such as air or water. Regular convection is caused by the tendency of most liquids to expand when heated—that is, to become less thick and rise due to the increased lightness.

41. Correct option: (B) Convection.

Convection is the primary technique through which the entire pot of water boils on a hot stove. The water in direct contact with the pot heats up when the bottom of the pot is heated on the burner. The water at the bottom becomes less thick and rises as it heats up. Cooler water from the pot's top travels down to replace it, creating a circular motion of fluid movement. Convection is the mechanism by which warm fluid rises and cooled fluid descends. This convection circulation is responsible for transferring heat throughout the pot of water, eventually leading to boiling.

42. Correct option: (A) Good.

Good heat absorbers are also good radiators.

This is based on Kirchhoff's heat radiation law.

The coefficient of absorption is equal to the coefficient of radiation at a given temperature, according to Kirchhoff's law of heat radiation.

43. Correct option: (A) Thermal convection.

Thermal convection occurs when heat is transferred from one location to another by the movement of molecules. Thermal convection is the transfer of heat via the movement of a fluid (liquid or gas), with warm sections rising and colder portions sinking. The movement of the fluid's molecules is what transports heat from one location to another. Thermal conduction (option B) on the other hand, involves the transfer of heat by direct contact between molecules without the substance moving in bulk. Thermal radiation (option C) uses electromagnetic waves to transmit heat by emitting and absorbing them. As a result, option A is the correct response.

44. Correct option: (C) Radiation.

Conduction and convection require a transport medium, such as a solid metallic rod or a fluid. There is no need for a medium in the radiation mode of heat transmission. Energy is transferred as electromagnetic waves, which may travel even in vacuum.

45. Correct option: (B) Clinical thermometer.

When the clinical thermometer is withdrawn from contact with the human body, a kink prevents mercury from falling below its recorded value. As a result, after taking the temperature, it is shaken to reset it to the default value.

46. Correct option: (A) 35°C -42°C.

The clinical thermometer has a temperature range of $35^{\circ}C$ to $42^{\circ}C$.

The clinical temperature range is between 35 and 42 degrees Celsius since our body temperature cannot normally fall below 35 degrees Celsius or rise over 42 degrees Celsius.

47. Correct option: (A) -10°C -110°C.

A laboratory thermometer is intended to measure a wide range of temperatures found in various scientific experiments. The temperature range of $-10^{\circ}C$ to $110^{\circ}C$ covers a wide range of temperatures often utilised in laboratory settings.

48. Correct option: (B) 37°C

A healthy person has a body temperature of $37^{\circ}C$, which is 98.6° Fahrenheit.

49. Correct option: (C) Radiation.

Heat can be transferred in three ways:

Conduction: the requirement for two bodies to be in touch.

Convection requires the presence of a material medium between the two bodies.

Radiation: occurs when there is a temperature differential, regardless of whether there is a material medium present or not.

When we sit in front of a room heater, we receive heat via the radiation process. There is no medium required for heat transfer by radiation.

50. Correct option: (A) Wood.

A material that does not carry electricity or heat well is known as an insulator.

Wood acts as an insulator. It does not let the free passage of electricity or heat through it. Iron (B), copper (C), and zinc (D) are all conductive and non-insulating materials.

51. Correct option: (A) Sea breeze.

With cooler dense air moving to occupy the area formed over the land, the air pressure over the ocean is higher. A sea breeze is the colder air that comes along with it. A sea breeze is the air that comes from the sea. It happens because of temperature variations during the day—land heats up faster than water, resulting in low pressure over land and high pressure over the sea. Air moves from high to low pressure, creating a cool breeze from the sea to the land.

52. Correct option: (C) Upwards.

Convection is constantly used to move heat vertically upwards.

This happens because medium particles near a heat source absorb heat from the source and begin to travel faster, causing the medium to expand.

As a result, the density of the medium around the source of heat decreases, making it lighter. The less dense media rises, while the dense medium from above descends to replace it.

As a result, a convection current is established in the medium.

53. Correct option: (D) Both. B and C

B) Denser liquid sinks: When a fluid is heated, it expands and loses density. As a result, the warmer, lighter fluid rises to the top, pushing the denser fluid to the bottom.

C) Lighter liquid rises to the surface: As the fluid at the bottom heats up and becomes less dense, it rises to the surface. To replace it, the colder, denser fluid at the top descends to the bottom.

As a result, both the denser fluid at the bottom and the lighter fluid at the top are involved in the convection current process. My previous explanation was inaccurate, and I appreciate your understanding.

54. Correct option: (B) White.

Because it absorbs more light, the darker the thing, the greater it emits heat. A white object, on the other hand, appears white because it reflects all wavelengths and absorbs little to no light. It absorbs little energy and emits little to no heat as a result. Heat energy can be

created by converting light energy. A black object absorbs all wavelengths of light and converts them to heat, resulting in the item becoming warm. Because a white object reflects all wavelengths of light, the light is not turned into heat and the object's temperature does not rise substantially.

55. Correct option: (C) Black.

Because they absorb more sunlight and transform it into heat, black-coloured things heat up faster than other colours. The dark colour absorbs a wider range of light wavelengths, both visible and non-visible (infrared), resulting in more absorbed energy and a faster temperature rise.

The colours we see in our daily lives are generated when objects absorb the majority of the light that falls on them and reflect a specific wavelength of light. That specific wavelength of light has a specific colour that we see as the colour of that object.

White and black are opposites in terms of light absorption; white absorbs no light and reflects everything that falls on it, whereas black absorbs everything that falls on it.

This means that the White absorbs the least energy while the Black absorbs the most.

56. Correct option: (B) Air.

Heat energy propagation is limited in the case of air due to conduction heat transfer due to its low thermal conductivity.

57. Correct option: (D) Land to sea.

After dusk, the breeze sweeps from the land to the sea. The land cools faster than the water, therefore cooling the air above it. The warmer air above the lake continues to ascend, while cooler air from above the land replaces it, resulting in a breeze.

A land breeze is a localised wind that blows from the land to the sea (typically at night). The temperature difference between the land and the sea throughout the night causes this occurrence. The land cools faster than the water, causing the air above it to become cooler and denser. As a result, a high-pressure region develops over the land. A land breeze is formed when air moves from a high-pressure area (land) to a low-pressure area (sea).

58. Correct option: (D) Keep us warm.

Woollen clothing is well-known for its insulating capabilities. They trap air near to the body, forming an insulating layer that helps us retain body heat and stay warm, especially in colder weather. Wool's natural fibres have the capacity to hold and keep heat, making it an excellent choice for cold-weather clothing.

59. Correct option: (B) Radiation.

Sunlight largely reaches the Earth's surface through radiation. Radiation is the transfer of thermal energy in the form of electromagnetic waves emitted by the Sun, such as visible light and infrared radiation. This energy travels through space and arrives on Earth, warming its surface and providing energy for numerous natural processes.

60. Correct option: (A) Melting point.

When a material is heated, the particles gain energy and begin to vibrate more rapidly. The particles shake so much at a certain temperature that their ordered structure collapses. The solid has melted into liquid at this moment. The melting point is the temperature at which this transition from solid to liquid occurs. At normal air pressure, each solid has a fixed melting point.

61. Correct option: (B) Freezing

Freezing is the process by which a liquid turns to a solid at a given temperature, known as the freezing point, with the release of heat energy. This is the inverse process of melting, in which a solid transform into a liquid.

62. Correct option: (B) Heat is absorbed.

Water is heated by the surrounding temperature during the evaporation process, and the molecules of water receive energy to enter the gaseous stage. And the temperature fluctuates in this energy shift process depending on the state of materials from which energy is transferred.

Water evaporation is an endothermic process, according to option **(D)**. Endothermic means absorbing heat, so this is the correct answer since heat is absorbed by the molecules throughout the evaporation process, allowing them to enter the gas stage.

63. Correct option: (B) Heat is absorbed

Heat is absorbed from the surroundings during melting.

A solid (such as ice) transforms into a liquid during the melting process. The molecules in the solid must absorb energy in order to transition from a stiff, ordered structure to a more fluid arrangement in the liquid state. This energy is provided by the environment in the form of heat, which causes the ice to melt and convert into water. As a result of absorbing heat, melting ice has a cooling impact on its surroundings.

64. Correct option: (B) Heat is released.

As heat is dissipated during the freezing process, the temperature of water drops. During freezing, the water molecule releases energy. The surrounding molecule absorbs the emitted energy.

65. Correct option: (B) Boiling point.

The temperature at which a liquid is transformed into vapour (gas) is known as the boiling point.

At boiling point, the vapour pressure of a liquid equals the pressure surrounding it, and the liquid transforms into vapour.

LEVEL – II

1. Correct option: (B) 160°C and 320°F.

If C represents the temperature on the Celsius scale and F represents the temperature on the Fahrenheit scale, then

In response to the question;

$F=2C$

The temperature on the Fahrenheit scale is now proportional to the temperature on the Celsius scale by;

$$\frac{C}{5}=\frac{F-32}{9}$$

Using the above relationship, we get,

$$\frac{C}{5}=\frac{2F-32}{9}$$

$9C=10C-160$

$10C-9C=160$

$C=160^{\circ}C$

As a result, the Fahrenheit temperature will be;

$F=2\times160$

$F=320^{\circ}F$

The needed temperature is $160^{\circ}C$ on a Celsius scale and $320^{\circ}F$ on a Fahrenheit scale.

2. Correct option: (C) $37.78C^o$.

For two scales of temperature,

$$\frac{C-0}{100-0}=\frac{\text{Reading}}{\text{UFP-LFP}}\text{-LFP}$$

where LFP and UFP are the scale's lower and upper fixed points

$$\frac{C}{100}=\frac{39\text{-}5}{95\text{-}5}=\frac{34}{90}=0.3778$$

$$C=0.6\times100=37.78^o$$

So, the temperature is $37.78C^o$.

3. Correct option: (A) 59.43^oC.

Lower fixed point = -3

Upper fixed point = 103

Temperature on the thermometer = 60

Utilise the formula below to translate the temperature from the specified thermometer scale to Celsius (°C).

$$C=\frac{\text{Reading -Lower fixed Point}}{\text{Upper fixed point-Lower fixed point}}$$

$$C=\frac{60-(-3)}{103-(-3)}\times100$$

$$C=59.43^o$$

So, the temperature is 59.43^oC.

4. Correct option: (B) 1.8^oF.

The conversion formula from Celsius to Fahrenheit is,

$$^oF=\frac{5}{9}\times C^o+32^o$$

$$^\circ F=\frac{5}{9}\times1^o+32^o$$

$$^oF=1.8^\circ F$$

So, the temperature of second system is 1.8^oF.

5. Correct option: (C)

6. Correct options: (A, D)

A healthy person has a body temperature of 37^oC, which is equal to 98.4^oF on average.

7. Correct options: (A, B, C, D).

(A) The SI unit of temperature is Kelvin (K).

(B) A scalar quantity is one that has simply magnitude, whereas a vector quantity has both magnitude and direction. Temperature is a scalar quantity since it is directionless at a location.

(C) The SI unit of heat is joule (J).

(D) 1 calorie (cal.) is approximately equal to 4.2 joules (J).

8. Correct options: (A, B, D).

Both a change in state and an increase in temperature can occur when a substance is heated. The nature of the substance, the quantity of heat delivered, and the original circumstances are just a few of the variables that affect the precise outcome.

Compressing a substance that expands when heated results in an increase in density and a decrease in volume for that specific substance.

The density of a substance reduces when it is compressed. To put it another way, when we compress something, the volume absorbs the heat and gradually decreases but the mass remains the same. Boyle's Law tells us that density and volume have an inverse relationship, meaning that as volume increases, density decreases and vice versa.
A substance's volume reduces and density rises as a result of compression.

9. Correct options (A, B)

A) A body's temperature drops when heat energy is lost. Temperature drops as a result of heat loss because the average kinetic energy of the body's particles decreases.
B) A body has a low temperature if it is cold. The average kinetic energy of the particles in a substance is measured by its temperature. When a body feels chilly, it means that its particles have less kinetic energy and are therefore warmer than the surrounding air.
C) A body's mass remains constant when heated. Heat does not modify a body's mass; instead, it may change a body's temperature, state of matter, or other attributes.
D) Calorimeters are not commonly used to monitor temperature. Calorimeters are instruments for measuring the energy or heat changes that occur during a chemical or physical process, frequently including reactions. Thermometers are typically used to measure temperature.

10. Correct options: (B, C, D)

B) Mercury does not adhere to a thermometer's glass walls, allowing for precise and reliable readings.
C) Mercury is appropriate for use in thermometers because of its high density, which causes it to expand and contract visibly in response to temperature variations.
D) Mercury has a comparatively high boiling point (356.73°C or 674.11°F) and a very low freezing point (about -38.83°C or -37.89°F). Mercury thermometers can measure a wide variety of temperatures thanks to this large temperature range.

11. correct options: (B, C, D)

B) Because conduction involves the direct contact of particles within a substance, it requires a medium.
C) Since radiation doesn't require a medium to transfer heat, it is possible to conduct radiation in a vacuum. It might be brought on by electromagnetic waves.
D) Convectional heat transfer also needs a medium. Convection is the movement of fluid particles (liquid or gas), which carries heat energy from one location to another to transfer heat.

12. Correct option: (D) $100^{o}C$.

The boiling point of water is $100^{o}C$.

13. Correct option: (A) $32^{o}F$.

The conversion formula from Celsius to Fahrenheit is,

$$^{o}F=\frac{9}{5}\times C^{o}+32^{o}$$

$$^{\circ}F=\frac{9}{5}\times 0^{o}+32^{o}$$

$$^{o}F=32^{\circ}F$$

So, the temperature of second system is $32^{o}F$.

14. Correct option: (B) 100°C.

The conversion formula from Fahrenheit to centigrade is,

$$^{\circ}C=\left[F-32\right]\times\frac{5}{9}$$

$$^{\circ}C=\left[212-32\right]\times\frac{5}{9}$$

$$^{\circ}C=100^{\circ}C$$

So, the $212^{\circ}F$ into centigrade is $100^{o}C$.

15. Correct option: (A) 75°C.

For two scales of temperature,

$$\frac{C-0}{100-0}=\frac{\text{Reading}}{\text{UFP-LFP}}\text{-LFP}$$

where LFP and UFP are the scale's lower and upper fixed points

$$\frac{C}{100}=\frac{73\text{-}4}{96\text{-}4}=\frac{34}{90}=0.75$$

$$C=0.75\times100=75^{o}$$

So, the correct temperature in Celsius scale $75^{o}C$.

16. Correct option: (B) 348K.

The conversion formula from Celsius to kelvin is,

$$T=273+C$$

$$T=273+75$$

$$T=348K$$

So, the $75^{\circ}C$ equal to 348K.

17. Correct option: (A) 167°F.

The conversion formula from Celsius to Fahrenheit is,

$$^{o}F=\frac{9}{5}\times C^{o}+32^{o}$$

$$^{\circ}F=\frac{9}{5}\times75^{o}+32^{o}$$

$$^{o}F=167^{\circ}F$$

So, the temperature of second system is $167^{o}F$.

18. Correct option: (A) Polishing the surface.

Because polished surfaces are poor radiators and heat absorbers, they can limit the amount of heat that is lost from a surface by radiation.

19. Correct option: (D) Conduction and convection.

A near vacuum is created as a result of the partial evacuation of the air space between the two flasks, considerably reducing heat transfer by conduction and convection.

20. Correct option: (C) Thermos flask.

A thermos flask is a device that minimises heat loss from conduction, convection, and radiation.

21. Correct option: (A) I and II are true.

I. A temperature increase of 1 Kelvin (K) corresponds to an increase in temperature of 1 degree Celsius (°C). The degree units on the Celsius and Kelvin scales have the same size, and the Kelvin scale begins at absolute zero, which is -273.15°C.

II: A temperature increase of 1 Kelvin is somewhat more than a temperature increase of 1 Celsius. This is thus because the Celsius scale begins at -273.15°C, but the Kelvin scale begins at absolute zero (0 K). The discrepancy between the two scales' beginning points is what causes the disparity.

22. Correct option: (A) I and II are true.

I. There are 180 degrees between the freezing point and boiling point of water on the Fahrenheit scale. This range uses a total of 180 degrees.

II. There is a 1.8 conversion factor between degrees Celsius (°C) and degrees Fahrenheit (°F). Consequently, 1°C is equivalent to 1.8°F.

23. Correct option: (C) I is true and II is false.

I. A liquid does turn into a gas during boiling, and this transformation usually takes place at a fixed temperature known as the boiling point. However, during this phase change, heat energy is absorbed rather than liberated.

II. A liquid turns into a solid during freezing, and this transformation typically takes place at a fixed temperature known as the freezing point. In reality, heat energy is released during freezing, which results in the liquid losing energy and changing into a solid state.

24. correct matching is (a-Q, b-P, c-S, d-R).

Column-I	**Column-II**
a) Celsius scale (in ^{o}C)	Q) Lower fixed point is 0.
b) Fahrenheit scale	P) Upper fixed point is 212.
c) Kelvin scale	S) Lower fixed point is 273.
d) Reaumur scale	R) Upper fixed point is 80.

(a) There is an upper fixed point and a lower fixed point on the Celsius scale. The lower fixed point is 0 degrees Celsius, and the upper fixed point is 100 degrees Celsius.

(b) In the Fahrenheit scale, the lower fixed point is 32 degrees, while the upper fixed point is 212 degrees. Since there is a 180 difference between the two, 180 divisions are required.

(c) The temperature of melting ice at normal pressure is +273K, and the temperature of boiling water at normal pressure is 373K, making these two points the lower and higher fixed points, respectively.

(d) There are 80 equal segments that make up the distance between the lower fixed point and the upper fixed point. This scale's divisions are each referred to as one degree Reaumur.

So, the correct matching is a)-(Q), (b)-(P), (c)-(S), (d)-(R).

25. Correct matching is (a-P, b-T, c-Q, d-P).

Column-I	**Column-II**
a) SI unit of temperature	S) Kelvin
b) Unit of heat	T) Joule
c) 1 calorie equals	Q) 4.2J
d) Absolute temperature K=	P) C + 273

(a) The SI unit of temperature is the Kelvin (K) scale. The equation $K = °C + 273$ relates it to the Celsius (°C) scale. The conversion between the two temperature scales is shown in this equation.

(b) The SI unit of energy, which includes heat energy, is the joule (J). Energy comes in the form of heat, and the joule, which serves as the measurement of energy, is the same as the unit of heat.

(c) Energy, specifically heat energy, is measured in calories. 4.2 joules are about equivalent to one calorie. This conversion factor is frequently used to translate between calories and the joule, the SI unit of energy.

(d) The Celsius temperature is multiplied by 273 to get the Kelvin temperature in order to relate the Kelvin (K) and Celsius (°C) scales. This conversion takes into account the fact that the Celsius scale begins at -273.15°C while the Kelvin scale begins at absolute zero (0 K).

So, the correct matching is a)-(P), (b)-(T), (c)-(Q), (d)-(P).

26. Correct matching is (a-Q, b-P, c-S, d-R).

Column-I	Column-II
a) Solid to liquid	Q) Melting
b) Liquid to gas	P) Evaporation
c) Solid to gas	S) Sublimation
d) Gas to liquid	R) Condensation

(a) Solid to liquid:

Melting: When a solid substance absorbs heat energy, melting occurs, turning it into a liquid. The melting point is the temperature at which melting starts to take place. For instance, ice (solid water) melts into liquid water as it absorbs heat.

(b) Liquid to gas:

Evaporation: A liquid substance turns into a gaseous state through the process of evaporation when heat energy is absorbed. It takes place at the liquid's surface and is independent of the liquid's boiling point. For instance, water will evaporate from a wet cloth or puddle's surface.

(c) Solid to gas:

Sublimation: Without going through the liquid phase, sublimation is the process by which a solid substance transforms straight into a gaseous state. This happens when a substance absorbs enough thermal energy to cause the solid structure's intermolecular connections to snap. A typical illustration is the sublimation of dry ice (solid carbon dioxide).

(d) Gas to liquid:

Condensation: A gas or vapour turns into a liquid state during condensation as a result of the release of heat energy. Evaporation is the opposite of it. Clouds or dew are created when water vapour cools and condenses into minute water droplets.

So, the correct matching is (a)-(Q), (b)-(P), (c)-(S), (d)-(R).

6. Electricity

Solutions

LEVEL-I

Simple Electric Circuits

1. Correct option: (C) Electric cell.

A cell or battery stores its energy in the form of chemical energy that is active in its constituent parts and can be instantly transformed into electric energy. The flow of electrical energy in circuits then takes the shape of a current.

2. Correct option: (D) Coil.

Inside the bulb, there is a spirally coiled wire that is supported by two stout wires at either end. A short, thin wire is supported by two larger wires within a lightbulb. This wire is referred to as a filament. The component of the light bulb that emits light is the filament. Tungsten makes up the filaments in incandescent light bulbs.

3. Correct option: (D) Tungsten.

A light bulb's filament is constructed of tungsten. White heat and long-lasting white light are two properties of tungsten.

4. Correct option: (A).

A circle with a semicircle inside it serves as the representation of an electric bulb, and the semicircle stands for the bulb's filament.

5. Correct option: (B) 2.

The terminals of an electric cell are two. endpoint for both positive and negative.

Electric Circuit

6. Correct option: (C) Switch (or) key.

A switch is a tool that you can use to open or close a circuit at a whim in order to control the flow of electric current through it. The switch shuts the circuit when it is activated, allowing electricity to flow. The circuit is opened when the switch is turned off, stopping the flow of power.

7. Correct option: (C) Metal cap.

A device called an electric cell transforms chemical energy into electrical energy. There are two terminals on every electric cell. Metal discs on electric cells are thought of as the negative terminals, whereas the metal caps on them are thought of as the positive terminals.

8. Correct option: (B) Closed.

An incomplete circuit prevents the flow of electricity. An open loop is what is used to describe a circuit that isn't complete. A closed loop is what we refer to after the circuit is finished. The light bulb lights in a closed loop because electrical current flows in a closed circuit.

9. Correct option: (C) Positive to negative.

The positive and negative terminals of the electric cell serve as the direction of current flow in an electric circuit.

Electric current flows uniformly, hence its direction is different from the flow of electrons. As a result, the positive terminal of the battery leads to the negative terminal in the direction of electric current in an electric circuit.

10. Correct option: (D) Electric circuit.

An enclosed path through which current can travel is called an electric circuit. A power source, connecting wires, and a load that absorbs the energy make up an electric circuit. If you connect the power cables to the lightbulb and point one end of the wire towards the power source, nothing will happen; however, if you connect the free cables to the battery and complete the circuit, you'll probably notice that the lightbulb starts to glow.

As a result, the circuit must always be run in an effort to ensure that the current flows.

11. Correct option: (C) Electric current.

A flow of electric charge is known as an electric current. This charge is frequently carried by moving electrons in a wire in electric circuits. It can also be transported by both ions and electrons, such as in an ionized gas (plasma), or by ions in an electrolyte.

12. Correct option: (B) Earth wire.

In the event of a short circuit, the earth wire bypasses excessive current to ground and safeguards us from electric shock.

Electric Resistance

13. Correct option: (D) Resistance.

Electric circuits' fundamental attribute of resistance prevents or inhibits the flow of electric current. In circuit equations, it is represented by the letter " R " and is measured in ohms (Ω $I \times R$). A conductor's ability to hinder the flow of electricity is referred to as resistance. Resistivity is the impediment that the conductor's unit area presents to the flow of current. In essence, resistivity is a property of the substance (copper), whereas resistance is a quality of the object (let's say a copper wire).

14. Correct option: (C) Nichrome.

Nichrome, being an alloy, will have a higher resistivity than silver, copper, and aluminium since alloys typically have higher resistivity than individual metals.

15. Correct option: (C).

The SI unit for electrical resistance is the ohm.

16. Correct option: (B) Maximum.

The sum of the individual resistances in a series circuit represents the total resistance. The total resistance is greater than the resistance of any individual bulb because the current must go sequentially through each one. As a result, a series circuit has the highest resistance.

17. Correct option: (D) Resistance.

Resistance is the obstruction a conductor material presents to the flow of an electric current. According to Ohm's law, resistance is given by as,

$$R = \frac{V}{I}$$

The SI unit for electrical resistance is the ohm.

Conductors & Inductors:

18. Correct option: (B) Conductors.

Conductors are substances with a large number of free electrons and very little resistance. Insulators are substances that, under normal circumstances, do not permit the passage of a current. A typical insulator has a high resistance.

19. Correct option: (D) Silver & Rubber.

Rubber and paper are not good electrical conductors. They are insulators as a result. Metals are generally effective conductors of electricity, as we all know. Silver and gold are hence electrical conductors.

20. Correct option: (B) Insulators.

Insulators or poor conductors are materials that do not allow an electric current to flow through them easily. For example, wood, paper, plastic, etc.

21. Correct option: (D) Rubber.

Materials that are insulators do not have any free electrons, hence they don't conduct electricity. Insulators are listed below: Rubber, Air, Paper, Ceramics, Mica, Bakelite, Polythene, Cotton, Sulphur, and Cotton

22. Correct option: (D) Plastic

Copper and aluminum are used to make electrical lines because they are conductors, while plastic (or rubber) is used to cover them since it is an insulator.

23. Correct option: (A) Having excess number of electrons.

Electric charges come in two flavors: positive and negative. An object is positively charged if it loses excess electrons or is uncharged, and is negatively charged if it obtains excess electrons. Therefore, a body can become negatively charged by receiving an excessive number of electrons.

24. Correct option: (C) Neutral body.

The earth is thought to have zero potential and limitless charges. The potential of the body is zero when it is connected to the earth. Since it must be electrically neutral, electrons travel to the body from the earth to balance the previously positively charged body. Electrical neutrality is commonly seen as being present on Earth. It is electrically neutral in the majority of practical circumstances because it possesses an equal number of positive and negative charges that balance one another out.

25. Correct option: (B) Negative to positive.

The convention of current flowing from the positive end to the negative terminal is all there is. It was determined that positive charges are transported by current flow. Because electrons have a negative charge, they are drawn to positive battery ends and repelled by negative ones. Electrons, therefore, flow from negative to positive when the battery is connected to a conductor that allows electrons to pass through it.

26. Correct option: (B) Statement (i) is false (ii) is true.

Statement (i) is untrue. While opposite charges actually attract, similar charges actually repel one another. Two positively charged things will, for instance, resist one another.

Statement (ii) is accurate. If the electrons are free to move around inside the substance, a substance with more electrons may be a good conductor. More free electrons in conductive materials like metals enable improved electrical conductivity.

27. Correct option: (D) Metals are conductors.

Metals are excellent electrical conductors because they have a significant amount of free valence electrons, whereas glass has none. Therefore, metals are good conductors of electricity while glass is a poor conductor.

28. Correct option: (C) Copper.

Because they are insulators—materials that do not conduct electricity well—glass, plastic, and wool are used to cover electrical wires to stop the flow of current. The highly conductive metal copper, on the other hand, is utilised in electrical wiring and other applications where good conductivity is necessary.

29. Correct option: (C) Metal.

Metals are typically effective electrical conductors. As a metal, silver contains this quality. In fact, among metals, it is one of the finest at conducting electricity. Insulators include plastic, rubber, and wood.

30. Correct option: (A) True.

It is impossible for pure water to have any ions because it has been totally deionized. For the energy to go through and across the liquid and through the water, it is now necessary. Pure water does not conduct electricity since it only includes water that is incapable of dissociating into ions due to its poor ion dissociation.

Additionally, the calcium and magnesium salts that are dissolved in pure water do not act as good electrical conductors. As a result, electricity does not conduct well through clean water. Pure water is therefore an insulator.

31. Correct option: (B) Both statements (i) and (ii) are true.

Statement (i) is true. Conductors are materials that permit the flow of electricity. They effectively conduct electric current and have minimal electrical resistance.

Statement (ii) is true. Materials that block the flow of electricity are known as insulators. They do not adequately conduct electricity and have considerable electrical resistance.

Thus, the characteristics of conductors and insulators are accurately described by both assertions.

Series & Parallel Connection of Bulbs

32. Correct option: (C) Parallel circuit.

In a series circuit, every element is joined end to end to create a single path for current to flow. Each component is linked across the other in a parallel circuit, creating exactly two sets of electrically connected points.

33. Correct option: (C) Electric current.

In a series circuit, the voltage across the circuit is the total of the individual voltage drops across each component, and the current flowing through each component is the same.

34. Correct option: (A) Do not glow.

When a bulb in a series connection is removed, an open circuit is made. The other bulb won't light up since the circuit is in series and is open at one point because current cannot pass through it.

If one of the lights in a series connection does not glow or fails, the other bulbs in the series connection will likewise not glow. This is so because each component in a series circuit receives the same amount of current. In the event that one bulb fails to conduct, the circuit as a whole experience an open circuit, which leaves all bulbs unlit.

35. Correct option: (B) Divided.

Electric current in a series connection is constant, and each component receives a portion of the voltage (potential difference). The same amount of electric current travels through each connected component in a series circuit. These components each experience a percentage of the entire voltage, but it is distributed among them.

36. Correct option: (C) Parallel circuit.

All of the parts of a series circuit are attached end to end, creating a single path for current flow. All of the components in a parallel circuit are linked across one another to create precisely two sets of electrically coupled points.

37. Correct option: (A) Voltage.

The voltage (potential difference) across each component in an electrical circuit connected in parallel is constant.

38. Correct option: (B) Minimum.

The total resistance of an electrical circuit in which components are connected in parallel is lower than the resistance of any one of the individual components.

39. Correct option: (C) Series.

Bulb connections for decoration are made end-to-end, or in series. This makes sure that the voltage is divided across the bulbs as needed for low-voltage operation and that the same amount of current flows through them. As a result, because the bulbs are connected in series, all of them stop lighting when one goes out.

40. Correct option: (A) Parallel.

In a home, electrical devices are linked in parallel.

When electrical appliances are connected in parallel, each one operates independently, regardless of whether the other item is on or off.

41. Correct option: (B) Match stick.

When a matchstick is used in place of a switch, the light does not illuminate.

Matchsticks cannot complete the circuit like a switch would because they do not offer a conductive path for electricity to flow. In other words, they block the passage of the electric current, which keeps the lightbulb from turning on.

42. Correct option: (D) Battery.

A battery is a collection of two or more cells. Three cells are united, electrically connected in series, and have positive and negative main terminals in the diagram. Torches, toys, alarms, and other devices all need batteries as their power source.

43. Correct option: (A) Electric pole.

Yes, a network of power lines that includes electric poles (Option A) as part of the distribution infrastructure is used to deliver electricity to residences. It's crucial to make clear that electric power plants, not neighbouring homes, are the main source of electricity. Electricity is produced in electric power plants and then delivered to residences and other users.

Electric power plants would therefore be the appropriate response if the question expressly asked about the source of electric energy; yet, if it was inquiring about the distribution network, then Option A (Electric pole) would be appropriate in the context of local distribution.

44. Correct option: (B) The thin filament.

Inside a bulb, there is a tiny filament that glows and emits light.

LEVEL-II

1. Correct option: (D) All bulbs will go together.

The answer that makes all of the bulbs glow simultaneously is (d). Once the circuit is complete, current will be present at every point. Because the bulbs are connected in series, they will all illuminate at the same time. When the switch is turned ON, an immediate setup of the electric current causes all the lamps to glow together.

2. Correct option: (D) Steel spoon.

If Ramu is out of connecting wires, he can utilise a steel spoon as a conductor to finish an electrical connection. Since steel is a good conductor of electricity, it can be used to close a circuit's gap and permit the flow of energy. The other alternatives—a glass bangle, a thick piece of thread, and a rubber pipe—are insulators and would prevent electricity from moving.

3. Correct option: (B) Bad conductors.

Bad conductors are substances that have the ability to conduct electricity but provide a significant resistance (opposition) to the flow of current. Electric insulators are substances that have a high resistance to electric current. Examples include the majority of non-metallic solids, including plastic, rubber, and wood. Since their atoms are closely bound to their electrons, electric current cannot freely pass through them. A further electrical insulator is dry air.

4. Correct option: (D) Circuit.

Electric Circuit refers to a continuous, closed route of an electric current. A load or resistance, wires, and a switch make up an electric circuit.

5. Correct options: (A) and (C).

You can use copper and iron nails in place of a switch to turn on a lightbulb. The electrical circuit can be finished with the copper and iron nail, allowing current to flow and lighting the bulb. Copper is a good electrical conductor, and an iron nail can be utilised as a point of contact. Rubber band and a plastic cap are insulators, therefore they prevent the passage of electricity from completing the circuit.

6. Correct options: (B) and (C).
When heated, it maintains its strength and has a high melting point. Tungsten is the primary component in light bulb filaments. Due to the high resistance of the filament in a light bulb, when current flows through it, the filament warms up, which causes the electrons to change energy levels and produce photons in the visible spectrum. greater losses would result in greater light. Because of this, the filament of a bulb has a high resistance.

7. Correct options: (B) and (C).
When heated, it maintains its strength and has a high melting point. Tungsten is the primary component in light bulb filaments.
Due to the fact that a bulb is lit by heating the filament, a filament is composed of a material with a high melting point, such as tungsten. The heated filament serves as the light source, and if the material that makes up the filament has a low melting point, it could melt.
Because a low resistance will enable all the current to pass through it quickly and prevent the filament from glowing a high resistance filament is necessary.

8. Correct options: (B) and (C).
A switch is used to create or interrupt a circuit. It can be utilized to open or interrupt a circuit (breaking it OFF) in order to halt the flow of current or to close or complete a circuit (turning it ON) in order to permit the flow of current. Therefore, B) To make and C) To break are the proper choices.

9. Correct options: (B) and (C).
Current division happens in a parallel electrical circuit when the resistance of each branch divides the total current entering a junction. A part of the total stream can pass through each branch. In contrast, voltage constancy refers to the fact that each component's voltage remains constant while being connected directly to the same power source points. Parallel circuits are perfect for applications where components must work at the same voltage while allowing various levels of current to flow through them, providing flexibility and adaptability in electrical designs. This topology guarantees that each component receives the full voltage.

10. Correct option: (B) 2.
A device called an electric cell transforms chemical energy into electrical energy. There are two terminals on every electric cell. Metal discs on electric cells are thought of as the negative terminals, whereas the metal caps on them are thought of as the positive terminals.

11. Correct option: (D) Chemical to electric.
Chemical energy is transformed into electrical energy in an electric cell. The flow of electrons produced by chemical reactions inside the cell is what we know as electric current.

12. Correct option: (D) Metal disc.
There are two terminals on every electric cell. Metal discs on electric cells are thought of as the negative terminals, whereas the metal caps on them are thought of as the positive terminals.

13. Correct option: (C) Electric current.
Electric current is the total charge that moves across a conducting wire over a specific period of time.

14. Correct option: (A) Resistance.
The amount of opposition to the flow of current, which is the flow of electrons, is referred to as resistance.

15. Correct option: (D) Resistor.
A resistor is the name of the instrument that measures electrical resistance.
An electronic component known as a resistor works to counter the passage of electric current in a circuit. It is frequently used to restrict or regulate the amount of current flowing

through a circuit and is measured in ohms. In applications including current limiting, voltage division, signal conditioning, and temperature sensing, resistors, which are frequently found in a variety of kinds including fixed and variable resistors, are essential components. For managing and regulating electrical currents inside of electronic devices and circuits, they are crucial. The power-handling capacity of resistors is specified in watts, and their values are denoted by color-coded bands.

16. Correct option: (B) Current.

The amount of current in a series electrical circuit is constant across all connected elements. The voltage drops across each component, yet the current across the circuit doesn't change.

17. Correct option: (C) Series.

A circuit is considered to be in a series configuration if each component is connected one after the other.

18. Correct option: (A) Voltage.

The voltage (option A) in a parallel electrical connection is constant across all components. This means that regardless of differences in current or resistance within specific branches of the parallel circuit, each component connected in parallel receives the same voltage as the source.

19. Correct option: (C) Conductors.

Conductors (or metals) are materials with a high number of free electrons and low current resistance.

20. Correct option: (D) Silver.

The finest electrical conductor available is silver. It has great electrical conductivity, which makes it simple for electric current to move through. Silver is the material of choice for electrical wiring, connectors, and other uses where effective electricity conduction is crucial.

21. Correct option: (B) Positively charged.

We can state that charged objects don't have an equal distribution of protons and electrons. Charged objects have an imbalanced charge, either more positive protons or electrons than electrons. Additionally, neutral objects have an equal quantity of protons and electrons, balancing their charge.

Therefore, an object acquires a positive charge when electrons are withdrawn.

22 Correct Matching is: (a)-(S),(b)-(T),(c)-(R),(d)-(Q).

(a) A gap or break in a circuit line that symbolizes an interruption in the electrical route is the open circuit symbol. It represents an imperfect circuit that prevents current flow. In electronic schematics and diagrams, this sign is used to indicate a disconnected or non-functioning component.

(b) The symbol for (C) is the same as for a resistor, but with an arrow through it.

(c) Electric lightbulbs are represented by the bulb sign in electrical diagrams. A cross or a zigzag line is enclosed inside a circle. It serves as a visual indicator or a lighting element when linked in a circuit because it indicates the existence of a light source that illuminates when current flows through it.

(d) Electrical diagrams show a closed circuit as a continuous line with no gaps or breaks. It symbolises a whole, linked electrical circuit, demonstrating that current can freely flow through all of the circuit's parts. With this symbol, the circuit is active and current can flow uninterruptedly along its whole length.

23 Correct Matching is: (a)-(Q),(b)-(R),(c)-(P),(d)-(S),(T).

(a) The voltage remains constant when bulbs are linked in parallel. Each bulb in a parallel circuit receives the same voltage because they are all directly connected across the voltage

source (such as a battery). However, each bulb's resistance will affect how much current flows through it.
(b) The positive terminal of the electric cell is the metal cap.
(c) The negative terminal of the electric cell is the metal disc.
(d) The current remains constant when bulbs are connected in series. Because there is only one path for the current to go in a series circuit, the same amount of current travels through every component. In a series circuit, the voltage across each bulb, however, is not constant; it increases as the current flows through each bulb.
The whole voltage delivered by the power supply is divided among the bulbs when they are linked in series. The voltage across each bulb is smaller than the total voltage because each bulb only receives a part of the total voltage. This is the reason why "voltage is divided" in a series circuit is a common phrase.

24 Correct Matching is: (a)-(T),(b)-(S),(c)-(R),(d)-(Q).

(a) Rubber is an example of an insulator since it prevents the easy flow of electric charges.
(b) Silver has high electrical conductivity and is the finest conductor of electricity.
(c) Connecting electrical systems to the Earth is the procedure of earthing, which is done to defend against electrical faults and prevent electric shocks.
(d) Electric charges may easily flow through conductors, and copper is one of the best conductors of all metals.

7. Magnetism

Solutions

LEVEL – I

1. Correct option: (D) All of these.

Artificial magnets are manufactured by humans using metals like iron, cobalt, or nickel alone or in combination with other metals. Artificial magnets can be made in a variety of shapes, including bar, horseshoe, cylindrical, and ball-ended. The strongest magnet in nature is magnetite, an iron mineral.

2. Correct option: (D) Generator.

Magnets can produce power in generators. Generators typically consist of a rotating coil of wire surrounded by a magnetic field that creates an electrical current in the wire. Many electrical generators operate in this manner.

3. Correct option: (A) Aluminium, nickel and cobalt.

Alnico is in fact an alloy that predominantly consists of the metals aluminium (Al), nickel (Ni), and cobalt (Co), as well as minor amounts of other elements like iron (Fe) and copper (Cu). An Alnico magnet, a powerful and stable permanent magnet, is produced by this combination of components.

4. Correct option: (A) Y must be a magnet.

When a magnet 'X' repels another metal 'Y,' this proves that 'Y' is likewise a magnet.

5. Correct option: (B) Magnetite.

One of the primary iron ores is the mineral magnetite. Using the chemical equation Fe_3O_4. One of the iron oxides is Fe_3O_4. Because it is ferromagnetic, magnetite attracts magnets. It can be made into a permanent magnet by applying magnetization. Of all the naturally occurring minerals on Earth, it has the strongest magnetic properties.

6. Correct option: (B) Fe_3O_4

The chemical formula for the substance found in natural magnets is Fe_3O_4.

7. Correct option: (C) Neodymium.

The strongest magnet is neodymium. Iron and boron were used to create the alloy.

8. Correct option: (B) Artificial magnets.

Artificial magnets are manufactured by humans using metals like iron, cobalt, or nickel alone or in combination with other metals. Artificial magnets can be made in a variety of shapes, including bar, horseshoe, cylindrical, and ball-ended. The strongest magnet in nature is magnetite, an iron mineral.

9. Correct option: (A) Magnetic materials.

Magnetic materials are those that are drawn to magnets, such as iron, steel, nickel, etc. Not all metallic substances, like sodium metal, are inherently magnetic. Certain conductors are magnetic, however there are also non-magnetic conductors.

10. Correct option: (C) Diamagnetic material.

Only paired electrons exist in diamagnetic materials, which are attracted to external magnetic fields. While the ferromagnetic substance is a permanent magnet and draws magnets strongly. Metals that contain a lot of free, unpaired electrons are known as ferromagnetic materials. In contrast, because unpaired electrons are present, paramagnetic material is only weakly attracted to magnetic materials.

11. Correct option: (A) Paramagnetic.

The propensity of paramagnetic materials is to travel from an area of weak magnetic field to one of high magnetic field or to become weakly attracted to a magnet.

12. Correct option: (C) Both A & B.
Platinum and manganese are both paramagnetic metals. Materials classified as paramagnetic exhibit a weak magnetic field attraction. This characteristic is present in manganese and platinum, and it means that these elements will align their magnetic moments with an applied magnetic field, producing a weak attraction towards the magnet. On the other hand, gold (D) is commonly categorised as a diamagnetic substance and is only marginally attracted to magnetic fields.
13. Correct option: (A) Hydrogen, mercury.
Indeed, two examples of diamagnetic materials are hydrogen and mercury. Materials that are diamagnetic are those that are only marginally attracted to magnetic fields. This characteristic, which is shared by hydrogen and mercury, causes them to align their magnetic moments in the opposite direction of an applied magnetic field, producing a weak repulsion.
14. Correct option: (A) Ferromagnetic.
Magnetic fields have a significant attraction for ferromagnetic materials.. Some materials, such as iron and cobalt, are very strongly drawn to magnetic fields. In addition to great attractivity, these materials can be made permanently magnetic.
15. Correct option: (A) Iron.
Ferromagnetic materials are those that are strongly drawn to a magnetic field. Among ferromagnetic materials are those made of iron, cobalt, and nickel.
16. Correct option: (C) Less than.
The effective length of a magnet: In accordance with the accepted definition, the effective length of a magnet is the distance between its two poles, which is never equal to the magnet's geometric length. However, the actual relationship between these two lengths depends on the magnet's shape.
17. Correct option: (D) Aluminium.
Aluminium is not a magnetic substance. When exposed to a magnetic field, it doesn't show very significant magnetic characteristics. Instead, aluminium is categorised as a paramagnetic or non-magnetic substance, which means that it only reacts weakly to magnetic fields.
18. Correct option: (B) Non-magnetic material.
A "Non-magnetic material" is a substance that cannot be made into an artificial magnet or attracted by a magnet.
19. Correct option: (A) Mica.
Usually non-magnetic, mica is a naturally occurring mineral. It is made up of layers of silicate with negligible to no magnetic characteristics. Mica is frequently used in electrical and electronic applications because it is non-magnetic and does not interact with magnetic fields or machinery.
20. Correct option: (A) North and South.
A magnet that is hanging freely will always point northward or southward. The earth is thought of as a massive magnet. The magnet's north pole points to the earth's south pole and vice versa.
21. Correct option: (B) Attract.
Magnets' like and unlike poles are always attracted to one another and repel one other. The north and south poles of two magnets will therefore pull towards one another.
22. Correct option: (A) Repel.
Magnetic repulsion occurs when like poles of magnets produce magnetic forces that push apart from one another because they have the same magnetic orientation.

23. Correct option: (B) Loses its.

The alignment of a magnet's atomic magnetic domains can be upset by heating it above a particular point (known as its Curie temperature) or by rough handling. The magnet becomes less magnetic as a result of this disruption. When something is heated, the heat energy disturbs the alignment of the atoms and causes them to move around. The atomic domains may be physically moved by rough handling. The outcome is a loss of magnetism in both situations, making the magnet less magnetic or fully non-magnetic.

24. Correct option: (B) Demagnetisation.

Demagnetization is the process through which a magnet loses its magnetic characteristics. Demagnetization can happen when a magnet is heated or hammered. Dropping the magnet from a height can also demagnetize it.

25. Correct option: (A) Magnetic compass.

A navigational tool is a magnetic compass. A magnetic compass points northward and southward. This is due to the fact that Earth behaves like a massive magnet, and its poles act like the poles of a massive bar magnet. As a result, a magnet that is hanging freely is drawn to the opposing poles and rotates in that direction. North-South is the direction in question.

26. Correct option: (B) Soft iron.

Typically, soft iron is used to make temporary magnets. Because soft iron loses its magnetic when the magnetising factor is eliminated, this is the reason. Consequently, soft iron is typically used to make temporary magnets.

27. Correct option: (D) Steel.

Steel is used to create permanent magnets because, when the steel bar is magnetised, the dipoles inside the domains are permanently realigned with their axes parallel and pointing in the same direction, changing the form of the domains. As a result, when steel is magnetised, it has permanent magnetism. A similar process occurs when a magnet is applied to an iron bar, but when the magnetism is removed, the mutual repulsive force between the parallel dipoles disturbs and rearranges them. Iron bar so lost its magnetic. As a result, iron briefly retains its magnetism.

28. Correct option: (B) Temporary magnet.

Soft magnets, commonly referred to as temporary magnets, have a low magnetic retention. When subjected to a magnetic field, they can be quickly magnetised, however once the external magnetic field is removed, they frequently lose their magnetism. On the other hand, permanent magnets may sustain their magnetism for a very long time and have strong magnetic retention.

29. Correct option: (B) Magnetic field.

The term "magnetic field" refers to the region around a magnet where its impact can be felt. The magnetic impact of electric currents and magnetic materials is known as a magnetic field. Both the direction and the magnitude (or strength) of the magnetic field at any one site are known.

30. Correct option: (C) Pole.

A magnet's poles are the two ends, yet they are never identical to one another. The north pole is the end that would point north if it were free to move. The south pole is at the other end. These magnetic poles act almost like charges of electricity. There are north and south poles on each magnet.

31. Correct option: (C) magnetic axis.

The magnetic axis is defined as the line that connects a magnet's two poles (just as the earth's magnetic poles are straight lines). A magnetised body's axis, specifically the one connecting the magnetic poles of the earth, unites its poles.

A fictitious line connecting a magnet's north and south poles is known as its magnetic axis. It designates the axis along which the magnetic field is strongest and the location of the strongest magnetic forces. Understanding how magnets interact with one another and with other magnetic objects is made easier by the magnetic axis notion.

32. Correct option: (A) $\text{Magnetic length} = \frac{5}{6} \times \text{Geometric length}$.

The relation between magnetic length and geometric length is given by,

$$\text{Magnetic length} = \frac{5}{6} \times \text{Geometric length}$$

33. Correct option: (C) Magnetic equator.

The magnetic equator is a line that runs through the centre of a magnet and parallel to its axis. The equatorial plane is the plane that contains all equators. The equatorial circle is the region on the equatorial plane where all points are equally spaced from the magnet's centre.

34. Correct option: (A) 15 cm.

Magnetic dipole of a bar magnet $= m \times 2l$

$$\text{Magnetic length}(2l) = \frac{5}{6}(\text{geometric})\text{length}$$

$$2l = \frac{5}{6} \times 18 = 15\text{cm}$$

So, the magnetic length is 15cm.

35. Correct option: (C) Magnetic length.

The magnet has two poles. The magnetic length is the shortest distance between the magnetic field's two poles. Every time, the magnetic length is shorter than the real or geometrical length.

36. Correct option: (C) Magnetic meridian.

The magnetic meridian is the vertical plane that runs through the magnetic axis of a freely suspended magnet at rest. The magnetic meridian, which connects the magnetic north and south poles equivalently, can be thought of as the earth's surface's magnetic force lines.

37. Correct option: (C) South pole.

The South Pole of a bar magnet is the end that faces the geometric south.

38. Correct option: (B) North pole.

The North Pole or North seeking end of the magnet is referred to as this. However, due to the attraction between opposite poles, the magnet's North Pole is actually pointed in the direction of the earth's magnetic South Pole.

39. Correct option: (B) Magnetic pole.

The poles of a magnet are the locations at either end of the magnet where the external magnetic field is strongest. North pole and south pole are the two poles of a magnet. So-called magnetic poles are the two ends of a magnet where the magnet's magnetic force is focused.

40. Correct option: (C) Artificial Magnet.

A piece of iron develops magnetic characteristics and turns into a magnet when it is rubbed with magnetite. Magnetization is the term for this action. The resulting iron object has been artificially made magnetic rather than being a naturally occurring magnet (like a lodestone). As a result, it is referred to as a synthetic magnet. It has magnetic qualities and the ability to draw other magnetic materials to it, but unlike a lodestone (a natural magnet), it was made with purpose.

41. Correct option: (D) All of the above.

A substance that is ferromagnetic is dysprosium. A type of ceramic material called ferrite (option B) is well-known for its potent magnetic characteristics. Since air (option C) is paramagnetic, magnetic fields only faintly attract it.

42. Correct option: (B) They can push or pull some objects without touching.

The primary property of a magnet is its ability to attract or repel other things without coming into contact. The property of the thing determines the attraction and repulsion. When a magnet is repelling an object, we experience a push, and when it is attracting, a pull.

43. Correct option: (B) North Pole to South Pole.

The strength and direction of the magnetic field surrounding a magnet are depicted by imaginary lines known as magnetic lines of force, or magnetic field lines. The field lines are typically shown to start at the North Pole of a magnet, extend outward, curve, and then enter the South Pole. This depicts the route a fictitious magnetic north pole would follow if it were inserted into the magnetic field. A magnetic line of force therefore travels from a magnet's north pole to its south pole.

44. Correct option: (C) Near the ends.

Because the magnetic strength is greatest at the ends of the magnet's poles, when a bar magnet is brought close to it, the most iron dust attaches to both ends of the magnet.

45. Correct option: (D) Magnetic sticker.

Electric fans, electric bells, and some television parts all frequently employ magnets in their construction. However, magnetic stickers are frequently formed of magnetic materials like ferromagnetic sheets or magnetic foils and do not typically rely on electromagnetism for their functionality.

46. Correct option: (B) Magnetisation.

The act of aligning magnetic domains within a material to create a magnetic field is known as magnetization. This is often accomplished by subjecting the material to a magnetic field or by making it magnetic by rubbing it with a magnet.

47. Correct option: (B) Slightly inside.

Although they are slightly within the magnet, the magnetic poles are not located exactly at the ends.

The North Pole and the South Pole are the two poles of the bar magnet. Although these poles are present at both ends of the bar magnet, they are absent from the very end. They are situated not too far from the bar magnet's extreme ends.

LEVEL – II

1. Correct option: (C) Pins sticks at the ends of the magnet.

The pins usually stick at the ends of the magnet when it is placed on a plastic plate with some pins scattered out on it.

The pins on a plastic plate with a magnet on it align with the magnetic field lines of the magnet when the magnet is placed on the plate. At the ends, where the magnetic pull is strongest, they frequently collect and adhere.

2. Correct option: (B) Repulsion.

Both two magnets with their opposing poles brought together and a magnet and a ferrous object like iron can attract one another. So, if both parts are attracted to one another, either both of them may be magnets, or one may be a magnet and the other may be a ferrous material.

Repulsion, on the other hand, is only conceivable when two magnets (with the same poles pulled together) are present. Repulsion is a reliable indicator of magnetism.

3. Correct option: (C) Platinum.

Materials that are only faintly drawn to a magnetic field are said to be paramagnetic. Manganese, Cobalt, and Iron are paramagnetic materials; Platinum, on the other hand, is not; it is often thought of as diamagnetic.

4. Correct option: (B) Closed curves.

Because they go from the North pole outside the magnet's body to the magnet's body and from the South pole to the North pole inside the magnet's body, forming a loop-like structure, magnetic field lines or magnetic lines of force are always closed curves. The magnetic field lines are therefore closed.

5. Correct option: (B) Artificial.

A chunk of iron (or other magnetic material) that has been artificially magnetised is referred to as an artificial magnet.

By encircling the soft iron core with a current-carrying conductor, artificial magnets are created.

Because natural magnets are not particularly magnetically powerful and have an odd and irregular shape, artificial magnets are necessary. Artificial magnets can be moulded into any shape and made extremely powerful.

6. Correct option: (A)

7. Correct option: (A) 5 cm.

A bar magnet's physical length is often thought to be equal to its magnetic geometric length. In this instance, the bar magnet's magnetic geometric length is equal to 5 cm if the distance between the magnetic poles is 5 cm.

8. Correct option: (D) B and C.

The bar magnet's two ends produce the most magnetic force. Both the north and south poles have equal strength. The centre of the magnet produces less force.

9. Correct option: (B) Used to find directions.

Always faces north and south. While magnetic compasses are useful for navigation and direction-finding, fluctuations in the Earth's magnetic field cause them to not always point precisely in a North-South direction. The compass may slightly deviate from true North due to magnetic declination. The compass is used to find directions, however it is not always properly aligned with the North-South compass needle.

10. Correct option: (B) Huge and weak magnet.

Although the Earth can be compared to a large magnet, its magnetic field is not as strong as that of ordinary magnets. It has a magnetic North and South Pole and operates like a bar magnet, although its magnetic field is weaker than that of normal permanent magnets.

11. Correct option: (C) Weakest, strongest.

A magnet's strength increases at its extremities and decreases in the centre. A bar magnet is encircled by three-dimensional lines of force on all sides. While unlike poles attract, like poles repel. The lines of force combine and the magnets pull together when the opposing poles of a magnet are brought together. The lines of force push apart from one another and magnets repel one another when like poles of a magnet are brought together. These forces of attraction and repulsion act on a bar magnet's poles (ends).

12. Correct option: (B) Hammering the magnet.

Demagnetization is the process through which a magnet loses its magnetic characteristics. Demagnetization can happen when a magnet is heated or hammered. Dropping the magnet from a height can also demagnetize it.

13. Correct option: (B) 12cm.

The magnet length of the bar magnet =10 cm

We know that the magnetic length is equivalent to $\frac{5}{6}$th of geometric length.

Thus, $\text{Magnetic length} = \frac{5}{6}(\text{geometriclength})$

$10 = \frac{5}{6}(\text{geometriclength})$

$10 \times \frac{6}{5} = \text{geometriclength}$

$\text{geometriclength} = 12\text{cm}$

So, the geometric length is 12 cm.

14. Correct option: (C) I and II are not correct.

Statement I: Magnet has only attraction property

"Magnet only has an attraction property." - This assertion is untrue. Both repulsion and attraction are characteristics of magnets. Magnets' like poles repel one another while their opposing poles attract.

Statement II: Iron is not a magnet

"Iron is not a magnet." - In general, this claim is untrue. Pure iron can be easily magnetised when in touch with a magnet despite the fact that it is normally not a powerful permanent magnet on its own. It can function as a momentary magnet.

Statement III: Magnet lines emerge from the South Pole

The third claim is that "magnet lines emerge from the South Pole." - This assertion is untrue. A magnet's North Pole produces magnetic field lines that penetrate the South Pole. This is how magnetic field lines convention is commonly described.

15. Correct option: (C) Both I and II are correct.

Statement I: Repulsion is the perfect test of magnetism.

The best way to examine a magnet is through repulsion because a magnetic substance also has attracting properties. Repulsion is the best indicator because anything made of iron, nickel, or cobalt will also be attracted to anything else but not repel it. A surefire indicator of magnetism is repulsion rather than attraction.

Statement II: Only similar magnet poles of two magnets cause repulsion.

"Only identical magnet poles between two magnets repel one another." - This statement is accurate in stating that opposite poles (North-South) attract one another whereas like poles (North-North or South-South) repel one another. It adequately explains magnetism's fundamental idea.

16. Correct option: (A) I is correct.

Statement I: Magnetic lines of force are always closed curves.

"Magnetic lines of force are always closed curves." - This assertion is true. From a magnet's North Pole to its South Pole, magnetic field lines always form closed loops both inside and outside the magnet.

Statement II: The materials that are attracted by magnets are called non-magnetic materials.

"Non-magnetic materials" refers to substances that are drawn to magnets. - This assertion is untrue. Typically, magnetic materials are those that are drawn to magnets. Materials that are not drawn to magnets are said to be non-magnetic.

17. Correct options: (A),(B) and (C).

18. Correct options: (A),(B) and (C).

Ebonite cannot be used to create magnets, but ferromagnetic elements like iron, cobalt, and steel can.

19. Correct options: (B), (C) and (D).

B) A comb won't often be drawn to a magnet unless it has ferromagnetic elements.

(C) A book usually isn't drawn to a magnet if it doesn't have any metal components, such staples or magnetic elements. This is due to the lack of magnetic characteristics and non-response to magnetic fields in paper and the majority of other non-metallic materials.
D) Wood - Since it is a non-magnetic substance, it is typically not drawn to magnets.

20. Correct options: (A) and (B).

Since ferrite magnets have powerful magnetic capabilities and are suitable for high-frequency applications, they are frequently employed in these electronic devices.

21. Correct option: (B) Repels.

A freely dangling magnetic needle will be repelled when the north pole of a bar magnet is brought close to that pole.
The opposite poles of a magnet's poles attract one another whereas the opposite poles repel one another, therefore the north pole repels the south pole and vice versa.

22. Correct option: (C) A force of repulsion.

Similar poles of two magnets repel one another when they come into contact.

23. Correct option: (A) A force of attraction

Each and every magnet has a north pole and a south pole. Magnets with opposite poles pull towards one another whereas those with similar poles repel one another. Therefore, if a ring magnet's north pole and south pole are placed close to one another, they will be attracted to one another.

24. Correct option: (B) North-South.

A magnet's south pole is drawn to the earth's magnetic north pole, which is also its geographic south pole. Therefore, magnets that are suspended freely align in a north-south direction.

25. Correct option: (A) Geometrical North.

Because the magnetic south pole of the earth is located at its geographic north pole, the north end of the freely dangling bar magnet always faces that direction.

26. Correct option: (B) Geometrical South.

The earth exhibits magnetic dipole behaviour. Therefore, a magnet that is freely suspended always points north-south because the south pole of the magnet attracts the north pole of the earth's magnet, which is the planet's geographical south pole.

27. Correct option: (D) Both A & B.

(A) Steel pins are drawn to magnets because iron, a ferromagnetic substance, is frequently present in them. The iron atoms in the pins are brought into alignment by the magnet's powerful magnetic field, which attracts them to the magnet.
(B) Iron nails are magnetically inducted to acquire the opposite polarity when they are brought close to one end of a magnet. Iron nails are drawn to the end of the magnet because dissimilar poles attract one another. Thus, the iron nail gets inducted into a magnet state before being attracted.

28. Correct option: (B) Iron nails.

Iron nails are magnetically inducted to acquire the opposite polarity when they are brought close to one end of a magnet. Iron nails are drawn to the end of the magnet because dissimilar poles attract one another. Thus, the iron nail gets inducted into a magnet state before being attracted.

29. Correct option: (B) Magnetic material.

Magnetic materials are those that are drawn to magnets, such as iron, steel, nickel, etc. Not all metallic substances, like sodium metal, are inherently magnetic.

30. Correct Matching is: (a)-(S),(b)-(P),(c)-(T),(d)-(R),(e)-(Q).

(a) The term "Horseshoe Magnet" refers to a magnet with poles that face one another in a curved configuration that resembles a horseshoe. Horseshoe magnets are made to focus the

magnetic field between their poles, making them useful for lifting and gripping objects among other things.
(b) Magnetic field lines, often referred to as magnetic lines of force, emerge from the North Pole and re-enter at the South Pole to form closed loops entirely within the magnet. They do not curve continuously from North to South outside of the magnet.
The direction and pattern of the magnetic field created by a magnet are typically represented by magnetic field lines. Please add more information so that it can be understood if you're referring to a particular situation or idea.
(c) The magnetic field is the region that surrounds a magnet where its influence can be felt and is defined by the presence of magnetic forces and magnetic characteristics. The interaction of magnets with other magnetic materials and moving charges is mostly explained by magnetic fields, which extend both within and outside the magnet.
(d) A tracing compass is a unique type of compass that can trace and depict the magnetic field lines created by magnets using a tiny, freely suspended magnetic needle. It is a helpful tool for educating people about magnetic field patterns and for doing so.
(e) By aligning with the Earth's magnetic field, a magnetic compass is a tool used by sailors and navigators to determine geographic direction, specifically the four cardinal directions (North, South, East, and West). It is a crucial tool for both on- and off-shore navigation.

31. Correct Matching is: (a)-(S),(b)-(P),(c)-(T),(d)-(R).

(a) Neodymium magnets are extremely powerful due to their high coercivity (resistance to demagnetization) and high degrees of magnetic saturation, which enable them to produce huge magnetic fields.
(b) Due to their modest repulsion when exposed to a magnetic field, gold and bismuth are regarded as diamagnetic materials. The atomic or molecular magnetic moments in these materials are aligned in the opposite direction to the applied magnetic field, which results in this behaviour.
-Individual atoms or molecules' magnetic moments are frequently coupled in diamagnetic materials so that, in the absence of an external magnetic field, their net magnetic moment is zero. These moments align in the field's anticlockwise direction when a magnetic field is applied, producing a weak repulsive force.
(c) Materials that are strongly drawn to a magnetic field are said to be ferromagnetic materials.
In ferromagnetic compounds include metal ions that are grouped in discrete areas known as domains. Every domain act as a tiny magnet. A ferromagnetic substance's domains are randomly oriented in their unmagnetized region, and their magnetic moments cancel. A substance experiences a large magnetic effect when a magnetic field is applied because all of the domains get oriented in that field's direction, as indicated in the diagram below. Even after the magnetic field is withdrawn and the ferromagnetic material turns into a permanent magnet, this domain ordering still exists. Nickel, cobalt, dysprosium, is an example of a ferromagnetic material.
(d) Aluminium and manganese are frequently regarded as paramagnetic metals. They are therefore only marginally drawn to magnetic fields. These materials exhibit a weak magnetic pull when exposed to a magnetic field because the magnetic moments of their atoms tend to line up with the field.
Nitrogen and oxygen, two substances that are likewise regarded as paramagnetic compounds, are found in air. Additionally, these materials exhibit mild magnetic field attraction.